TPE2 (D), TPE3 (D)

Installation and operating instructions



TPE2 (D), TPE3 (D)
Installation and operating instructions
(all available languages)
http://net.grundfos.com/qr/i/93024260

GRUNDFOS

TPE2 (D), TPE3 (D)

English (GB)
Installation and operating instructions 4
Appendix A 116
Appendix B 119

Original installation and operating instructions

Table of contents

1. General information 5
1.1 Hazard statements 5
1.2 Notes 5

2. Abbreviations and definitions 6

3. Product introduction 7
3.1 Nameplate 7
3.2 Type key 8

4. General description 10
4.1 Radio module 10
4.2 Battery 10
4.3 Cybersecurity 10

5. Receiving the product 12
5.1 Delivery 12
5.2 Transporting the product 12
5.3 Inspecting the product 12
5.4 Handling the product 12

6. Applications 13
6.1 Pumped liquids 13

7. Installing the product 13
7.1 Location 13
7.2 Mechanical installation 15
7.3 Electrical connection 19

8. Startup 37
8.1 Flushing the pipe system 37
8.2 Priming the pump 37
8.3 Starting the pump 38
8.4 Shaft seal run-in 38

9. Operating conditions 39
9.1 Maximum number of starts and stops 39
9.2 Alternating operation of twin-head pumps . . 39
9.3 Liquid temperature 39
9.4 Ambient temperature 39
9.5 Operating pressure or test pressure 39
9.6 Inlet pressure 39
9.7 Electrical data 39
9.8 Sound pressure level 39
9.9 Environment 39
9.10 Installation altitude 40
9.11 Humidity 40
9.12 Cooling the motor 40

10. User interfaces 41
10.1 Operating panels, HMI 200 and 201 41
10.2 Operating panels, HMI 300 and 301 44

11. Grundfos GO 50
11.1 Communication 50
11.2 Menu overview for Grundfos GO 51

12. Setting the product 55
12.1 Heat energy monitor 55
12.2 Setpoint 55
12.3 Operating mode 55
12.4 Set manual speed 56
12.5 Set user-defined speed 56
12.6 Control mode 56
12.7 Setting the proportional pressure 63
12.8 FLOWLIMIT 63
12.9 Automatic night setback 63
12.10 Analog inputs 64
12.11 Grundfos Direct Sensor 66
12.12 Pt100/1000 inputs 68
12.13 Digital inputs 69
12.14 Digital inputs/outputs 70
12.15 Signal relay (Relay outputs) 72
12.16 Analog output 73
12.17 Controller (Controller settings) 74
12.18 Operating range 75
12.19 External setpoint function 76
12.20 Predefined setpoints 77
12.21 Limit-exceeded function 78
12.22 Ramps 80
12.23 Skip band 80
12.24 Standstill heating 80
12.25 Alarm handling 80
12.26 Motor bearing monitoring 81
12.27 Service 81
12.28 Communication 81

13. Selecting control mode 89

14. Priority of settings 92

15. Grundfos Eye 93

16. Identification of the motor model 95

17. Changing the position of the operating panel 95

18. Grundfos GO Link 97
18.1 Communication 97
18.2 Ethernet 97
18.3 Grundfos MI 301 97
18.4 Grundfos PC Tool Link 97

19. Servicing the product 98
19.1 Maintenance 98
19.2 Service 98

20. Cleaning the product 99

21. Fault finding 100
21.1 The motor does not run when started . . 100
21.2 Motor alarm when the power supply is switched on 100
21.3 Occasional motor alarm 100
21.4 No motor alarm, but the pump does not run 100
21.5 The pump performance is not constant . 100
21.6 The pump runs but delivers no water . . 101
21.7 The pump runs backwards when switched off 101
21.8 Leakage in shaft seal 101
21.9 Noise 101

22. Technical data 102
22.1 Technical data, three-phase motors . . . 102
22.2 Inputs and outputs 105

23. Other technical data 106
23.1 Ecodesign Directive 106
23.2 EMC (electromagnetic compatibility) . . . 106
23.3 Enclosure class 108
23.4 Insulation class 108
23.5 Standby power consumption 108
23.6 Cable entry sizes 109
23.7 Torques 109
23.8 Sound pressure level 110

24. Factory settings 113

25. Disposing of the product 115

26. Document quality feedback 115

1. General information



Read this document before you install the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1.1 Hazard statements

The symbols and hazard statements below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The hazard statements are structured in the following way:



SIGNAL WORD

Description of the hazard

Consequence of ignoring the warning

- Action to avoid the hazard.

1.2 Notes

The symbols and notes below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



Observe these instructions for explosion-proof products.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.

Tips and advice that make the work easier.

2. Abbreviations and definitions

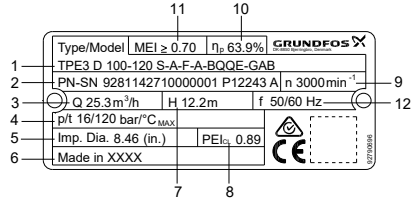
AI	Analog input
AL	Alarm, out of range at lower limit
AO	Analog output
AU	Alarm, out of range at upper limit
CIM	Communication interface module
Current sinking	The ability to draw current into the terminal and guide it towards earth in the internal circuitry.
Current sourcing	The ability to push current out of the terminal and into an external load which must return it to earth.
DI	Digital input
DO	Digital output
ELCB	Earth-leakage circuit breaker
FM	Functional module
GDS	Grundfos Digital Sensor Factory-fitted sensor in some Grundfos pumps
GENIbus	Proprietary Grundfos fieldbus standard
GFCI	Ground fault circuit interrupter USA and Canada
GND	Earth
Grundfos Eye	Status indicator light
LIVE	Low voltage with the risk of electric shock if the terminals are touched
OC	Open collector: Configurable open-collector output
PE	Protective earth
PELV	Protective extra-low voltage A voltage that cannot exceed ELV under normal conditions and under single-fault conditions, except earth faults in other circuits.
RCD	Residual-current device
SELV	Safety extra-low voltage A voltage that cannot exceed ELV under normal conditions and under single-fault conditions, including earth faults in other circuits.
TPE2	Single-head pump without factory-fitted differential-pressure and temperature sensor
TPE2 D	Twin-head pump without factory-fitted differential-pressure and temperature sensor
TPE3	Single-head pump with factory-fitted differential-pressure and temperature sensor
TPE3 D	Twin-head pump with factory-fitted differential-pressure and temperature sensor

3. Product introduction

These installation and operating instructions apply to the Grundfos TPE2, TPE2 D and TPE3, TPE3 D pumps.

The pumps are fitted with frequency-controlled permanent-magnet motors for three-phase mains connection.

3.1 Nameplate



TM083495



TM083487

Example of nameplate

Pos.	Description	
1	Type designation	
2	Identification code	
	92811427	Product number
	10000001	Serial number
	P1	Production site code
	2243	Production year and week (YYWW)
	A	Service model
3	Nominal flow rate	
4	Pressure rating and maximum temperature	
5	Actual impeller diameter	
6	Country of origin	
7	Nominal pump head	
8	WRAS approval or Pump Energy Index (PEI) PEI_{CL}: constant load PEI_{VL}: variable load	

Pos.	Description
9	Rated pump speed
10	Hydraulic efficiency at best efficiency point
11	Minimum efficiency index
12	Frequency

3.2 Type key

Example: TPE3 D 65-120 S-A-F-A-BQQE-FAB

Code	Type	Explanation
TPE3		Pump range, electronically speed-controlled pump
	TPE2	Without built-in sensor
	TPE3	Built-in differential-pressure and temperature sensor
D		Twin-head pump
		Nominal diameter of inlet and outlet ports [mm]
65		Maximum head [decimeters (dm)]
-120	S	Built-in differential-pressure and temperature sensor
S	N	Without built-in sensor
-A		Code for pump version:
	A	Basic version
	I	PN 6 flange
	X	Special version
-F		Code for pipe connection:
	F	DIN flange
-A		Code for materials:
	A	Basic version (Cast-iron pump housing with Cast-iron/ stainless steel 1.1301/Composite PES/PP 30 % GF impeller/Composite PES/PP 40 % GF impeller)
	B	Cast iron pump housing with bronze impeller
	I	Stainless steel 1.4308 pump housing and motor stool
	O	Ductile cast-iron pump housing with cast-iron impeller
	S	Cast-iron pump housing with stainless steel 1.4408 impeller
	Y	Ductile cast-iron pump housing with bronze impeller
-BQQE		Code for shaft seal including other plastic and rubber pump parts, except the neck ring.
-F		Code for NEMA motors [kW (hp)]:
	C	0.25 (0.33)
	D	0.37 (0.5)
	E	0.55 (0.75)
	F	0.75 (1)
	G	1.1 (1.5)
	H	1.5 (2)
	I	2.2 (3)
	J	3 (4)
	K	3.7 (5)
	L	5.5 (7.5)
	M	7.5 (10)
	N	11 (15)
	O	15 (20)

Code	Type	Explanation
	P	18.5 (25)
	Q	22 (30)
A		Code for phase & voltage and other information:
	A	1 × 200-240 V
	B	3 × 200-240 V
	C	3 × 440-480 V
	D	3 × 380-500 V
	W	Not for sale in North America
	Y	Out of DOE scope (department of energy)
B		Code for speed variant [rpm]:
	A	1450-2200
	B	2900-4000
	C	4000-5900

3.2.1 Codes for shaft seal

Example: BQQE

Code	Type	Explanation
B		Grundfos type designation
	A	O-ring seal with fixed seal driver
	B	Rubber bellows seal
	D	O-ring seal, balanced
	G	Bellows seal with reduced seal faces
	R	O-ring seal with reduced seal faces
Q		Material of rotating face
	A	Carbon, antimony-impregnated
	B	Carbon, resin-impregnated
	Q	Silicon carbide
Q		Material of stationary seat
	B	Carbon, resin-impregnated
	Q	Silicon carbide
	U	Tungsten carbide
E		Material of secondary seal
	E	EPDM
	P	NBR rubber
	V	FKM
	F	FXM

4. General description

The pumps are fitted with frequency-controlled permanent-magnet motors for three-phase mains connection.

The product must only be connected to protected network subnets with strict access control.

4.1 Radio module

4.1.1 GLoWpan

CAUTION
Radiation
Minor or moderate personal injury

- Locate the product at a minimum distance of 20 cm from any body parts. Human tissue may be heated by RF energy.

Installers and end users must be provided with these installation and operating instructions and operating conditions for satisfying RF exposure compliance.

The product incorporates a class 1 radio module for remote control. You can use the module anywhere in the EU without restrictions.

For installation in the USA and Canada, see the appendix.

Via the built-in radio module, the product can communicate with other MGE motors.

The product contains a class 1 radio. Grundfos will support the product with security updates for at least 2 years from production of the unit.

GLoWpan information

Frequency of operation	2405-2480 MHz
Modulation type	GP O-QPSK
Data rate	1 Mbps
Transmit power	5 dBm EIRP with internal antenna

4.1.2 Bluetooth

The product incorporates a Bluetooth (BLE) module for remote control. You can use the module anywhere in the EU without restrictions.

For installation in the USA and Canada, see the appendix.

Via the built-in Bluetooth module, the product can communicate with Grundfos GO.

The product contains a Bluetooth (BLE) module. Grundfos will support the product with security updates for at least 2 years from production of the unit.

Bluetooth information

Frequency of operation	2400 - 2483.5 MHz
Modulation type	GFSK
Data rate	2 Mbps
Transmit power	5 dBm EIRP with internal antenna

4.2 Battery

A Li-ion battery is fitted in the FM310 and FM311 functional modules.

The Li-ion battery complies with the Battery Directive (2006/66/EC). The battery does not contain mercury, lead or cadmium.

WARNING
Intoxication or risk of chemical burn
Death or serious personal injury

- The battery can cause severe or fatal injuries in 2 hours or less if it is swallowed or placed inside any part of the body. In such an event, seek medical attention immediately.

- The replacement or servicing of batteries must be carried out by a qualified person.
- The battery contained within this product, whether new or used, is hazardous and is to be kept away from children.

4.3 Cybersecurity

The product must only be connected to protected network subnets with strict access control.

According to the standard EN 18031-1:2024, the RJ45 connector used for connecting to IP networks is considered an exposed network interface, along with the GENIbus TCP protocol used on it, which is considered an exposed service.

4.3.1 Network interfaces and services

In the factory default state, the product exposes the following network interfaces:

Interface	Description
RJ45	Wired Ethernet / IP connectivity

The following services are exposed by the product over its network interfaces in the factory default state.

Interface	Service	Description
RJ45	GENIpro TCP	It is used for connecting the product to a service tool on an engineering workstation. Pairing is required before communication can take place. The GENIpro TCP pairing state is persistent.

5. Receiving the product

5.1 Delivery

The pump is delivered from the factory in a carton with a wooden bottom, specially designed for transport by fork-lift truck or a similar vehicle.

5.2 Transporting the product

WARNING
Falling objects
Death or serious personal injury
- Secure the product during transportation to prevent it from tilting or falling down.

CAUTION
Crushing of feet
Minor or moderate personal injury
- Wear safety shoes when moving the product.

5.3 Inspecting the product

Before you install the product, do the following:

1. Check that the product is as ordered.
2. Check that no visible parts have been damaged.

If parts are damaged or missing, contact your local Grundfos sales company.

5.4 Handling the product

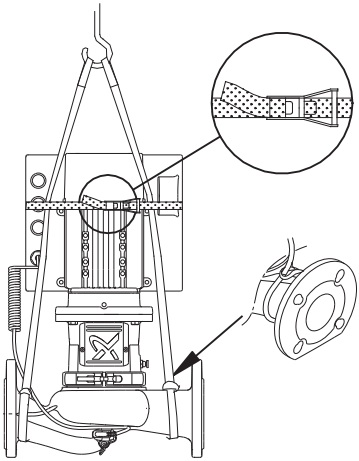
Observe local regulations setting limits for manual lifting or handling. The motor weight is stated on the nameplate.

CAUTION
Back injury
Minor or moderate personal injury
- Use lifting equipment.

CAUTION
Crushing of feet
Minor or moderate personal injury
- Wear safety shoes and attach lifting equipment to the motor eyebolts when handling the product.

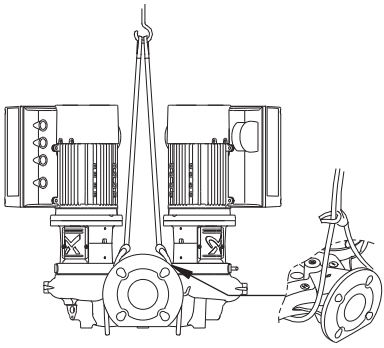
Do not lift the product by the terminal box.

Lift the pump by means of nylon straps. See figures TPE2, TPE3 and TPE2 D, TPE3 D.



TM057914

TPE2, TPE3



TM057915

TPE2 D, TPE3 D

6. Applications

The pumps are designed to circulate hot or cold water in residential, institutional and industrial applications:

- heating systems
- district heating plants
- central heating systems for blocks of flats
- air-conditioning systems
- cooling systems.

In addition, the pumps are used for liquid transfer and water supply in for instance:

- washing systems
- domestic hot-water systems
- industrial systems in general.

To ensure optimum operation, the dimensioning range of the system must fall within the performance range of the pump.

6.1 Pumped liquids

WARNING
Contact with hazardous liquids
Death or serious personal injury

- If the pump is used for a liquid which is injurious to health, it will be classified as contaminated. In such cases, take proper precautions to avoid injury to health when operating or working on the pump.
- Wear personal protection equipment.

The pump is suitable for thin, clean, non-aggressive and non-flammable liquids, not containing solid particles or fibres that may attack the pump mechanically or chemically.

Examples:

- Central heating system water. The water must meet the requirements of accepted standards on water quality in heating systems.
- Cooling liquids.
- Domestic hot water.
- Industrial liquids.
- Softened water.

The pumping of liquids with a density and/or kinematic viscosity higher than that of water will have the following effects:

- a considerable pressure drop
- a drop in hydraulic performance
- a rise in power consumption.

In such cases, fit the pump with a bigger motor. If in doubt, contact Grundfos.

The EPDM O-rings fitted as standard are primarily suitable for water.

If the water contains mineral or synthetic oils or chemicals or if other liquids than water are pumped, chose the O-rings accordingly.

7. Installing the product

7.1 Location

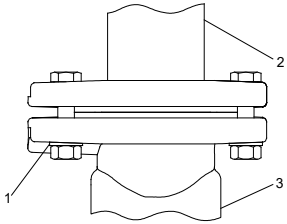
WARNING
Hot or cold surface
Death or serious personal injury

- Make sure that no one can accidentally come into contact with hot or cold surfaces.

In order to maintain the UL mark, additional requirements apply to the equipment. See section Installation in the USA and Canada.

The pump must be sited in a dry, well-ventilated, but frost-free position.

When installing pumps with oval bolt holes in the pump flange (PN 6/10), use washers as shown in the figure below.



TM010683

Use of washers for oval bolt holes

Pos.	Description
1	Washer
2	Installation side
3	Pump side

Arrows on the pump housing show the direction of flow of liquid through the pump.

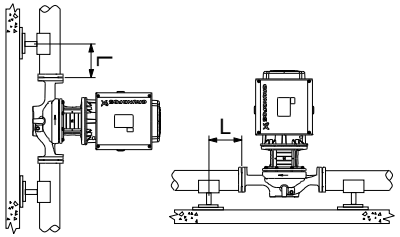
Pumps with motors smaller than 11 kW can be installed in horizontal or vertical pipes.

The motor must never fall below the horizontal plane.

Pumps with motors of 11 kW and up may only be installed in horizontal pipes with the motor in vertical position.

However, some TP, TPE pumps of 11 kW and up may be suspended directly in the pipes (horizontally or vertically). See the section TP, TPE pumps from 11 kW and up suspended in the pipes.

In installations where the pump is suspended directly in the pipes, the pump can support the pipe length L on both sides of the pump (L less than 3 × DN), see the figure below. In installations where the pump is suspended directly in the pipes, the pump must be lifted and held in correct position by means of ropes or similar until both pump flanges are completely fastened to the pipe flanges.



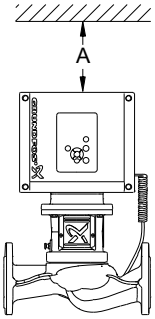
TM086591

Pump suspended directly in the pipes

The motor must never be installed below the horizontal plane.

For inspection and motor or pump head removal, the following clearance is required above the motor:

- 300 mm for motors up to and including 4.0 kW.
- 1 m for motors of 5.5 kW and up.

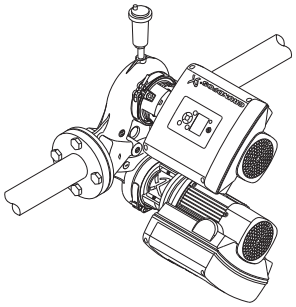


TM086590

Required clearance above the motor

Motor size	A
0.25 - 4.0 kW	≥ 300 mm
5.5 kW and up	≥ 1 m

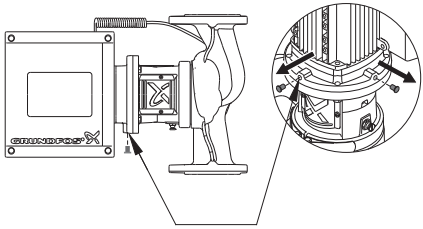
Twin-head pumps installed in horizontal pipes must be fitted with an automatic vent in the upper part of the pump housing. The automatic vent is not supplied with the pump.



TM057983

Automatic vent

If the liquid temperature falls below the ambient temperature or if the pump is installed outside, condensation may form in the motor during inactivity. In this case, make sure that the drain hole in the motor flange is open and points downwards.



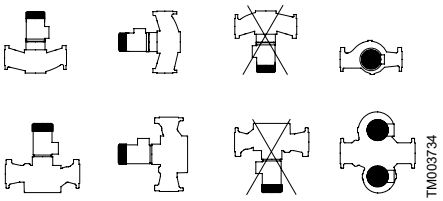
TM057917

Drain hole in motor flange

If twin-head pumps are used for pumping liquids with a temperature below 0 °C (32 °F), condensed water may freeze and cause the shaft to get stuck. The problem can be remedied by installing heating elements. For pumps with motors smaller than 11 kW, the pump must be installed with the motor shaft in horizontal position.

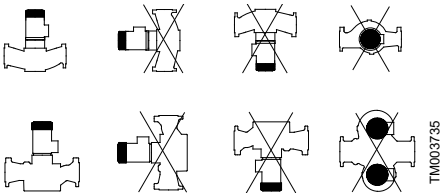
7.2 Mechanical installation

You can install TP pumps with motors smaller than 11 kW in horizontal or vertical pipes.



Installation of motor sizes smaller than 11 kW

You can suspend pumps with motors smaller than 11 kW directly in the pipes, provided the pipes can support the pump. If not, install the pump on a mounting bracket or base plate. Only install TP pumps with motors of 11 kW and up in horizontal pipes with the motor in vertical position. Always install the pump on an even and rigid foundation.



Installation of motor sizes of 11 kW and up

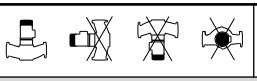
Note: The motor must never point downwards.

The product must be installed in a place with access control to prevent unauthorized access to the product.

Install the pumps in such a way that strain from the pipes is not transferred to the pump housing.

However, you can suspend some TP, TPE pumps of 11 kW and up directly in the pipes, either horizontally or vertically. See the table below:

TP, TPE pumps from 11 kW and up suspended in the pipes

Pump type	PN 6	PN 16	P2 [kW]		
50 Hz					
TPE2/TPE3 40-630	-	●	11	●	-
TPE2/TPE3 50-760	-	●	11	●	-
TPE2/TPE3 50-810	-	●	15	●	-
TPE2/TPE3 50-820	-	●	18.5	●	-
TPE2/TPE3 50-830	-	●	22	●	-
TPE2/TPE3 65-310	-	●	2.2	-	●
TPE2/TPE3 65-760	-	●	11	-	●
TPE2/TPE3 65-810	-	●	15	-	●
TPE2/TPE3 65-820	-	●	18.5	-	●
TPE2/TPE3 65-830	-	●	22	-	●
TPE2/TPE3 80-460	-	●	11	-	●
TPE2/TPE3 80-510	-	●	15	-	●
TPE2/TPE3 80-560	-	●	18.5	●	-
TPE2/TPE3 80-610	-	●	22	●	-
TPE2/TPE3 100-360	-	●	11	-	●
TPE2/TPE3 100-460	-	●	15	-	●
TPE2/TPE3 100-510	-	●	18.5	-	●
TPE2/TPE3 100-560	-	●	22	-	●
TPE2/TPE3 125-260	-	●	11	-	●
TPE2/TPE3 125-310	-	●	22	●	-
TPE2/TPE3 125-320	-	●	15	-	●
TPE2/TPE3 125-410	-	●	18.5	-	●
TPE2/TPE3 125-460	-	●	22	-	●
TPE2/TPE3 150-230	-	●	11	●	-
TPE2/TPE3 150-240	-	●	15	-	●
TPE2/TPE3 150-250	-	●	22	-	●
TPE2/TPE3 150-270	-	●	18.5	-	●
TPE2/TPE3 150-280	-	●	22	●	-
TPE2/TPE3 200-130	-	●	11	●	-
TPE2/TPE3 200-150	-	●	15	●	-
TPE2/TPE3 200-190	-	●	18.5	●	-
TPE2/TPE3 200-200	-	●	22	●	-

7.2.1 Pipes

Fit isolating valves on either side of the pump to avoid draining the system if the pump needs to be cleaned or repaired.

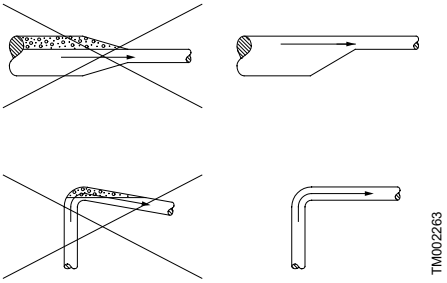
The pump is suitable for pipeline mounting, provided that the pipes are adequately supported either side of the pump.

When installing the pipes, make sure that the pump housing is not stressed by the pipes.

The inlet and outlet pipes must be of an adequate size, taking the pump inlet pressure into account.

To avoid sediment buildup, do not fit the pump at the lowest point of the system.

Install the pipes so that air pockets are avoided, especially on the inlet side of the pump.



Correct pipes on the inlet side of the pump

DANGER
Pump can explode
Death or serious personal injury

- The pump is not allowed to run against a closed valve except during startup. Operating against a closed valve at an extended period of time will cause an increase in temperature and the formation of steam and may result in damages to or explosion of the pump housing. The valve must be kept open during operation.

If there is any danger of the pump running against a closed outlet valve, ensure a minimum liquid flow through the pump by connecting a bypass or a drain to the outlet pipe. The drain can for instance be connected to a tank. The minimum flow rate must be at least 10 % of the maximum flow rate. The flow rate and head are stated on the pump nameplate.

7.2.2 Changing terminal box positions

DANGER
Electric shock
Death or serious personal injury

- Switch off the power supply to the motor and to the signal relays. Wait at least 5 minutes before you make any connections in the terminal box. Make sure that the power supply cannot be accidentally switched on.

The terminal box can be turned to four positions, in 90° steps.

Change the terminal box position as follows:

1. If necessary, remove the coupling guards using a screwdriver. Do not remove the coupling.
2. Remove the screws securing the motor to the pump.
3. Turn the motor to the required position.
4. Replace and tighten the screws.
5. Replace the coupling guards.

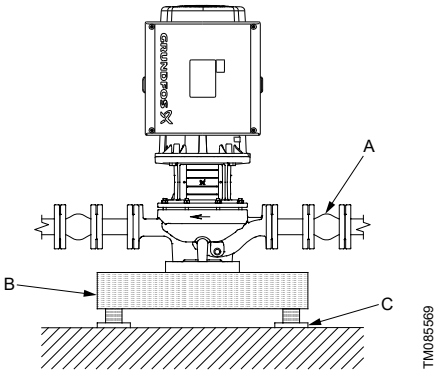
7.2.3 Elimination of noise and vibrations

In order to achieve optimum operation and minimum noise and vibration, consider vibration damping of the pump. Generally, always consider this for pumps with motors of 11 kW and up.

Smaller motor sizes, however, may also cause undesirable noise and vibration.

Noise and vibration are generated by the revolutions of the motor and pump and by the flow in pipes and fittings. The effect on the environment is subjective and depends on correct installation and the state of the rest of the system.

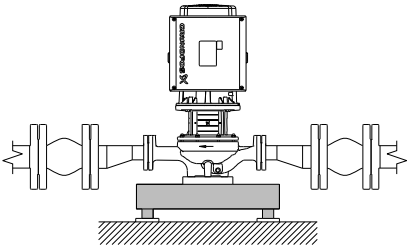
Elimination of noise and vibrations is best achieved by means of a concrete foundation, vibration dampers and expansion joints.



Foundation

Pos.	Description
A	Expansion joint
B	Concrete pedestal
C	Vibration damper

At high liquid velocities (greater than 5 m/s), we recommend that you fit larger expansion joints matching the pipes.



Pump installed with larger expansion joints

7.2.4 Foundation

We recommend that you install the pump on a concrete foundation which is heavy enough to provide permanent and rigid support for the entire pump. The foundation must be capable of absorbing any vibration, normal strain or shock. As a rule of thumb, the weight of the concrete foundation must be 1.5 times the weight of the pump. Place the pump on the foundation and fasten it.

7.2.5 Base plate

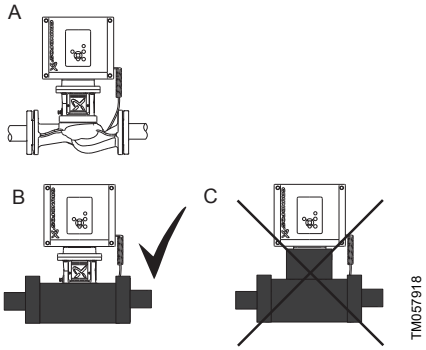
Single-head pumps have two or three tapped holes in the bottom of the pump housing which can be used for fitting a Grundfos base plate to the pump. The base plate is available as an optional extra.

Twin-head pumps have four tapped holes in the bottom of the pump housing. For some twin-head pumps, a base plate consisting of two halves is available.

7.2.6 Insulation

Do not insulate the motor stool as this will trap any vapour escaping from the shaft seal, thus causing corrosion. Covering the motor stool with insulation will also make inspection and service difficult.

Follow the guidelines in figure below when insulating the pump.



Insulation of TPE2, TPE3 pumps

Pos.	Description
A	Without insulation
B	Correct insulation
C	Incorrect insulation

7.2.7 Frost protection

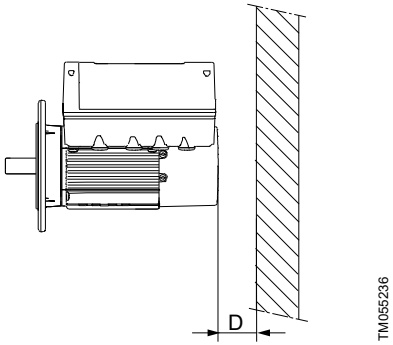
Drain pumps which are not being used during periods of frost to avoid damage.

7.2.8 Cable entries

See the size of the cable entries in the section on other technical data.

7.2.9 Ensuring motor cooling

Leave at least 50 mm between the end of the fan cover and a wall or other fixed objects.



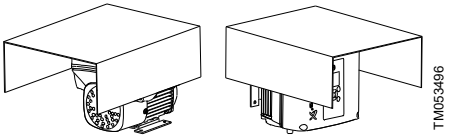
Minimum distance, D, from the motor to a wall or other fixed objects

7.2.10 Outdoor installation

If you install the motor outdoors, provide the motor with a cover and open the drain holes to avoid condensation on the electronic components. See figure Examples of covers, not supplied by Grundfos.

When fitting a cover to the motor, observe the guidelines in section Ensuring motor cooling.

The cover must be sufficiently large to ensure that the motor is not exposed to direct sunlight, rain or snow. Grundfos does not supply covers. We therefore recommend that you have a cover built for the specific application. In areas with high humidity, we recommend that you connect the motor permanently to the mains supply and activate the built-in standstill heating function. See section Standstill heating.



Examples of covers, not supplied by Grundfos

7.3 Electrical connection

WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box. Make sure that the power supply cannot be switched on accidentally.
- Check that the supply voltage and frequency correspond to the values stated on the nameplate.

CAUTION
Sharp element
Minor or moderate personal injury

- When installing the wiring in the terminal box, wear protective gloves to avoid cutting your hands on sharp edges.

The user or the installer is responsible for correct earthing and protection according to local regulations.

All electrical connections must be carried out by qualified persons.

7.3.1 Electrical supply systems

Power supply network and earthing systems

If you want to supply the product through an IT network, make sure that you have a suitable product variant. If you are in doubt, contact Grundfos.

The internal EMC filter remains connected, and subsequently no reduced leakage current variant is available.

Supply line types

Model J:

The product is not suitable for use on corner earthed grids in installations more than 2000 m above sea level.

Model K:

For applications in accordance with IEC 61800-5-1, the maximum voltage to ground must not exceed 277 V.

The product is not suitable for use for corner earthed grids.

Maximum installation altitude: 3500 m.

- TN-S earthing system
- TN-C earthing system

- TN-C-S earthing system
- TT earthing system

7.3.2 Protection against electric shock, indirect contact

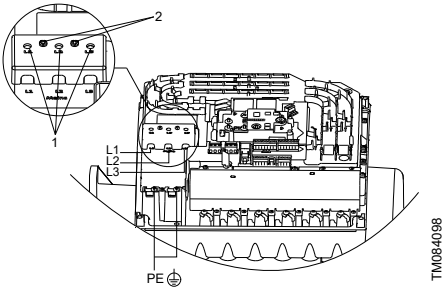
WARNING
Electric shock
Death or serious personal injury
- Connect the product to protective earth and provide protection against indirect contact in accordance with local regulations.

Protective-earth conductors must have a yellow and green (PE) or yellow, green and blue (PEN) colour marking.

7.3.3 Cover for the power cables

Model K is equipped with a cover for the power cables.

The cover is attached to the insulation cover with 2 screws (2) and is equipped with 3 voltage measuring holes (1) for the respective phases (L1, L2, L3).



The cover for the power cables must be installed before switching on the product.

7.3.4 Protection against mains voltage transients

The product is protected against mains voltage transients in accordance with EN 61800-3.

7.3.5 Motor protection

The product incorporates thermal protection against slow overloading and blocking. No external motor protection is required.

Model J: The product includes load- and speed-sensitive motor-overload protection.

Model K: The product includes load- and speed-sensitive motor-overload protection with thermal memory retention.

7.3.6 Connecting an external switch

We recommend that you connect the product to an external switch.

1. Connect the switch via terminals 2 (DI1) and 6 (GND).
   A jumper is added from factory.
2. Enable the **External stop** function.
   Default setting from factory.

7.3.7 Cable requirements

7.3.7.1 Cable cross-section

WARNING
Electric shock
Death or serious personal injury
- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box.
- Follow the wiring diagrams and local regulations.
- Use branch-circuit protection fuses.
- Comply with local regulations as to cable cross-sections.
- Use the recommended fuse size.
- Connect the cables to terminals by applying the recommended tightening torque.

WARNING
Fire hazard
Death or serious personal injury
- Comply with local regulations as to cable cross-sections.
- Use the recommended fuse size.
- Connect the cables to terminals by applying the recommended tightening torque.

Make sure that the cables are secured with cable glands providing strain relief.

Recommended cable type for Model K: H07RN-F.

Cable cross-section data for MGE motors

3 × 380-500 V, 50/60 Hz, Model J

Speed [rpm]	Power P2 [kW]	Supply voltage [V]	Nominal current [A]	Cable cross-section [mm²]
1450-2200	2.2	3 × 380-500	4.3 - 3.6	0.5
	3.0	3 × 380-500	5.8 - 4.6	1
	4.0	3 × 380-500	7.7 - 6.0	1.5
	5.5	3 × 380-500	10.5 - 8.4	2
	7.5	3 × 380-500	14.1 - 11.1	2
2900-4000	3.0	3 × 380-500	5.8 - 4.8	1
	4.0	3 × 380-500	7.6 - 6.2	1.5
	5.5	3 × 380-500	10.3 - 8.2	2
	7.5	3 × 380-500	14.1 - 11.2	2
	11.0	3 × 380-500	20.3 - 16.0	3.5
4000-5900	3.0	3 × 380-500	5.9 - 4.8	1
	4.0	3 × 380-500	7.6 - 6.1	1.5
	5.5	3 × 380-500	10.5 - 8.3	2
	7.5	3 × 380-500	14.2 - 11.3	2
	11.0	3 × 380-500	20.3 - 16.0	3.5

3 × 200-240 V, 50/60 Hz, Model J

Speed [rpm]	Power P2 [kW]	Supply voltage [V]	Nominal current [A]	Cable cross-section [mm²]
3400-4000	2.2	3 × 200-240	7.8 - 6.5	1.5
	3	3 × 200-240	10.5 - 8.8	2
	4	3 × 200-240	14.1 - 11.8	2
	5.5	3 × 200-240	19.6 - 16.3	3.5

3 × 380-480 V, 50/60 Hz, Model K

Speed [rpm]	Power P2 [kW]	Supply voltage [V]	Nominal current [A]	Cable cross-section [mm²]	Cable cross-section [AWG]
1450-2200	11	3 × 380-480	20.2 - 16.4	6	10
	15	3 × 380-480	26.7 - 21.8	6	8
	18.5	3 × 380-480	33.2 - 26.9	10	8
	22	3 × 380-480	39.2 - 31.5	10	8

Speed [rpm]	Power P2 [kW]	Supply voltage [V]	Nominal current [A]	Cable cross-section [mm^2]	Cable cross-section [AWG]
2900-4000	15	3 × 380-480	26.7 - 22	6	8
	18.5	3 × 380-480	33 - 27.8	10	8
	22	3 × 380-480	39.2 - 31.5	10	8

3 × 200-240 V, 50/60 Hz, Model K

Speed [rpm]	Power P2 [kW]	Supply voltage [V]	Nominal current [A]	Cable cross-section [mm^2]	Cable cross-section [AWG]
3400-4000	7.5	3 x 200-240	25.6 - 21.4	10	8
	11	3 x 200-240	37.4 - 31.4	10	8

3 × 400-480 V, 50/60 Hz, Model K

Speed [rpm]	Power P2 [kW]	Supply voltage [V]	Nominal current [A]	Cable cross-section [mm^2]	Cable cross-section [AWG]
3500-4000	26	3 × 400-480	43.8 - 37.6	16	6

Conductors

Conductor types

Model J: Use stranded or solid copper conductors only.

Model K: Use stranded copper conductors only.

Conductor temperature ratings

Model J: Use minimum 60 °C copper conductors.

Model K: Use minimum 75 °C copper conductors. The wire sizes for the mains supply must be sized for a wire size which is suitable for at least 125% of the rated input current of the motor drive units.

7.3.7.2 Three-phase connections

The cables in the terminal box must be as short as possible. However, the separated protective-earth conductor must be so long that it is the last one to be disconnected in case the cable is inadvertently pulled out of the cable entry.

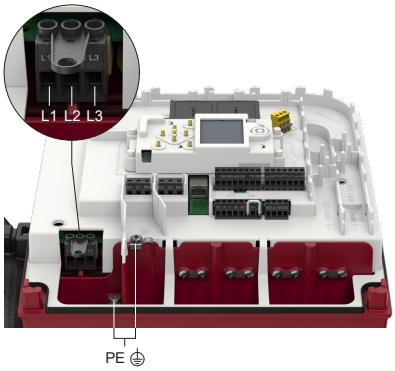
To maintain the cURus mark, additional requirements apply to the equipment. See the appendix concerning installation in the USA and Canada.

Model J: To avoid loose connections, ensure that the terminal block for L1, L2 and L3 is pressed home in its socket when the power cable has been connected.

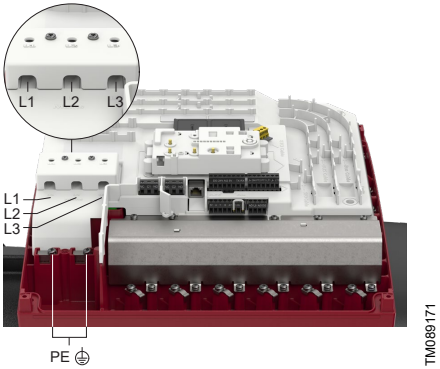
Model K: To avoid loose connections, ring terminals must be used. Ensure that ring terminals are short enough to stay within the terminal cover.

Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Power supply connection on a three-phase product



Model J



Model K

Pos.	Description
L1	Phase 1
L2	Phase 2
L3	Phase 3
PE	Protective earth

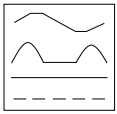
7.3.8 Additional protection

7.3.8.1 Residual-current circuit breakers

WARNING
Electric shock
Death or serious personal injury

- This product can cause a DC current in the protective-earth conductor. If a residual current-operated protective (RCD) or monitoring (RCM) device is used for protection in case of direct or indirect contact, only an RCD or RCM of Type B is allowed on the supply side of this product.

The residual-current circuit breaker must be marked.

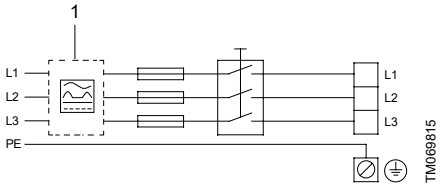


Take into account the total leakage current of all the electrical equipment in the installation.

This product may cause a direct current in the protective-earth conductor.

Connection example for three-phase supply

The figure shows an example of a mains-connected three-phase motor with a main switch, a backup fuse and a residual-current circuit breaker, type B.



Pos.	Description
1	Residual-current circuit breaker, type B
L1	Phase 1
L2	Phase 2
L3	Phase 3
PE	Protective earth

7.3.8.2 Overvoltage and undervoltage protection

Overvoltage and undervoltage may occur in case of unstable power supply or a faulty installation. The product stops if the voltage falls outside the permissible voltage range. The product restarts automatically when the voltage is within the permissible voltage range. The product requires no additional protection relay.

The product is protected against transients from the power supply according to EN 61800-3. In areas with high lightning intensity, we recommend external lightning protection.

Overvoltage category:

The product is approved for Overvoltage category III rating.

7.3.8.3 Overload protection

The motor-current protection settings are fixed for each motor variants. The settings ensure that the motor is protected against overtemperature in all operating states with regard to supply voltage and shaft load, including a blocked shaft.

The motors are current controlled and will respond by reducing the speed if the shaft load increases more than 10 % of the nominal load.

If the shaft load forces the speed down to minimum speed, the motor shuts down.

A sudden increase in the motor current caused by a fault where the peak of the motor current is increased 60 % above nominal will cause the motor to shut down within 0.5 ms.

The product requires no additional protection.

7.3.8.4 Overtemperature protection

The motor is thermally protected by a temperature measurement in the drive. It can handle the lack of airflow over the motor in case the fan cover is blocked. It also means that the protection has a built-in memory retention.

The time from start to shutdown due to overtemperature is therefore always longer when starting at a motor temperature close to the ambient temperature compared with restarting after a shutdown due to overtemperature.

7.3.8.5 Protection against phase unbalance

Phase unbalance on the power supply must be minimised. The three-phase motor must be connected to a power supply with a quality corresponding to IEC 60146-1-1, class C. This also ensures long life of the components.

7.3.8.6 Short-circuit current

The product's electronic power output short-circuit protection circuitry meets the requirements of IEC 60364-4-41:2005/AMD1:-, Clause 411.

Model J: If a short circuit occurs, the pump can be used on a power supply delivering not more than 5000 RMS symmetrical amperes, 600 V maximum.

Model K: Suitable for use on a circuit capable of delivering not more than 5000 rms symmetrical amperes, when protected by gG fuses. See the section on fuse sizes.

7.3.8.7 Functional modules

The functional modules are different types of add-on boards containing various types of input and output terminals for the user to connect different types of sensors, for example switches and relays.

The product can only contain one functional module at the time.

The following functional modules are available:

- FM110 (option)
- FM310 (standard)
- FM311 (option) [1)]

1) Without Bluetooth (BLE).

The following functional modules are available:

- FM310 (standard)
- FM311 (option) [2)]

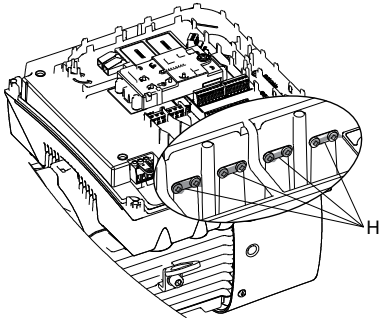
2) Without Bluetooth (BLE).

The selection of module depends on the application and the required number of inputs and outputs.

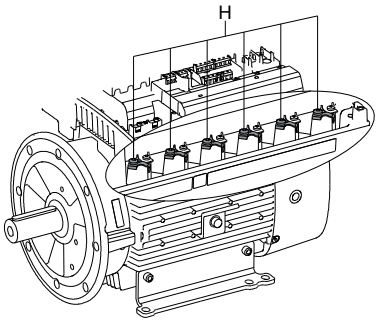
Cable connections

The screen of signal cables and bus connection cables must be connected to ground via one of the earth clamps (H).

See the section on signal cables and bus connection cables.



Model J



Model K

Functional module, FM110

Inputs and outputs

The module has these connections:

- two analog inputs
- two digital inputs or one digital input and one open-collector output
- Grundfos Digital Sensor input and output
- one signal relay output
- GENIbus/Modbus connection
- two Safe Torque Off (STO) inputs
- Bluetooth (BLE) connection.

Signal relay 1

LIVE: You can connect supply voltages up to 250 VAC to the output.

SELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or safety extra-low voltage to the output as desired.



WARNING
Electric shock
Death or serious personal injury

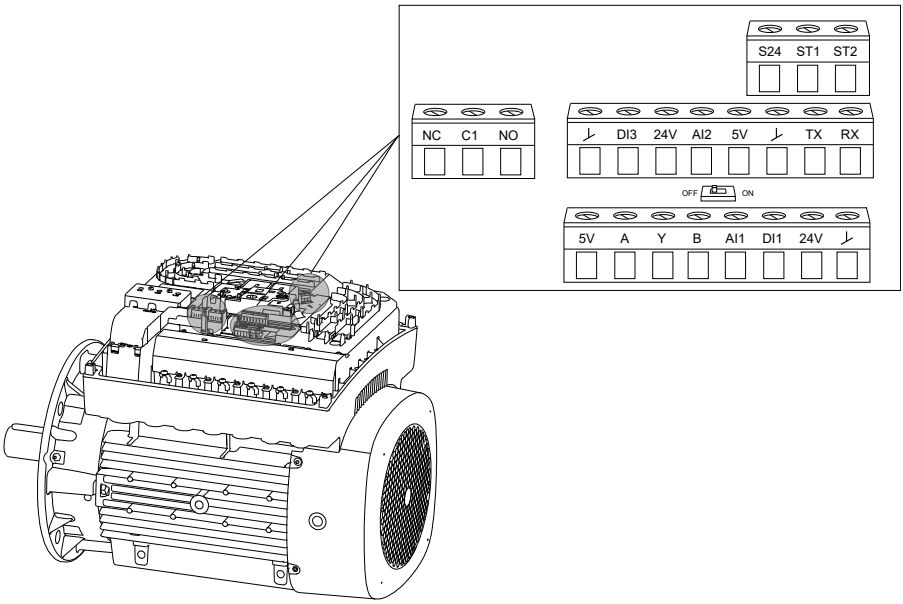
- Make sure that the wires to be connected to the connection relays below are separated from each other by reinforced insulation in their entire lengths.

The inputs and outputs are internally separated from the mains-conducting parts by reinforced insulation and galvanically separated from other circuits. All control terminals are supplied with safety extra-low voltage (SELV), ensuring protection against electric shock.

Cables for the relays must be double insulated or reinforced and rated at least 250V/2A.

The Ethernet cable must be rated at least Cat5e/Cat6 with screening.

The 250V contacts of the alarm relays of these devices must not be connected directly to the supply mains, but energized by an isolated power supply or transformer with galvanic isolation.



TM082861

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1: LIVE or SELV
C1	Common	
NO	Normally open contact	
GND	GND	Signal ground
DI3	DI3/OC1	Digital input/output, configurable Open collector: Maximum 24 V resistive or inductive
24V	+24 V	Power supply
AI2	AI2	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
5V	+5 V	Power supply to a potentiometer or sensor
GND	GND	Signal ground
TX	GDS TX	Grundfos Digital Sensor output
RX	GDS RX	Grundfos Digital Sensor input
5V	+5 V	Power supply to a potentiometer or sensor
A	GENIbus, A	GENIbus, A (+) / Modbus, D1 (+)
Y	GENIbus, Y	GENIbus, GND / Modbus, GND

Terminal	Type	Function
B	GENIbus, B	GENIbus, B (-) / Modbus, D0 (-)
AI1	AI1	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
DI1	DI1	Digital input, configurable  Digital input 1 is factory-set to be start or stop input where an open circuit results in stop. A jumper has been factory-fitted between terminals DI1 and GND. Remove the jumper if digital input 1 is to be used as external start or stop or any other external function.
24V	+24 V	Power supply
GND	GND	Signal ground
S24	+24 V (STO)	Power supply to the Safe Torque Off inputs
ST1	STO1	Safe Torque Off - Input 1
ST2	STO2	Safe Torque Off - Input 2

Functional module, FM310 and FM311

Inputs and outputs



The FM311 functional module does not include Bluetooth connection.

The module has these connections:

- three analog inputs
- one analog output
- two dedicated digital inputs
- two configurable digital inputs or open-collector outputs
- Grundfos Digital Sensor input and output
- two Pt100/1000 inputs
- two LiqTec sensor inputs
- two signal relay outputs
- GENIbus/Modbus connection
- two Safe Torque Off (STO) inputs
- Ethernet connection
- Bluetooth (BLE) connection. [3)]

3) FM311 is without Bluetooth.

Signal relay 1

LIVE: You can connect supply voltages up to 250 VAC to the output.

SELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or safety extra-low voltage to the output as desired.

Signal relay 2

SELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or safety extra-low voltage to the output as desired.

Connection terminals for inputs and outputs



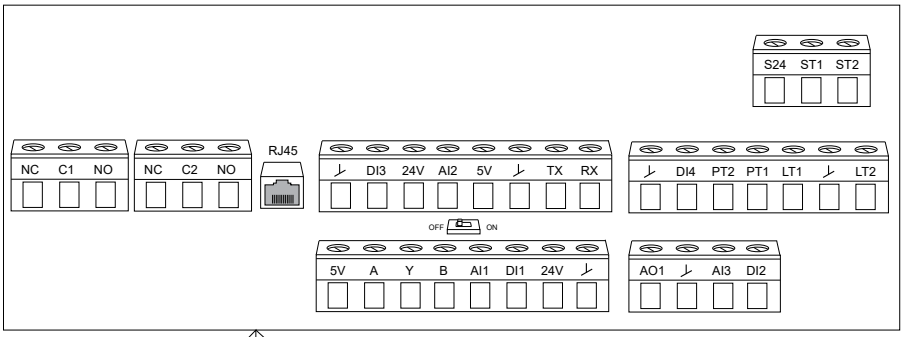
WARNING
Electric shock
Death or serious personal injury

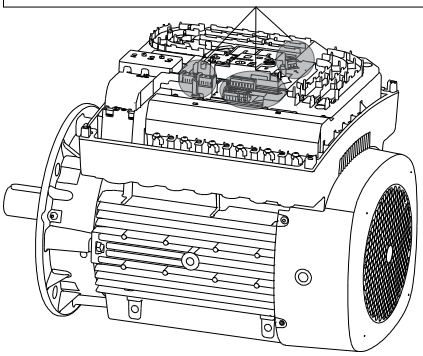
- Make sure that the wires to be connected to the connection groups below are separated from each other by reinforced insulation in their entire lengths.

The inputs and outputs are internally separated from the mains-conducting parts by reinforced insulation and galvanically separated from other circuits. All control terminals are supplied with safety extra-low voltage (SELV), ensuring protection against electric shock.

Cables for the relays and the Ethernet cable must be rated at least 250 V/2 A.

The relays are approved for overvoltage category II, whether power is supplied from a transformer or the power supply.





TM082862

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1: LIVE or SELV
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2: SELV only
C2	Common	
NO	Normally open contact	
RJ45	Ethernet	Ethernet communication
GND	GND	Signal ground
DI3	DI3/OC1	Digital input/output, configurable Open collector: Maximum 24 V resistive or inductive
24V	+24 V	Power supply
AI2	AI2	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
5V	+5 V	Power supply to a potentiometer or sensor
GND	GND	Signal ground
TX	GDS TX	Grundfos Digital Sensor output

Terminal	Type	Function
RX	GDS RX	Grundfos Digital Sensor input
GND	GND	Signal ground
DI4	DI4/OC2	Digital input/output, configurable Open collector: Maximum 24 V resistive or inductive
PT2	Pt100/1000 input 2	Pt100/1000 sensor input 2
PT1	Pt100/1000 input 1	Pt100/1000 sensor input 1
LT1	LiqTec sensor input 1	LiqTec sensor input 1 White conductor
GND	GND	Signal ground Brown and black conductors
LT2	LiqTec sensor input 2	LiqTec sensor input 2 Blue conductor
5V	+5 V	Power supply to a potentiometer or sensor
A	GENIbus, A	GENIbus, A (+) / Modbus, D1 (+)
Y	GENIbus, Y	GENIbus, GND / Modbus, GND
B	GENIbus, B	GENIbus, B (-) / Modbus, D0 (-)
AI1	AI1	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
DI1	DI1	Digital input, configurable  Digital input 1 is factory-set to be start or stop input where an open circuit results in stop. A jumper has been factory-fitted between terminals DI1 and GND. Remove the jumper if digital input 1 is to be used as external start or stop or any other external function.
24V	+24 V	Power supply
GND	GND	Signal ground
AO1	AO	Analog output: • 0-20 mA or 4-20 mA • 0-10 V.
GND	GND	Signal ground
AI3	AI3	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
DI2	DI2	Digital input, configurable
S24	+24 V (STO)	Power supply to the Safe Torque Off inputs
ST1	STO1	Safe Torque Off - Input 1
ST2	STO2	Safe Torque Off - Input 2

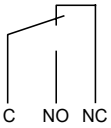
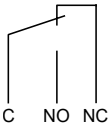
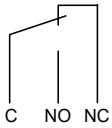
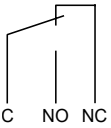
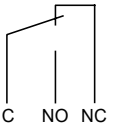
7.3.8.8 Signal relays

The motor has two outputs for potential-free signals via two internal relays. You can set the signal outputs to **Operation**, **Pump running**, **Ready**, **Alarm** and **Warning**.

The functions of the two signal relays are shown in the table below.

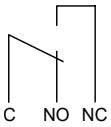
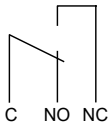
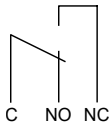
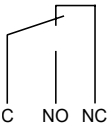
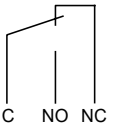
Grundfos Eye is off

The power is off.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					-

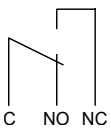
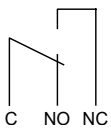
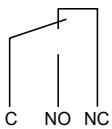
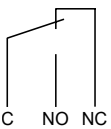
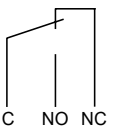
Grundfos Eye is rotating green

The pump runs in **Normal** mode in open or closed loop.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Normal** **Min.** or **Max.**

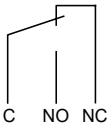
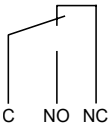
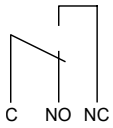
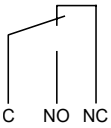
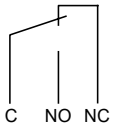
Grundfos Eye is rotating green

The pump runs in **Manual** mode.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Manual**

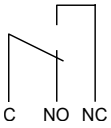
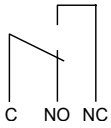
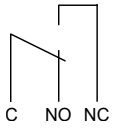
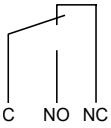
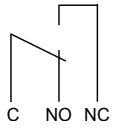
Grundfos Eye is permanently green

The pump is ready for operation but is not running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Stop**

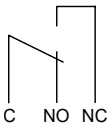
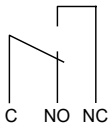
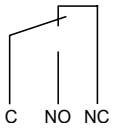
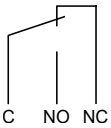
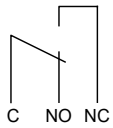
Grundfos Eye is rotating yellow

There is a warning, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Normal** **Min.** or **Max.**

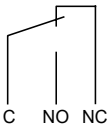
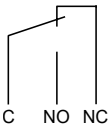
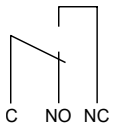
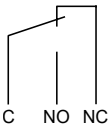
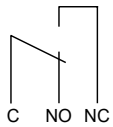
Grundfos Eye is rotating yellow

There is a warning, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Manual**

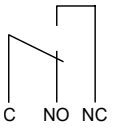
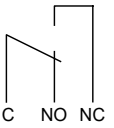
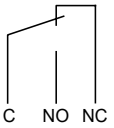
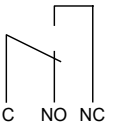
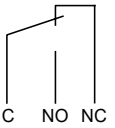
Grundfos Eye is permanently yellow

There is a warning, and the pump is stopped via a **Stop** command.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Stop**

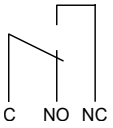
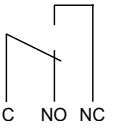
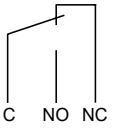
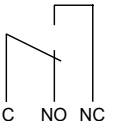
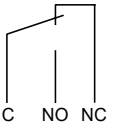
Grundfos Eye is rotating red

There is an alarm, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Normal** **Min.** or **Max.**

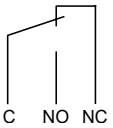
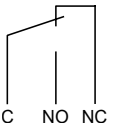
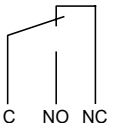
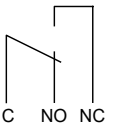
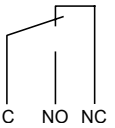
Grundfos Eye is rotating red

There is an alarm, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Manual**

Grundfos Eye is flashing red

The pump is stopped due to an alarm.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Stop**

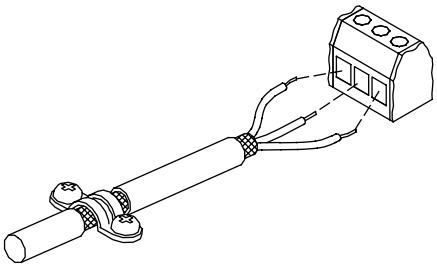
7.3.8.9 Signal cables

Use screened cables with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2 for the external on/off switch, digital inputs, setpoint and sensor signals.

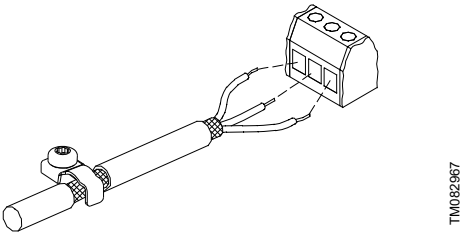
The wires in the motor terminal box must be as short as possible.

Connecting signal cables

1. Connect the screens of the cables to the frame at both ends with good connection. The screens must be as close as possible to the terminals.



Model J



Model K

2. Connect the signal cables to the terminals.
3. Depending on the model, tighten one or two earth clamp screws.

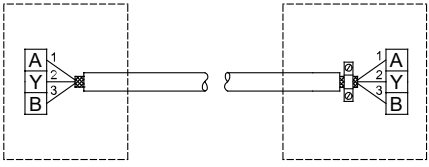
   See the section on functional modules.

7.3.8.10 Bus connection cable

Connecting a 3-core bus cable, GENIbus

For the bus connection, use a screened 3-core cable with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2.

- If the motor is connected to a unit with a cable clamp identical to the one on the product, connect the screen to the cable clamp.
- If the unit has no cable clamp, leave the screen unconnected at this end.



Connecting a 3-core bus cable, Modbus

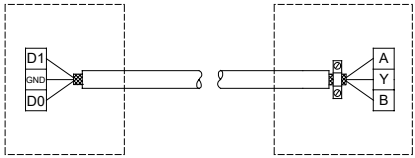
A screened, twisted-pair cable must be used. The cable screen must be connected to protective earth at both ends.

Recommended connection

Terminal	Modbus	Colour code	Data signal
A	D1	Yellow	Positive
B	D0	Brown	Negative
Y	Common/ GND	Grey	Common/ GND

Fitting the cable

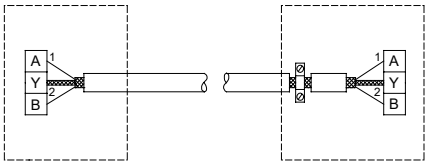
1. Connect the yellow conductor to terminals D1 and A.
2. Connect the brown conductor to terminals D0 and B.
3. Connect the grey conductor to terminals Common/GND and Y.
4. Connect the cable screens to protective earth via the earth clamp.



It is important to connect the screen to protective earth through the earth clamp and to connect the screen to protective earth in all units connected to the bus line.

Connecting a 2-core bus cable

- Connect a screened 2-core bus cable as follows:



Bus signal

The product enables serial communication via an RS-485 input. The communication is carried out according to the Grundfos GENIbus protocol and enables connection to a building management system or another external control system.

Via a bus signal, you can remote-set operating parameters, such as setpoint and operating mode. At the same time, the product can provide status information about important parameters, such as the actual value of the control parameter, input power and fault indications, via the bus.

Contact Grundfos for further information.

If you use a bus signal, the local settings made via Grundfos GO or the HMI 300 or 301 operating panel will be overruled. In case the bus signal fails, the product will run with the local settings made via Grundfos GO or the HMI 300 or 301 operating panel.

7.3.8.11 Installing a communication interface module

WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box. Make sure that the power supply cannot be switched on accidentally.

Always use an antistatic service kit when handling electronic components. This prevents static electricity from damaging components.

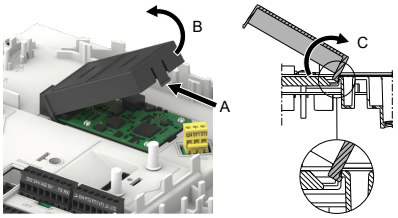
When unprotected, the component must be placed on the antistatic cloth.



TM051590

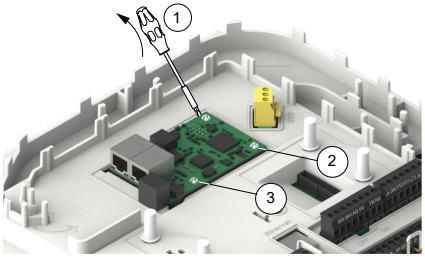
Antistatic service kit

1. Disconnect the power supply.
2. Remove the terminal box cover.
3. Remove the plug connection for the CIM module.
4. Remove the cover of the CIM module by pressing the locking tab (A) and lifting the end of the cover (B). Then lift the cover off the hooks (C).



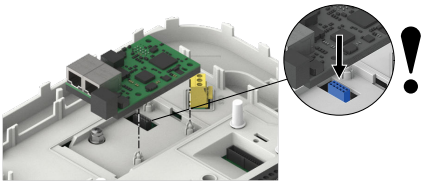
TM087823

5. Remove the screw connecting the frame connection of the CIM module to the functional module.
6. Insert a screwdriver at the three plastic holders and loosen the CIM module from the isolation cover.



TM087824

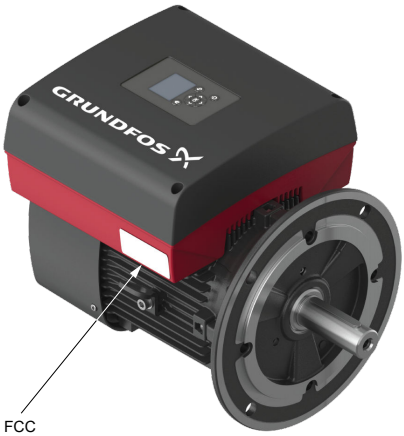
7. Gently lift the CIM module away from the isolation cover so that the connecting plug is not damaged.
8. Fit the new CIM module by aligning it with the plastic holders and the connecting plug. Press home the module using your fingers.



TM087825

9. Fit the frame screw in the CIM module.

10. Fit the cover by leading the slits into the end with the plug connection and clicking the locking tabs onto the isolation cover.
11. Press down the module using your fingers. Check that the plug has been pressed home.
12. If the module is supplied with an FCC label, fix the label on the terminal box.



13. Fit the terminal box cover.

8. Startup

8.1 Flushing the pipe system

Before starting up the pump for the first time, clean the pipe system thoroughly by flushing it and filling it with clean water.

Do not use the pump for flushing the pipe system.

The warranty does not cover any damage caused by flushing the pipe system by means of the pump.

8.2 Priming the pump

Fill the pump with liquid and vent the pump before starting the pump. To ensure correct venting, the vent screw must point upwards.

Closed systems or open systems where the liquid level is above the pump inlet

1. Close the outlet isolating valve and loosen the vent screw in the motor stool. See figure Position of vent screw.

WARNING
Escaping hot or cold liquids
Death or serious personal injury

- Pay attention to the direction of the vent hole, and make sure that the escaping hot or cold liquid does not cause injury to persons or damage to the equipment.
- Wear personal protection equipment.

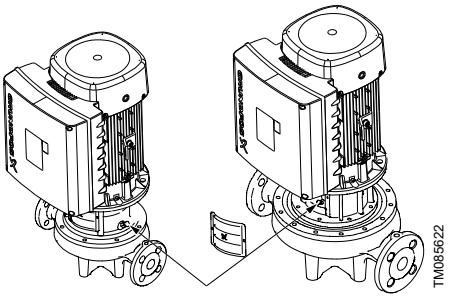
2. Slowly open the isolating valve in the inlet pipe until a steady stream of liquid runs out of the vent hole.
3. Tighten the vent screw and completely open the isolating valve(s).

Open systems where the liquid level is below the pump inlet

Make sure that the inlet pipe and the pump is filled with liquid and vented before you start the pump.

1. Close the outlet isolating valve and open the isolating valve in the inlet pipe.
2. Loosen the vent screw. See figure Position of vent screw.
3. Remove the plug from one of the pump flanges, depending on the pump location.
4. Pour liquid through the priming port until the inlet pipe and the pump are filled with liquid.
5. Replace the plug and tighten securely.
6. Tighten the vent screw.

You can fill the inlet pipe with liquid and vent it before you connect it to the pump. You can also install a priming device before the pump.



Position of vent screw

8.3 Starting the pump

1. Before starting the pump, completely open the isolating valve on the inlet side of the pump and leave the outlet isolating valve almost closed.
2. Start the pump. See section User interfaces.
3. Vent the pump by loosening the vent screw in the motor stool until a steady stream of liquid runs out of the vent hole. See figure Position of vent screw.

WARNING
Escaping hot or cold liquids
Death or serious personal injury

- Pay attention to the direction of the vent hole, and make sure that the escaping hot or cold liquid does not cause injury to persons or damage to the equipment.
- Wear personal protection equipment.

4. When the pipe system has been filled with liquid, slowly open the outlet isolating valve until it is completely open.

Related information

10. User interfaces

8.4 Shaft seal run-in

The seal faces are lubricated by the pumped liquid, meaning that there may be a certain amount of leakage from the shaft seal.

When the pump is started up for the first time, or when a new shaft seal is installed, a certain run-in period is required before the leakage is reduced to an acceptable level. The time required for this depends on the operating conditions, i.e. every time the operating conditions change, a new run-in period will be started.

Under normal conditions, the leaking liquid will evaporate.

As a result, no leakage will be detected.

However, liquids such as kerosene will not evaporate.

The leakage may therefore be seen as a shaft seal failure.

9. Operating conditions

9.1 Maximum number of starts and stops

The number of starts and stops via the power supply must not exceed four times per hour.

When switched on via the power supply, the pump will start after approximately 5 seconds.

If a higher number of starts and stops is desired, use the input for external start/stop when starting or stopping the pump.

When started via an external on/off switch, the pump will start immediately.

9.2 Alternating operation of twin-head pumps

On twin-head pumps, the duty and backup pumps must be alternated on a regular basis, i.e. once a week, to ensure an even distribution of the operating hours on both pumps. The pumps alternate automatically. See section Multipump setup (Setup of multi-pump system).

If twin-head pumps are used for pumping domestic hot water, the duty and backup pumps must be alternated on a regular basis, i.e. once a day, to avoid blocking of the backup pump due to deposits such as calcareous deposits. The pumps alternate automatically. See section Multipump setup (Setup of multi-pump system).

If twin-head pumps are alternated by an external controller, it is recommended to alternate the pumps at least once every 24 hours to avoid seizing shaft seals.

Related information

12.28.26 Multipump function

9.3 Liquid temperature

-25 °C (~ -13 °F) up to +140 °C (~ +284 °F).

The maximum liquid temperature depends on the shaft seal type and the pump type.

Depending on the cast-iron version and the pump application, the maximum liquid temperature may be limited by local regulations and laws.

The maximum liquid temperature is stated on the pump nameplate.

If the pump is used for liquids at high temperatures, the life of the shaft seal may be reduced.

9.4 Ambient temperature

9.4.1 Ambient temperature during storage and transportation

Minimum: -30 °C.

Maximum: 60 °C.

9.4.2 Ambient temperature during operation

	3 × 200-240 V	3 × 380-500 V
Minimum	-20 °C	-20 °C
Maximum	40 °C	50 °C

The motor can operate with the rated power output, P2, at 50 °C, but continuous operation at higher temperatures reduces the expected product life. If the motor is to operate at ambient temperatures between 50 and 60 °C, select an oversized motor. Contact Grundfos for further information.

9.5 Operating pressure or test pressure

The pressure test has been made with water containing anti-corrosive additives at a temperature of 20 °C (~ 68 °F).

Pressure stage	Operating pressure		Test pressure	
	[bar]	[MPa]	[bar]	[MPa]
PN 6	6	0.6	10	1.0
PN 6 / PN 10	10	1.0	15	1.5
PN 16	16	1.6	24	2.4

9.6 Inlet pressure

To ensure optimum and quiet pump operation, the inlet pressure, system pressure, must be adjusted correctly. See the table in the Appendix.

For the calculation of specific inlet pressures, contact the local Grundfos company or see the data booklet for TP, TPD, TPE, TPED, TPE2, TPE2 D, TPE3 and TPE3,D on Grundfos Product Center (https://product-selection.grundfos.com/).

9.7 Electrical data

See section Technical data.

For specific motor data, see the motor nameplate.

9.8 Sound pressure level

See section Sound pressure level.

9.9 Environment

Non-aggressive and non-explosive atmosphere.

9.10 Installation altitude

The installation altitude is the height above sea level of the installation site.

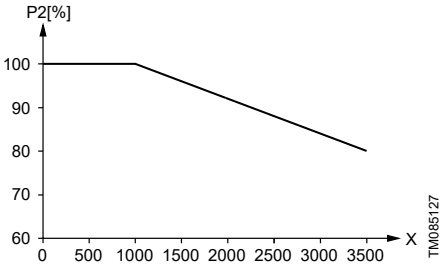
Products installed up to 1000 m above sea level can be loaded 100 %.

- **Model J:** The product is not suitable for use on corner earthed grids in installations more than 2000 m above sea level.
- **Model K:** The product is not suitable for use on corner earthed grids.

  The motors can be installed up to 3500 m above sea level.

Products installed more than 1000 m above sea level must not be fully loaded due to the low density and consequent low cooling effect of the air.

The motor output power (P2) in relation to the altitude above sea level is shown in the graph.



Pos.	Description
P2	Motor output power [%]
X	Altitude [m]

9.11 Humidity

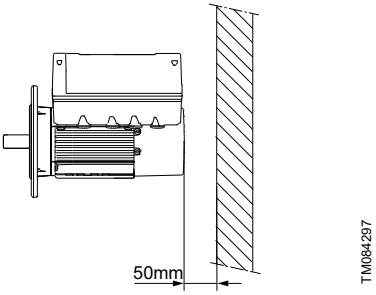
Description	Percentage
Maximum humidity (non-condensing)	95 %

If the humidity is constantly high and above 85 %, open the drain holes in the drive-end flange to vent the motor.

If you install the motor in moist surroundings or areas with high humidity, ensure that the bottom drain hole is open. As a result, the motor becomes self-venting, allowing water and humid air to escape. When you open the drain hole, the enclosure class of the motor will be lower than standard.

9.12 Cooling the motor

- Install the motor allowing a distance of minimum 50 mm between the end of the fan cover and the wall or another fixed object.



MGE

- Position the product with sufficient space around.
- Make sure that the temperature of the cooling air does not exceed 50 °C.
- Keep cooling fins and fan blades clean.

10. User interfaces



WARNING
Hot surface
Death or serious personal injury
- Touch only the buttons on the operating panel. The product may be very hot.



WARNING
Electric shock
Death or serious personal injury
- If the operating panel is cracked or perforated, replace it immediately. Contact the nearest Grundfos sales company.

You can change the settings by means of the following user interfaces:

- HMI 200 operating panel
- HMI 201 operating panel [4)]
- HMI 300 operating panel
- HMI 301 operating panel [4)]
- Grundfos GO application.

4) HMI without a radio module.

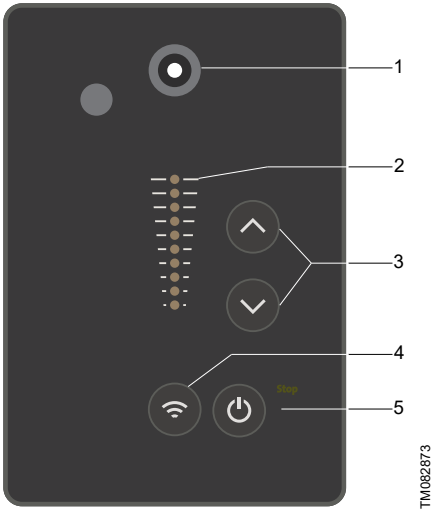
All settings are saved if the power supply is switched off.

Related information

8.3 Starting the pump

10.1 Operating panels, HMI 200 and 201

Pump variant	Fitted as standard	Option
TPE2, TPE2 D	●	-
TPE3, TPE3 D	-	●



Pos.	Symbol	Description
1		**Grundfos Eye:** The indicator light shows the operating status of the product.
2	-	Light fields for indication of the setpoint.
3		**Up**/**Down**: The buttons change the setpoint.
4		**Communication:** The button enables communication with Grundfos GO and other products of the same type.
5		**Start/Stop**: Press the button to make the product ready for operation or to start and stop the product. **Start**: If you press the button when the product is stopped, the product starts if no other functions with higher priority have been enabled. **Stop**: If you press the button when the product is running, the product always stops. When you press the button, the stop icon appears at the bottom of the display.

10.1.1 Setting the setpoint in constant parameter mode

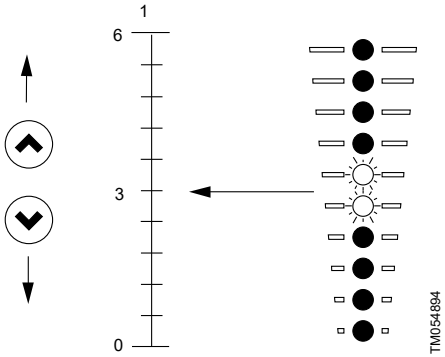
The following applies for motors set to operate in **Const. other val.**

Set the desired setpoint by pressing the **Up** or **Down** buttons.

The green light fields on the operating panel indicate the setpoint set.

The following example applies to a pump or motor in an application where a pressure sensor gives a feedback to the pump or motor. The sensor has been set manually, and the pump or motor does not automatically register a connected sensor.

Light fields 5 and 6 are activated, indicating a desired setpoint of 3 bar with a sensor measuring range from 0 to 6 bar. The setting range is equal to the sensor measuring range.

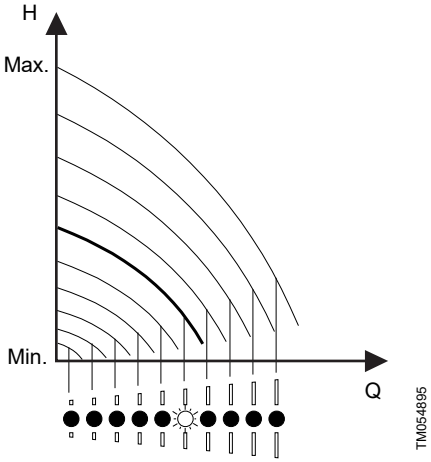


10.1.2 Setting the setpoint in constant curve mode

Set the desired setpoint by pressing the **Up** or **Down** buttons.

The green light fields on the operating panel indicate the setpoint set.

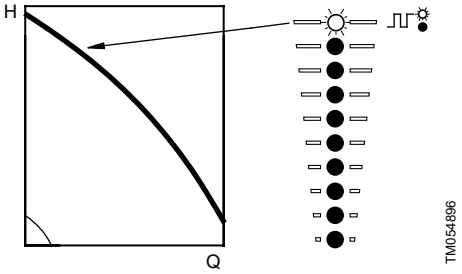
Example: In **Constant curve** mode, the motor output is between minimum and maximum speed defined by **Operating range**.



10.1.3 Setting to maximum speed

The motor must not be in operating mode **Stop**.

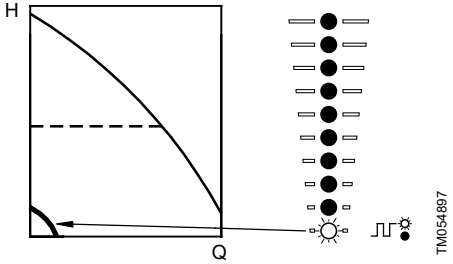
- Press and hold the **Up** button until the top light field is on and starts flashing.



10.1.4 Setting to minimum speed

The motor must not be in operating mode **Stop**.

- Press and hold the **Down** button until the bottom light field is on and starts flashing.



10.1.5 Starting the pump

How you start the pump depends on how it was stopped.

Start the pump in one of the following ways:

- If the pump was stopped by pressing the **Start/Stop** button, start the pump by pressing the **Start/Stop** button.
- If the pump was stopped by pressing and holding the **Down** button, start the pump by pressing and holding the **Up** button.

10.1.6 Stopping the pump

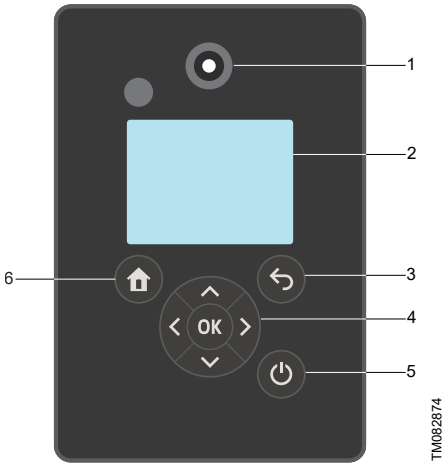
- Stop the pump in one of the following ways:
  - Press the **Start/Stop** button.
  - Press and hold the **Down** button until all light fields are off.
  - Use Grundfos GO.
  - Use a digital input set to **External stop**.

10.1.7 Resetting alarms and warnings in products with the HMI 200 or HMI 201 operating panel

- You can reset a fault indication in one of the following ways:
  - Briefly press the **Up** or **Down** button.
    This is not possible if the buttons have been locked.
    This does not change the setting of the motor.
  - Switch off the power supply until the indicator lights are off.
  - Switch the external start and stop input off, and then on again.
  - Use Grundfos GO.
  - Use the digital input if you have set it to **Alarm resetting**.

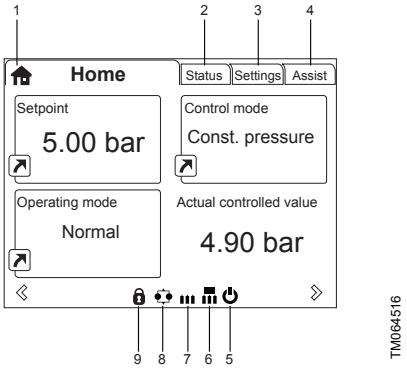
10.2 Operating panels, HMI 300 and 301

Pump variant	Fitted as standard	Option
TPE2, TPE2 D	-	●
TPE3, TPE3 D	●	-



Pos.	Symbol	Description
1		**Grundfos Eye:** The indicator light shows the operating status of the product.
2	-	Graphical colour display.
3		**Back**: Press the button to go one step back.
		Left/**Right**: Press the buttons to navigate between main menus, displays and digits. When you change the menu, the display shows the top display of the new menu.
		Up/**Down**: Press the buttons to navigate between submenus or change the value settings. If you have disabled the possibility to make settings with the **Enable/ disable settings** function, you can enable it again temporarily by pressing these buttons simultaneously for at least 5 seconds.
4	OK	**OK**: Press the button to do as follows: • save changed values, reset alarms and expand the value field • enable communication with Grundfos GO and other products of the same type. When you try to establish radio communication between the product and Grundfos GO or another product, the green indicator light in Grundfos Eye flashes. In the controller display, a note states that a device wants to connect to the product. Press **OK** on the product operating panel to allow communication with Grundfos GO or Grundfos GO Link and other products of the same type.
5		**Start/Stop**: Press the button to make the product ready for operation or to start and stop the product. **Start**: If you press the button when the product is stopped, the product starts if no other functions with higher priority have been enabled. **Stop**: If you press the button when the product is running, the product always stops. When you press the button, the stop icon appears at the bottom of the display.
6		**Home**: Press the button to go to the **Home** menu.

10.2.1 Home display



Pos.	Symbol	Description
1		**Home**: This menu shows up to four user-defined parameters. You can access each parameter directly from this menu.
2	-	**Status**: This menu shows the status of the product and system, warnings and alarms.
3	-	**Settings**: This menu gives access to all setting parameters. The menu also allows you to make detailed settings.
4	-	**Assist**: This menu enables assisted setup, provides a short description of the control modes and offers fault-finding advice.
5		**Start/Stop**: The icon indicates that the product was stopped with the **Start/Stop** button.
6		**Master:** The icon indicates that the product is functioning as the master in a multipump system.
7		**Slave:** The icon indicates that the product is functioning as a slave in a multipump system.
8		**Multioperation:** The icon indicates that the product is operating in a multipump system.
9		**Lock:** The icon indicates that the possibility to make settings has been disabled for protective reasons.

10.2.2 Startup guide

The function is only available in the HMI 300 and HMI 301 operating panels.

The startup guide starts at the first startup and guides you through the settings needed for the product to operate in the given application. When the startup guide has been completed, the main menus appear in the display.

You can always run the startup guide at a later time.

10.2.3 Menu overview for the HMI 300 and 301 operating panels

		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Home		●	●	●

Status		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Operating status		●	●	●
	Operating mode, from	●	●	●
	Control mode	●	●	●
Pump performance		●	●	●
	Actual control. value	●	●	●
	Max. curve and duty point	●	●	●
	Resulting setpoint	●	●	●
	Liquid temp.	●	●	●
	Speed	●	●	●
	Acc. flow and specific energy	●	●	●
Power and energy consumption		●	●	●
Measured values		●	●	●
	Analog input 1	●	●	●
	Analog input 2	●	●	●
	Analog input 3 [5)]	●	●	●
	Grundfos Direct Sensor	●	●	●
	Pt100/1000 input 1 [5)]	●	●	●
	Pt100/1000 input 2 [5)]	●	●	●
Analog output [5)]		●	●	●
Warning and alarm		●	●	●
	Actual warning or alarm	●	●	●
	Warning log	●	●	●
	Alarm log	●	●	●
Heat energy monitor		-	●	●
	Heat power	-	●	●
	Heat energy	-	●	●
	Flow rate	-	●	●
	Volume	-	●	●

Status		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
	Hours counter	-	●	●
	Temperature 1	-	●	●
	Temperature 2	-	●	●
	Differential temp.	-	●	●
Operating log		-	●	●
	Operating hours	●	●	●
Fitted modules		●	●	●
Date and time [5)]		●	●	
Product identification		●	●	●
Motor bearing monitoring		●	●	●
Multi-pump system		-	-	●
	System operating status	-	-	●
	System performance	-	-	●
	System input power and energy	-	-	●
	Pump 1, multi-pump system	-	-	●
	Pump 2, multi-pump system	-	-	●
	Pump 3, multi-pump system	-	-	●
	Pump 4, multi-pump system	-	-	●

5) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Settings	TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Setpoint	●	●	●
Operating mode	●	●	●
Set manual speed	●	●	●
Set user defined speed	●	●	●
Control mode	●	●	●
FLOWLIMIT	-	-	●
Automatic night setback	-	●	●
Setting the proportional pressure	-	●	●

Settings		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Analog inputs		●	●	●
	Analog input 1, setup	●	●	●
	Analog input 2, setup	●	●	●
	Analog input 3, setup [5]	●	●	●
	Grundfos Direct Sensor	●	●	●
Pt100/1000 inputs [5]		●	●	●
	Pt100/1000 input 1, setup [5]	●	●	●
	Pt100/1000 input 2, setup [5]	●	●	●
Digital inputs		●	●	●
	Digital input 1, setup	●	●	●
	Digital input 2, setup	●	●	●
Digital inputs/ outputs		●	●	●
	Digital input/output 3, setup	●	●	●
	Digital input/output 4, setup [5]	●	●	●
Relay outputs		●	●	●
	Relay output 1	●	●	●
	Relay output 2	●	●	●
Analog output [5]		●	●	●
	Output signal [5]	●	●	●
	Function of analog output [5]	●	●	●
Controller settings		●	●	●
Operating range		●	●	●
Setpoint influence		●	●	●
	Ext. setpoint infl.	●	●	●
	Predefined setpoint [5]	●	●	●
Temperature influence		-	●	●
Monitoring functions		●	●	●
	Motor bearing monitoring	●	●	●
	Alarm handling	●	●	●
	Motor bearing maintenance	●	●	●

Settings		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
	Limit-exceeded function	●	●	●
Special functions		●	●	●
	Pulse flowmeter setup	●	●	●
	Ramps	●	●	●
	Standstill heating	●	●	●
Communication		●	●	●
	Pump number	●	●	●
	Enable/disable radio comm.	●	●	●
	Enable/disable Bluetooth comm.	●	●	●
	Initiate Bluetooth connection	●	●	●
	Setup of AYB terminals	●	●	●
	Setup of Ethernet	●	●	●
General settings		●	●	●
	Language	●	●	●
	Set date and time	●	●	●
	Units	●	●	●
	Enable/disable settings	●	●	●
	Delete history	●	●	●
	Define Home display	●	●	●
	Display settings	●	●	●
	Store actual settings	●	●	●
	Recall stored settings	●	●	●
	Run start-up guide	●	●	●

5) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Assist	TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Assisted pump setup	●	●	●
Setup, analog input	●	●	●
Setting of date and time	●	●	●
Setup of multi-pump system	●	●	●
Description of control mode	●	●	●
Assisted fault advice	●	●	●

11. Grundfos GO

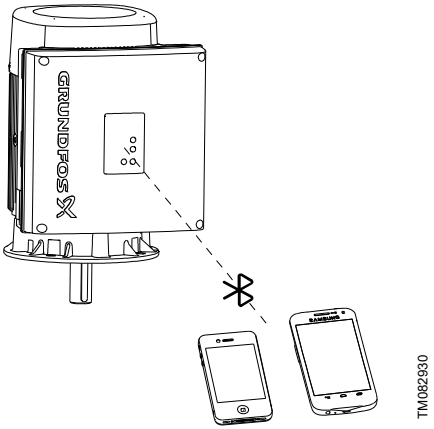
CAUTION
Radiation
Minor or moderate personal injury

- Locate the product at a minimum distance of 20 cm from any body parts. Human tissue may be heated by RF energy.

Installers and end users must be provided with these installation and operating instructions and operating conditions for satisfying RF exposure compliance.

The product is designed for wireless communication with Grundfos GO using Bluetooth (BLE).

Grundfos GO enables you to set functions and gives you access to status overviews, technical product information and current operating parameters.



11.1 Communication

When Grundfos GO initiates communication with the product, the indicator light in the centre of Grundfos Eye flashes green.

On products fitted with the HMI 100 or HMI 200 operating panel, you can enable communication by pressing the **Communication** button.

On products fitted with the HMI 300 operating panel, the display indicates that a wireless device is trying to connect to the product. Press **OK** on the operating panel to connect the product with Grundfos GO, or press the **Home** button to reject connection.

Symbol	Description
OK	Press **OK** on the operating panel to connect the product with Grundfos GO.
	Press the **Home** button to reject connection.

11.1.1 Bluetooth communication

Bluetooth communication can take place at distances up to 10 m. The first time Grundfos GO communicates with the product, you enable communication by pressing the **Communication** button or **OK** on the operating panel.

Later when communication takes place, the product is recognised by Grundfos GO, and you can select the product from the **List** menu.

11.2 Menu overview for Grundfos GO

		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Dashboard		●	●	●

View all metrics		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Pump and application		●	●	●
	Actual controlled value	●	●	●
	Acc. flow, specific energy	●	●	●
	Energy consumption	●	●	-
	Energy consumption, system	-	-	●
	Power consumption	●	●	-
	Power consumption, system	-	-	●
	Motor bearing service	●	●	-
	Resulting setpoint	●	●	-
	Resulting system setpoint	-	-	●
	Motor speed	●	●	-
	Trend data	-	●	-
	Heat energy monitor	-	●	-
	Pump 1	-	-	●
	Pump 2	-	-	●
	Pump 3	-	-	●
	Pump 4	-	-	●
Operating Log		●	●	●
	Operating hours	●	●	-
	Operating hours, system	-	-	●
	Motor current	●	●	-
	Number of starts	●	●	-
	Liquid temperature	-	●	-
Inputs/outputs		●	●	-
	Analog input 1	●	●	-
	Analog input 2	●	●	-
	Analog input 3 [6)]	●	●	-
	Grundfos Direct Sensor	●	●	-

View all metrics		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
	Analog, Output [6]	●	●	-
	Pt100/1000 input 1 [6]	●	●	-
	Pt100/1000 input 2 [6]	●	●	-
	Digital input 1	●	●	-
	Digital input 2 [6]	●	●	-
	Digital input/output 3	●	●	-
	Digital input/output 4 [6]	●	●	-
Monitored metrics		●	●	●
	Ambient temperature	●	●	●
	Differential pressure	●	●	●
	Differential pressure, inlet/outlet	●	●	●
	Differential temperature, external	●	●	●
	External pressure 1	●	●	●
	External pressure 2	●	●	●
	Feed tank pressure	●	●	●
	Flow rate	●	●	●
	Pressure : inlet	●	●	●
	Pressure : outlet	●	●	●
	Other parameter	●	●	●
	Tank pressure, external	●	●	●
	Temperature 1	●	●	●
	Temperature 2	●	●	●
	Conductivity	●	●	●
Fitted modules		●	●	-
	Functional module	●	●	-
	Power board	●	●	-
	CIM module	●	●	-
	Operating panel	●	●	-

[6] Only available if an advanced functional module, type FM310 or FM311, is fitted.

Settings		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Pump and application		●	●	●
	Pump name	●	●	●
	Control mode	●	●	●

Settings		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
	FLOWLIMIT	-	●	-
	Automatic night setback	-	●	-
	Temperature influence	-	●	-
	Operating mode	●	●	●
	Setpoint	●	●	●
	Set user-defined speed	●	●	●
	Operating range	●	●	●
	Controller	●	●	●
	External setpoint funct.	●	●	-
	Predefined setpoint	●	●	●
	Setting the proportional pressure	●	●	-
	Buttons on product	●	●	-
	Service	●	●	-
	Alternating operation, time	-	-	●
	Sensor to be used	-	-	●
	Time for pump changeover [7)]	-	-	●
Inputs/outputs		●	●	-
	Analog input 1	●	●	-
	Analog input 2	●	●	-
	Analog input 3 [7)]	●	●	-
	Grundfos Direct Sensor	●	●	-
	Analog output [7)]	●	●	-
	Pt100/1000 input 1 [7)]	●	●	-
	Pt100/1000 input 2 [7)]	●	●	-
	Digital input 1	●	●	-
	Digital input 2 [7)]	●	●	-
	Digital input/output 3	●	●	-
	Digital input/output 4 [7)]	●	●	-
	Relay output 1	●	●	-
	Relay output 2	●	●	-
Monitoring functions		●	●	●
	Alarm handling	●	●	-

Settings		TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
	Limit 1 exceeded	●	●	●
	Limit 2 exceeded	●	●	●
	Limit 3 exceeded [7)]	●	●	●
	Limit 4 exceeded [7)]	●	●	●
	Motor bearing monitoring	●	●	-
Special functions		●	●	●
	Pulse flow meter	●	●	-
	Ramps	●	●	-
	Standstill heating	●	●	-
Communication		●	●	-
	Bluetooth communication	●	●	-
	Radio communication	●	●	-
	GENIbus Number	●	●	-
	Connectivity and port settings	●	●	-
General		●	●	-
	Connection code	●	●	-
	Date and time [7)]	●	●	-
	Firmware	●	●	-
	Store settings	●	●	-
	Recall settings	●	●	-
	Undo	●	●	-
	Unit configuration	●	●	-

[7)] Only available if an advanced functional module, type FM310 or FM311, is fitted.

Alarms and warnings	TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Alarm log	●	●	●
Warning log	●	●	●

Setup	TPE2, TPE2 D	TPE3, TPE3 D	Multipump system
Assisted pump setup	●	●	-
Assisted fault advice	●	●	-
Application wizard	●	●	-
Multi-pump setup	●	●	●

12. Setting the product

You can set control functions via Grundfos GO, Grundfos GO Link or the HMI 300 or HMI 301 operating panel.

- If only one function name is mentioned, it refers to both Grundfos GO and the operating panel.
- If a function name is mentioned in a parenthesis, it refers to a function on the operating panel.

12.1 Heat energy monitor

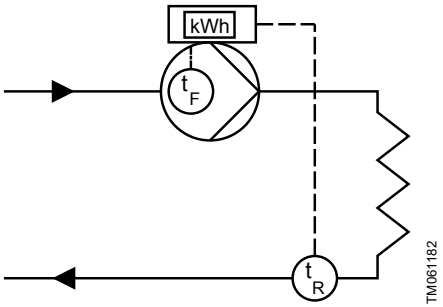
Pump variant	Heat energy monitor
TPE2, TPE2 D	-
TPE3, TPE3 D	•

The heat energy monitor is a monitoring function that calculates the heat energy consumption within a system. The built-in flow estimation needed for the calculation has an inaccuracy of ± 10 % of the maximum flow rate in the area down to 10 % flow and down to 12.5 % of the maximum head. The calculations are based on water at a temperature of 20 °C. Also, the temperature measurements needed for the calculation have some inaccuracy depending on the sensor type. Therefore, you cannot use the heat energy value for billing purposes. However, the value is perfect for optimisation purposes in order to prevent excessive energy costs caused by system imbalances.

The heat energy monitor requires an additional temperature sensor installed in the flow pipe or return pipe depending on where the pump is installed.

Use the analog inputs and/or Pt100/1000 inputs for measuring the temperatures used for calculation by the heat energy monitor.

The used inputs must not be set to **Not active** and one of the measuring parameters must be set to **Temperature 2**.



Example: *Pump installed in the flow pipe and additional temperature sensor installed in the return pipe*

Pos.	Description
t_F	Flow-pipe temperature
t_R	Return-pipe temperature

12.2 Setpoint

Pump variant	Setpoint
TPE2, TPE2 D	•
TPE3, TPE3 D	•

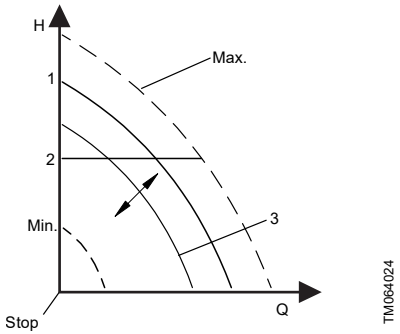
When you have selected the desired control mode, set the setpoint.

12.3 Operating mode

Pump variant	Operating mode
TPE2, TPE2 D	•
TPE3, TPE3 D	•

Possible operating modes

Normal	The product runs according to the selected control mode.
Stop	The product stops.
Min.	The product runs at minimum speed. You can use the minimum curve mode in periods in which a minimum flow is required. When operating according to the minimum curve, the pump is operating like an uncontrolled pump.
Max.	The product runs at maximum speed. You can use the maximum curve mode in periods in which a maximum flow is required. When operating according to the maximum curve, the pump is operating like an uncontrolled pump.
Manual	The product is operating at a manually set speed, and the setpoint via bus and setpoint influence function are overruled.
User-defined speed	The product is operating at a speed set by the user.



Pos.	Description
1	**Normal**
2	**Normal**
3	**Manual**

12.4 Set manual speed

Pump variant	Set manual speed
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The function is only available in the HMI 300 and 301 operating panels.

Use this function to set the speed in percentage of the maximum speed. When you have set the operating mode to **Manual**, the product starts running at the set speed.

With Grundfos GO, you can set the speed via the **Setpoint** menu.

12.5 Set user-defined speed

Use this function to set the motor speed in percentage of the maximum speed. When you set the operating mode to **User-defined speed**, the motor starts running at the set speed.

12.6 Control mode

Pump variant	Control mode
TPE2, TPE2 D	●
TPE3, TPE3 D	●

You can choose between the following control modes:

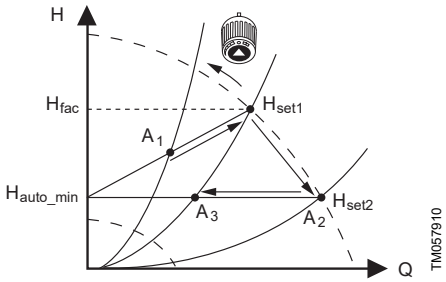
- **Prop. pressure** (proportional pressure)
- **Const. pressure** (constant pressure)
- **Const. temp.** (constant temperature)
- **Con. diff. press.** (constant differential pressure)
- **Con. diff. temp.** (constant differential temperature)
- **Const. flow rate** (constant flow rate)
- **Const. level** (constant level)
- **Const. other val.** (constant other value)
- **Const. curve** (constant curve).

12.6.1 AUTOADAPT

Pump variant	AUTOADAPT
TPE2, TPE2 D	-
TPE3, TPE3 D	●

The AUTOADAPT control mode continuously adapts the pump performance according to the actual system characteristic.

Manual setting of the setpoint is not possible.



AUTOADAPT

When the AUTOADAPT control mode has been enabled, the pump will start with the factory setting, H_{fac} is equal to H_{set1}, and then adjust its performance to A_1. See the figure above.

When the pump registers a lower head on the maximum curve, A_2, the AUTOADAPT function automatically selects a correspondingly lower control curve, H_{set2}. If the valves in the system close, the pump adjusts its performance to A_3.

A_1:	Original duty point.
A_2:	Lower registered head on the maximum curve.
A_3:	New duty point after AUTOADAPT control.
H_{set1}:	Original setpoint setting.
H_{set2}:	New setpoint after AUTOADAPT control.
$H_{fac.}$:	Factory setting.
H_{auto_min}:	A fixed value of 1.5 m.

The AUTOADAPT control mode is a form of proportional-pressure control where the control curves have a fixed origin, H_{auto_min}.

The AUTOADAPT control mode has been developed specifically for heating systems and we do not recommend that you use it for air-conditioning and cooling systems.

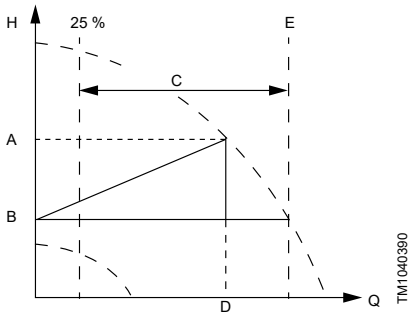
12.6.2 FLOWADAPT

Pump variant	FLOWADAPT
TPE2, TPE2 D	-
TPE3, TPE3 D	•

When you select FLOWADAPT, the pump runs AUTOADAPT and ensures that the flow rate never exceeds the entered FLOWLIMIT value.

The setting range for FLOWLIMIT is 25 to 90 % of the maximum flow rate of the pump.

The factory setting of the FLOWLIMIT is the flow rate where the AUTOADAPT factory setting meets the maximum curve.



FLOWADAPT

Pos.	Description
A	H_{fac}
B	H_{auto_min}
C	Setting range
D	Q_{fac}
E	90 % Q_{max}

12.6.3 Proportional pressure

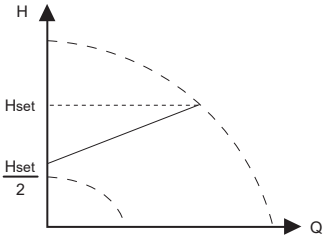
Pump variant	Proportional pressure
TPE2, TPE2 D	-
TPE3, TPE3 D	●

The head of the pump is reduced at decreasing water demand and increased at rising water demand. See the figure below.

This control mode is especially suitable in systems with relatively large pressure losses in the distribution pipes. The head of the pump increases proportionally to the flow in the system to compensate for the large pressure losses in the distribution pipes.

The setpoint can be set with an accuracy of 0.1 m. The head against a closed valve is half the setpoint. The setting range is between 25 % and 90 % of maximum head.

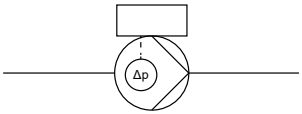
For more information about settings, see section on setting the proportional pressure.



Proportional pressure

Example:

- Factory-fitted differential-pressure sensor.



Proportional pressure

Controller settings

For the recommended controller settings, see the section on the controller.

Related information

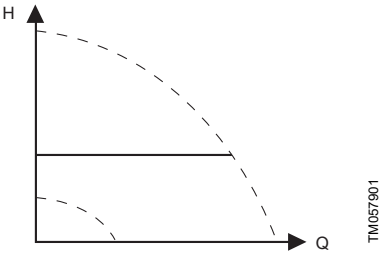
12.7 Setting the proportional pressure

12.17 Controller (Controller settings)

12.6.4 Constant pressure

Pump variant	Constant pressure
TPE2, TPE2 D	●
TPE3, TPE3 D	●

We recommend this control mode if the pump is to deliver a constant pressure, independently of the flow in the system. The pump maintains a constant pressure independently of the flow rate.

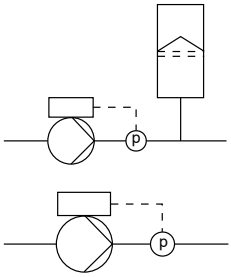


Constant pressure

This control mode requires an external pressure sensor as shown in the examples below. You can set the pressure sensor in the **Assist** menu. See the section on assisted pump setup. The setting range is between 12.5 % and 100 % of maximum head.

Example:

- One external pressure sensor



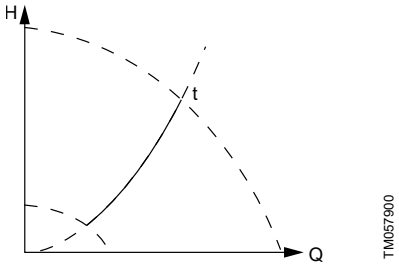
Controller settings

For recommended controller settings, see the section on the controller.

12.6.5 Constant temperature

Pump variant	Constant temperature
TPE2, TPE2 D	•
TPE3, TPE3 D	•

This control mode ensures a constant temperature. Constant temperature is a comfort control mode that you can use in domestic hot-water systems to control the flow to maintain a constant temperature in the system.

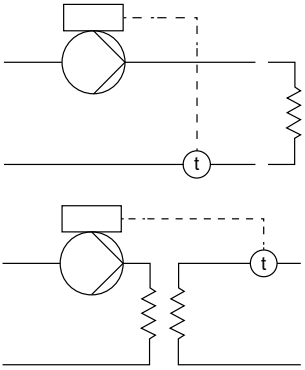


Constant temperature

This control mode requires an external temperature sensor as shown in the example below.

Example:

- One external temperature sensor



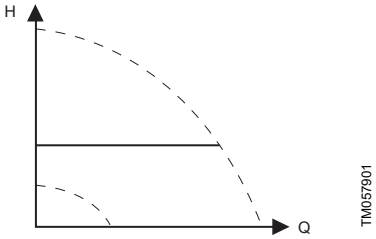
Controller settings

For the recommended controller settings, see the section on the controller.

12.6.6 Constant differential pressure

Pump variant	Constant differential pressure
TPE2, TPE2 D	•
TPE3, TPE3 D	•

The pump maintains a constant differential pressure, independently of the flow rate in the system. This control mode is primarily suitable for systems with relatively small pressure losses.

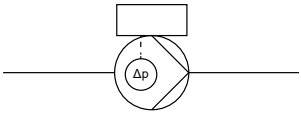


Constant differential pressure

The setting range is between 12.5 % and 100 % of maximum head. This control mode requires either an internal or external differential-pressure sensor or two external pressure sensors as shown in the examples below.

Examples:

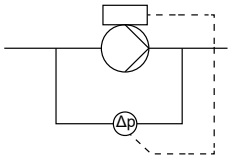
- Factory-fitted differential-pressure sensor.

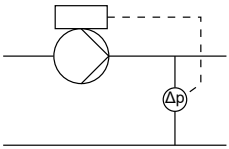


- One external differential-pressure sensor.

  The pump uses the input from the sensor to control the differential pressure.

  You can set the sensor manually or by using the **Assist** menu. See the section on assisted pump setup.

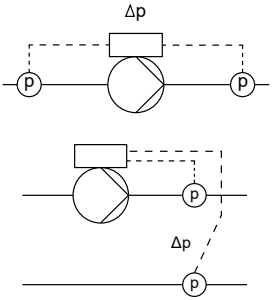




- Two external pressure sensors.

  Constant differential-pressure control is achievable with two individual pressure sensors. The pump uses the inputs from the two sensors and calculates the differential pressure.

  The sensors must have the same unit and must be set as feedback sensors. You can set the sensors manually, sensor by sensor, or by using the **Assist** menu. See the section on assisted pump setup.



Controller settings

For recommended controller settings, see the section on the controller.

Related information

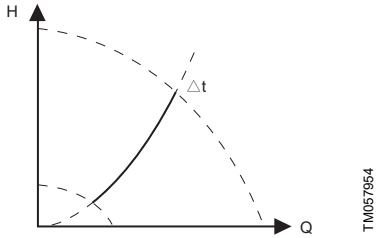
12.17 Controller (Controller settings)

12.28.23 Assisted pump setup

12.6.7 Constant differential temperature

Pump variant	Constant differential temperature
TPE2, TPE2 D	•
TPE3, TPE3 D	•

The pump maintains a constant differential temperature in the system and the pump performance is controlled according to this.



Constant differential temperature

This control mode requires either two temperature sensors or one external differential-temperature sensor. See the examples below.

The temperature sensors can either be analog sensors connected to two of the analog inputs or two Pt100/1000 sensors connected to the Pt100/1000 inputs, if these are available on the specific pump.

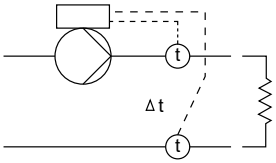
Set the sensor in the **Assist** menu under **Assisted pump setup**. See the section on assisted pump setup.

Examples:

- Two external temperature sensors.

  Constant differential-temperature control is achievable with two temperature sensors. The pump uses the inputs from the two sensors and calculates the differential temperature.

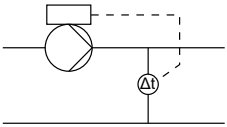
  The sensors must have the same unit and must be set as feedback sensors. You can set the sensors manually, sensor by sensor, or by using the **Assist** menu. See the section on assisted pump setup.



- One external differential-temperature sensor.

The pump uses the input from the sensor to control the differential temperature.

You can set the sensor manually or by using the **Assist** menu. See the section on assisted pump setup.



Controller settings

For recommended controller settings, see the section on the controller.

Related information

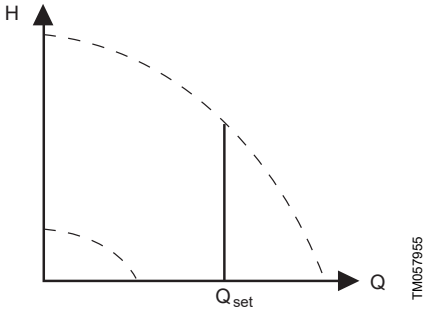
12.17 Controller (Controller settings)

12.28.23 Assisted pump setup

12.6.8 Constant flow rate

Pump variant	Constant flow rate
TPE2, TPE2 D	•
TPE3, TPE3 D	•

The pump maintains a constant flow in the system, independently of the head.

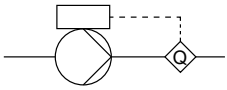


Constant flow rate

This control mode requires an external flow sensor. See the example below.

Example:

- One external flow sensor.



Constant flow rate

Controller settings

For recommended controller settings, see the section on the controller.

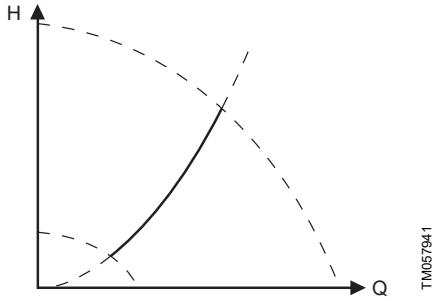
Related information

12.17 Controller (Controller settings)

12.6.9 Constant level

Pump variant	Constant level
TPE2, TPE2 D	•
TPE3, TPE3 D	•

The pump maintains a constant level, independently of the flow rate.



Constant level

This control mode requires an external level sensor.

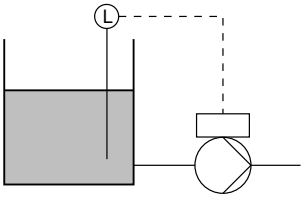
The pump can control the level in a tank in two ways (see the figure above):

- As an emptying function where the pump draws the liquid from the tank.
- As a filling function where the pump pumps the liquid into the tank.

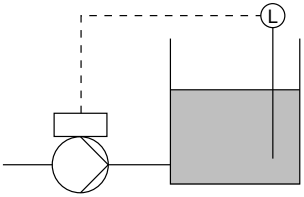
The type of level control function depends on the setting of the built-in controller.

Example:

- One external level sensor with emptying function.



- One external level sensor with filling function.



Controller settings

For recommended controller settings, see the section on the controller.

Related information

12.17 Controller (Controller settings)

12.6.10 Constant other value

Pump variant	Constant other value
TPE2, TPE2 D	•
TPE3, TPE3 D	•

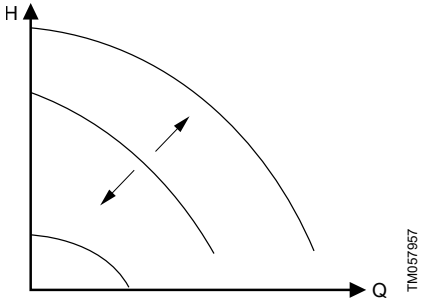
Use this control mode to control a value which is not available in the **Control mode** menu. To measure the controlled value, connect a sensor to one of the analog inputs. The controlled value is shown in percentage of the sensor range.

12.6.11 Constant curve

Pump variant	Constant curve
TPE2, TPE2 D	•
TPE3, TPE3 D	•

Use this control mode to control the motor speed.

You can set the desired speed in percentage of the maximum speed in the range from user-set minimum speed to user-set maximum speed.



Pos.	Description
P	Pressure
Q	Flow rate

12.7 Setting the proportional pressure

Pump variant	Setting the proportional pressure
TPE2, TPE2 D	-
TPE3, TPE3 D	•

Related information

12.6.3 Proportional pressure

12.7.1 Control curve function

You can set the proportional curve either to quadratic or linear to match the system curve.

12.7.2 Zero flow head

You can set this value in percentage of the setpoint, and define how much the setpoint must be reduced at a closed valve. With a setting of 100 %, the control mode is equal to the constant differential pressure.

12.7.3 Fixed inlet pressure

This menu enables the use of a fixed inlet pressure.

12.7.4 Inlet pressure

Enter the fixed inlet pressure that is to be supplied to the pump.

12.7.5 Pump data

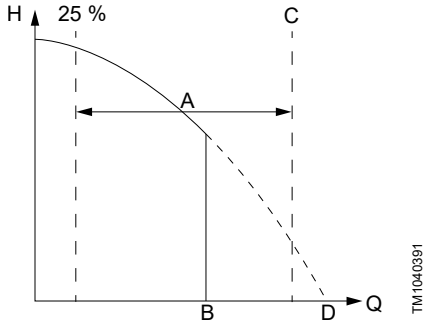
To enable the pump to operate in proportional pressure, the controller needs to process the pump curve. Enter the maximum head, rated head and rated flow rate from the pump nameplate.

12.8 FLOWLIMIT

Pump variant	FLOWLIMIT
TPE2, TPE2 D	-
TPE3, TPE3 D	•

FLOWLIMIT

- Enable the FLOWLIMIT function.
- Set the FLOWLIMIT.



FLOWLIMIT

Pos.	Description
A	Setting range
B	Q_{limit}
C	90 % Q_{max}
D	Q_{max}

You can combine the FLOWLIMIT function with the following control modes:

- Proportional pressure
- Constant differential pressure
- Constant differential temperature
- Constant temperature
- Constant curve.

A flow-limiting function ensures that the flow rate never exceeds the entered FLOWLIMIT value.

The setting range for FLOWLIMIT is 25 % to 90 % of the Q_{max} of the pump.

12.9 Automatic night setback

Pump variant	Automatic night setback
TPE2, TPE2 D	-
TPE3, TPE3 D	•

Once you have enabled automatic night setback, the pump automatically changes between normal duty and night setback, duty at low performance.

Changeover between normal duty and night setback depends on the flow-pipe temperature.

The pump automatically changes over to night setback when the built-in sensor registers a flow-pipe temperature drop of more than 10 to 15 °C within approximately two hours. The temperature drop must be at least 0.1 °C/min.

Changeover to normal duty takes place without a time lag when the temperature has increased by approximately 10 °C.

Note: You cannot enable automatic night setback when the pump is in constant-curve mode.

12.10 Analog inputs

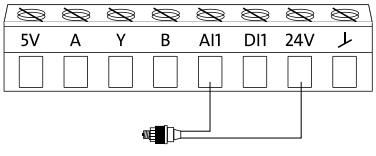
Pump variant	Analog inputs
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The inputs and outputs available depend on the functional module fitted in the motor.

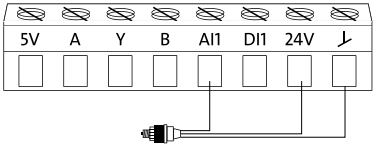
Functional module	Analog input 1 (Terminal AI1)	Analog input 2 (Terminal AI2)	Analog input 3 (Terminal AI3)
FM110	●	●	-
FM310	●	●	●
FM311	●	●	●

Wiring examples:

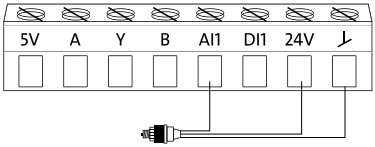
These connection scenarios are also valid for connection to analog input 2 and analog input 3.



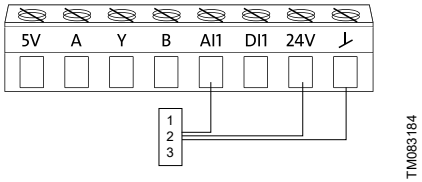
2-wire sensor, 0/4-20 mA



3-wire sensor, 0/4-20 mA



3-wire sensor, 0.5 - 3.5 V, 0-5 V, 0-10 V



Setpoint influence, 0.5 - 3.5 V, 0-5 V, 0-10 V; 0/4-20 mA

Pos.	Description
1	Potentiometer
2	PLC
3	External controller

To set the input, make the settings below:

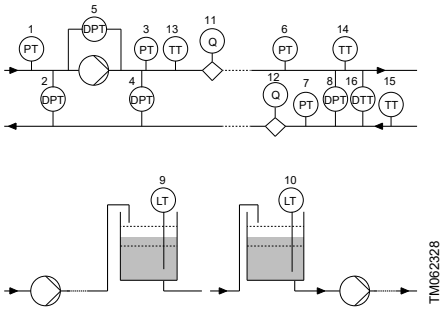
Function

You can set the inputs to these functions:

- **Not active**
- **Feedback sensor**
  The sensor is used for the selected control mode.
- **Setpoint influence**
  The input signal is used for influencing the setpoint.
- **Other function**
  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.



Pos.	Sensor function/measured parameter
1	**Inlet pressure**
2	**Diff. press., inlet**
3	**Outlet pressure**
4	**Diff. press.,outlet**
5	**Diff. press.,pump**
6	**Press. 1, external**
7	**Press. 2, external**
8	**Diff. press., ext.**
9	**Storage tank level**
10	**Feed tank level**
11	**Pump flow**
12	**Flow, external**
13	**Liquid temp.**
14	**Temperature 1**
15	**Temperature 2**
16	**Differential temp.**
Not shown	**Ambient temp.**
Not shown	**Conductivity**
Not shown	**Other parameter**

Unit

Parameter	Available units
Pressure	bar, m, kPa, psi, ft
Level	m, ft, in
Pump flow	m^3/h, l/s, yd^3/h, gpm
Liquid temperature	°C, °F
Other parameter	%

Electrical signal

Available signal types:

- 0.5 - 3.5 V
- 0-5 V
- 0-10 V
- 0-20 mA
- 4-20 mA.

Sensor range, minimum value

Set the minimum value of the connected sensor.

Sensor range, maximum value

Set the maximum value of the connected sensor.

12.10.1 Setting two sensors for differential measurement

Two analog sensors must be installed and connected electrically to measure a parameter at two different locations in a system.

The pressure, temperature and flow parameters can be used for differential measurement.

- Set the analog inputs according to the measured parameter:

Parameter	Sensor 1, measured parameter	Sensor 2, measured parameter
Pressure, option 1	**Inlet pressure**	**Outlet pressure**
Pressure, option 2	**Press. 1, external**	**Press. 2, external**
Flow	**Pump flow**	**Flow, external**
Temperature	**Temperature 1**	**Temperature 2**

If you want to use the **Con. diff. press.**, **Con. diff. temp.** or **Const. flow rate** control modes, you must configure both sensors as **Feedback sensor**.

12.11 Grundfos Direct Sensor

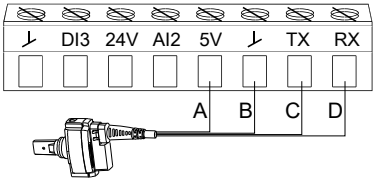
Pump variant	Grundfos Direct Sensor
TPE2, TPE2 D	-
TPE3, TPE3 D	•

Grundfos Direct Sensor is a digital sensor that auto detects range and unit.

Grundfos Direct Sensor always has the capability to also measure the media temperature. The pump will automatically detect range and unit of the temperature sensor.

For information about the functions and measured parameters of each sensor, see the sections on the sensor, temperature and dry-running protection.

Wiring example:



Designation	Colour
A	Brown
B	Green
C	Yellow
D	White

Factory setting

See the section on factory settings.

Related information

12.28.23 Assisted pump setup

24. Factory settings

12.11.1 Sensor

Function

You can set the sensor to the following functions:

- **Not active**
- **Feedback sensor**

  The sensor is used for the selected control mode.
- **Setpoint influence**

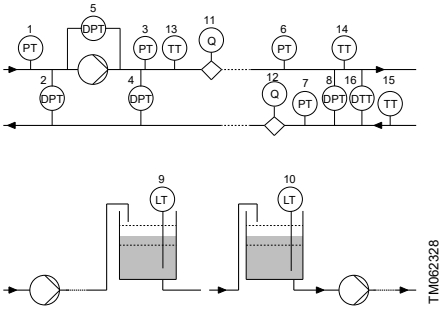
  The input signal is used for influencing the setpoint.
- **Other function**

  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.

Note that the list will be reduced to match the installed sensor.



Pos.	Sensor function/ measured parameter
1	**Inlet pressure**
2	**Diff. press., inlet**
3	**Outlet pressure**
4	**Diff. press.,outlet**
5	**Diff. press.,pump**
6	**Press. 1, external**
7	**Press. 2, external**
8	**Diff. press., ext.**
9	**Storage tank level**
10	**Feed tank level**
11	**Pump flow**
12	**Flow, external**
13	**Liquid temp.**
14	**Temperature 1**
15	**Temperature 2**
16	**Differential temp.**
Not shown	**Ambient temp.**
Not shown	**Other parameter**

12.11.2 Temperature

Function

You can set the sensor to the following functions:

- **Not active**
- **Feedback sensor**

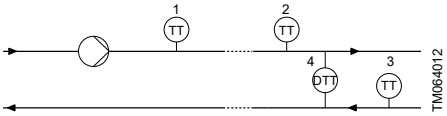
  The sensor is used for the selected control mode.
- **Setpoint influence**

  The input signal is used for influencing the setpoint.
- **Other function**

  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.



Pos.	Sensor function/ measured parameter
1	**Liquid temp.**
2	**Temperature 1**
3	**Temperature 2**
4	**Differential temp.**
Not shown	**Ambient temp.**

12.11.3 Dry-running protection

Use this function to set dry-running protection to **Enabled** or **Disabled**.

The function requires that a CPS sensor has been fitted in the pump head and connected to the pump. When you have enabled the dry-running protection function, it stops the pump if dry running occurs. Restart the pump manually if it has been stopped due to dry running.

Dry-running detection delay

You can set a detection delay to make sure that the pump is given a chance to start up and pump the air in the pump out before the dry-running protection function again detects dry running and stops the pump.

More than 10 seconds of dry running can damage the shaft seal and can reduce the lifetime of the product.

Range: 0-254 seconds.

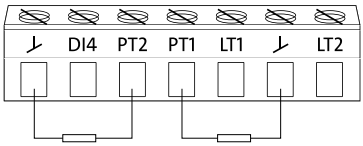
12.12 Pt100/1000 inputs

Pump variant	Pt100/1000 inputs
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Pt100/1000 input 1 (Terminals PT1, GND)	Pt100/1000 input 2 (Terminals PT2, GND)
FM110	-	-
FM310	●	●
FM311	●	●

Wiring example:



Pt100/1000

To set the input, choose one of the below settings.

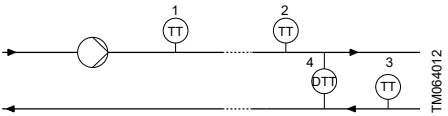
Function

You can set the inputs to these functions:

- **Not active**
- **Feedback sensor**
  The sensor is used for the selected control mode.
- **Setpoint influence**
  The input signal is used for influencing the setpoint.
- **Other function**
  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.



Pos.	Sensor function/measured parameter
1	**Liquid temp.**
2	**Temperature 1**
3	**Temperature 2**
4	**Differential temp.**
Not shown	**Ambient temp.**

Measuring range

-50 to +204 °C.

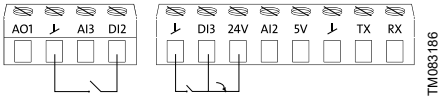
12.13 Digital inputs

Pump variant	Digital inputs
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Digital input 1 (Terminals DI1, GND)	Digital input 2 (Terminals DI2, GND)
FM110	●	-
FM310	●	●
FM311	●	●

Wiring example:



Digital input

To set the input, make the settings below:

Function

You can set the inputs to these functions:

- **Not active**

  When set to **Not active**, the input has no function.
- **Ext. stop**

  When the input is deactivated, open circuit, the motor stops.
- **Min.** (minimum speed)

  When the input is activated, the motor runs at the set minimum speed.
- **Max.** (maximum speed)

  When the input is activated, the motor runs at the set maximum speed.
- **User defined speed**

  When the input is activated, the motor runs at a speed set by the user.
- **External fault**

  When the input is activated, a timer is started. If the input is activated for more than 5 seconds, the motor stops and a fault is indicated. The function depends on input from external equipment.
- **Alarm resetting**

  When the input is activated, a fault indication, if any, is reset.
- **Dry running**

  When this function is selected, a lack of inlet pressure or water shortage (dry running) can be detected. When this happens, the pump stops. The pump cannot restart as long as the input is activated. This requires the use of an accessory such as these:
  - a pressure switch installed on the inlet side of the pump
  - a float switch installed on the inlet side of the pump.
- **Accumulated flow**

  When this function is selected, the accumulated flow can be registered. This requires the use of a flowmeter which can give a feedback signal as a pulse per defined volume of water.
- **Reverse rotation**

  This function reverses the direction of rotation of the motor.
- **Predefined setpoint 1**

  The function applies only to digital input 2.

  When you set digital inputs to a predefined setpoint, the pump operates according to a setpoint based on a combination of the activated digital inputs.
- **Activate output**

  When this function is selected, the related digital output is activated. This is done without any changes to pump operation.
- **Local motor stop**

  When the function is selected, the given motor in a multimotor system setup stops without affecting the performance of the other motors in the system.

The priority of the selected functions are interdependent.

A stop command always has the highest priority.

Activation of digital inputs

You can set the digital inputs to trigger on either Closed contact or Open contact. Selecting the trigger function can only be set via Grundfos GO Link.

The digital inputs can be activated either as active low or active high.

The digital inputs will react as described in the table below:

Activate/ Closed contact	Deactivate/Open contact
GND/0V	Floating/3-24V

12.13.1 Timer function for a digital input

Activation delay

The activation delay (T1) is the time between the digital signal and the activation of the selected function.

Range: 0-6000 seconds.

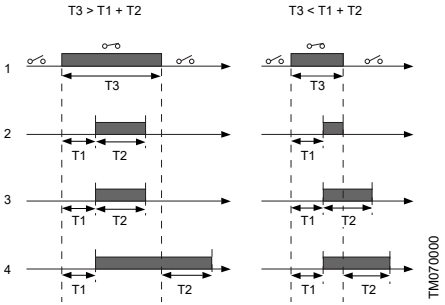
Duration time

Available modes:

- **Not active**
- **Active with interrupt**
- **Active without interrupt**
- **Active with after-run**.

The duration time (T2) is the time which, together with the mode, determines how long the selected function is active.

Range: 0 - 15,000 seconds.



Pos.	Description
1	**Digital input**.
2	**Active with interrupt**.
3	**Active without interrupt**.
4	**Active with after-run**.
T1	**Activation delay**
T2	**Duration time**
T3	The period of time when the digital input is activated.

12.14 Digital inputs/outputs

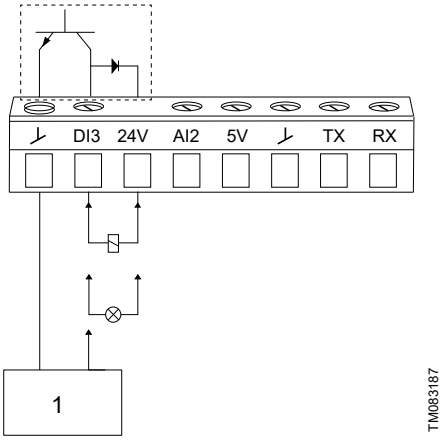
Pump variant	Digital inputs/outputs
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Digital input/ output 3 (Terminals DI3, GND)	Digital input/ output 4 (Terminals DI4, GND)
FM110	●	-
FM310	●	●
FM311	●	●

You can select whether the interface is to be used as an input or output. The output is an open collector. You can connect the open collector to, for example, an external relay or a controller such as a PLC.

Wiring example:



Digital output, open collector

Pos.	Description
1	External controller

Mode

You can set the digital input or output 3 and 4 to act as a digital input or digital output.

Functions if the digital input or output is set to input:

- **Not active**
- **Ext. stop**
- **Min.**
- **Max.**
- **User defined speed**
- **External fault**
- **Alarm resetting**
- **Dry running**
- **Accumulated flow**
- **Reverse rotation**
- **Predefined setpoint 2** (digital input/output 3)
- **Predefined setpoint 3** (digital input/output 4)
- **Local motor stop**
- **Activate output**

Functions if the digital input or output is set to output:

- **Not active**
- **Ready**
- **Alarm**
- **Operation**
- **Pump running**
- **Warning**
- **Limit 1 exceeded**
- **Limit 2 exceeded**
- **Limit 3 exceeded** [8)]
- **Limit 4 exceeded** [8)]
- **Digital input 1, state**
- **Digital input 2, state**
- **Digital input 3, state**
- **Digital input 4, state**

8) Only available if an advanced functional module, type FM310 or FM311, is fitted.

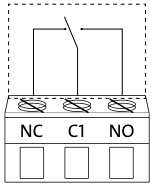
12.15 Signal relay (Relay outputs)

Pump variant	Signal relay
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The motor has two outputs for potential-free signals via two internal relays.

Functional module	Signal relay 1 (Terminals NC, C1, NO)	Signal relay 2 (Terminals NC, C2, NO)
FM110	●	-
FM310	●	●
FM311	●	●

Wiring example:



TM083188

Relay output

FunctionsYou can configure the signal relays to be activated when the product changes to one of the following states:

- **Not active**
  The relay has been deactivated.
- **Ready**
  The motor may be running or is ready to run, and no alarms are active.
- **Alarm**
  There is an active alarm, and the motor is stopped.
- **Operating** (**Operation**)
  Operating equals **Running**, but the motor is still in operation when it is stopped, for example, by the **Stop function** or **Limit exceeded**.
- **Running** (**Pump running**)
  The motor shaft is rotating.
- **Warning**
  There is an active warning.
- **Limit 1 exceeded**
  When you have set this function and the limit is exceeded, the signal relay is activated.
- **Limit 2 exceeded**
  When you have set this function and the limit is exceeded, the signal relay is activated.
- **Limit 3 exceeded** [9)]
  When you have set this function and the limit is exceeded, the signal relay is activated.
- **Limit 4 exceeded** [9)]
  When you have set this function and the limit is exceeded, the signal relay is activated.
- **External fan control** (**Control of external fan**)
  When you select this function, the relay is activated if the internal temperature of the motor electronics reaches a preset limit value. In this way the relay activates external cooling to add additional cooling to the motor.
- **Digital input 1, state**
  Follow digital input 1. If digital input 1 is triggered, the digital output is also triggered.
- **Digital input 2, state**
  Follow digital input 2. If digital input 2 is triggered, the digital output is also triggered.
- **Digital input 3, state**
  Follow digital input 3. If digital input 3 is triggered, the digital output is also triggered.
- **Digital input 4, state**
  Follow digital input 4. If digital input 4 is triggered, the digital output is also triggered.

9) Only available if an advanced functional module, type FM310 or FM311, is fitted.

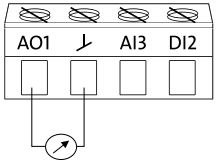
12.16 Analog output

Pump variant	Analog output
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Analog output (Terminals AO, GND)
FM110	-
FM310	●
FM311	●

Wiring example:



TM083185

Analog output, 0/4-20 mA, 0-10 V

The analog output enables external control systems to read specific operating data.

To set the analog output, make the following settings.

Output signal

Possible signal types:

- 0-10 V
- 0-20 mA
- 4-20 mA.

Function of analog output

Actual speed	
0 %	**100 %**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Sensor value	
Minimum	**Maximum**
0 V	10 V

Sensor value	
Minimum	**Maximum**
0 mA	20 mA
4 mA	20 mA

Resulting setpoint	
0 %	**100 %**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Motor load	
0 %	**100 %**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Motor current		
0 %	**100 %**	**200 %**
0 V	5 V	10 V
0 mA	10 mA	20 mA
4 mA	12 mA	20 mA

Limit-exceeded function	
Output not active	**Output active**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

12.17 Controller (Controller settings)

Pump variant	Controller (Controller settings)
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The pumps have a factory default setting of gain (K_p) and integral time (T_i).

However, if the factory setting is not the optimum setting, you can change the gain and the integral time:

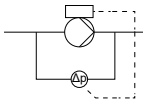
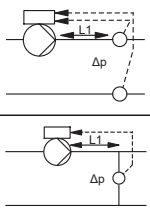
- Set the gain within the range from 0.1 to 20.
- Set the integral-action time within the range from 0.1 to 3600 seconds. If you select 3600 seconds, the controller functions as a P controller.

Furthermore, you can set the controller to inverse control.

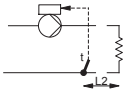
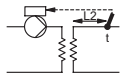
This means that if you increase the setpoint, the speed is reduced. In the case of inverse control, you must set the gain within the range from -0.1 to -20.

Guidelines for setting of PI controller

The tables below show the recommended controller settings:

Constant differential pressure	K_p	T_i
	0.5	0.5
	0.5	L1 < 5 m: 0.5 L1 > 5 m: 3 L1 > 10 m: 5

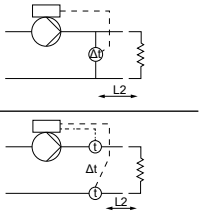
L1: Distance in metres between the pump and the sensor.

Constant temperature	K_p		T_i
	Heating system	Cooling system	
	0.5	-0.5	10 + 5L2
	0.5	-0.5	30 + 5L2

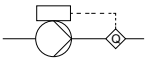
10) In heating systems, an increase in pump performance results in a rise in temperature at the sensor.

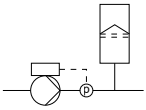
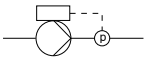
11) In cooling systems, an increase in pump performance results in a drop in temperature at the sensor.

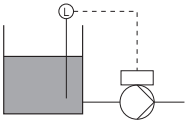
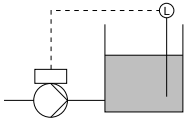
L2: Distance in metres between the heat exchanger and the sensor.

Constant differential temperature	K_p	T_i
	-0.5	10 + 5L2

L2: Distance in metres between the heat exchanger and the sensor.

Constant flow rate	K_p	T_i
	0.5	0.5

Constant pressure	K_p	T_i
	0.5	0.5
	0.5	0.5

Constant level	K_p	T_i
	-10	0
	10	0

General rules of thumb:

If the controller is too slow-reacting, increase the gain.

If the controller is hunting or unstable, dampen the system by reducing the gain or increasing the integral time.

Factory setting

See the section on factory settings.

Related information

12.6.3 Proportional pressure
12.6.6 Constant differential pressure
12.6.7 Constant differential temperature
12.6.8 Constant flow rate
12.6.9 Constant level
24. Factory settings

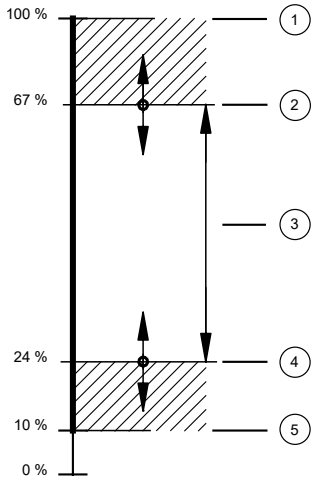
12.18 Operating range

Pump variant	Operating range
TPE2, TPE2 D	•
TPE3, TPE3 D	•

Set the operating range as follows:

1. Set the minimum speed within the range from fixed minimum speed (5) to user-set maximum speed (2).
2. Set the maximum speed within the range from user-set minimum speed (4) to fixed maximum speed (1).

The range between the user-set minimum and maximum speed is the operating range (3).



Pos.	Description
1	Fixed maximum speed
2	User-set maximum speed
3	Operating range
4	User-set minimum speed
5	Fixed minimum speed

12.19 External setpoint function

Pump variant	External setpoint function
TPE2, TPE2 D	•
TPE3, TPE3 D	•

Use this function to influence the setpoint by an external signal via one of the analog inputs.

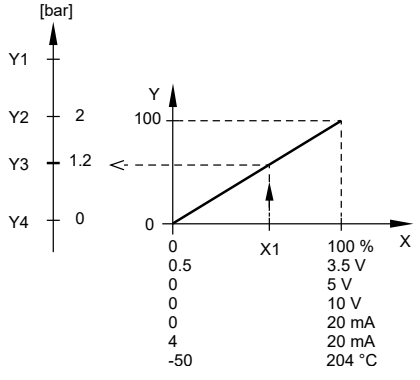
If the FM310 or FM311 functional module is fitted, you can also influence the setpoint via one of the Pt100/1000 inputs.

To enable the function, set one of the analog inputs or Pt100/1000 inputs to **Setpoint influence** with Grundfos GO or to **Ext. setpoint infl.** with the HMI 300 or 301 operating panel.

Example of setpoint influence in control mode Const. pressure

Actual setpoint: actual input signal × setpoint.

At a setpoint of 2 bar and an external setpoint of 60 %, the actual setpoint is 0.60 × 2 = 1.2 bar.

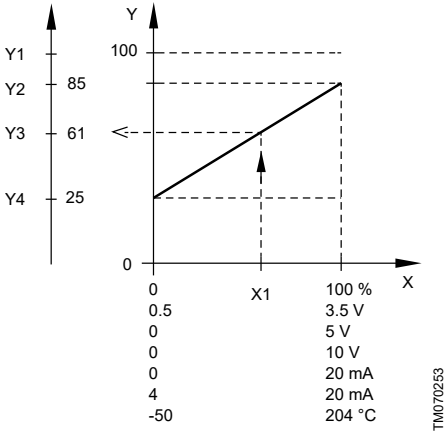


Pos.	Description
X:	External input signal from 0 to 100 %
Y:	Setpoint influence from 0 to 100 %
X1:	Actual input signal, 60 %
Y1:	Sensor maximum
Y2:	Setpoint
Y3:	Actual setpoint
Y4:	Sensor minimum

Example of a constant curve with linear influence function

Actual setpoint: actual input signal × (setpoint - user-set minimum speed) + user-set minimum speed.

At a user-set minimum speed of 25 %, a setpoint of 85 % and an external setpoint of 60 %, the actual setpoint is 0.60 × (85 - 25) + 25 = 61 %.

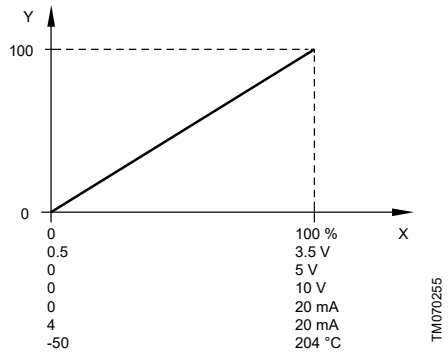


Pos.	Description
X:	External input signal from 0 to 100 %
Y:	Setpoint influence from 0 to 100 %
X1:	Actual input signal, 60 %
Y1:	Fixed maximum speed in percentage
Y2:	Setpoint speed in percentage
Y3:	Actual setpoint speed in percentage
Y4:	User-set minimum speed in percentage

12.19.1 Setpoint influence functions

12.19.1.1 Linear function

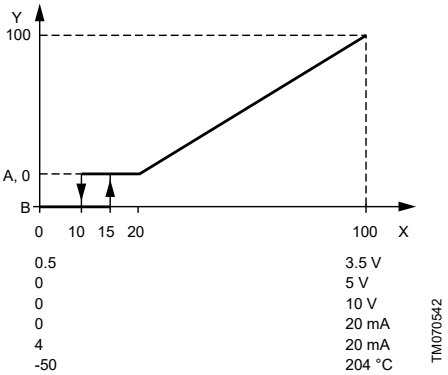
The setpoint is influenced linearly from 0 to 100 %.



Pos.	Description
X	External input signal from 0 to 100 %
Y	Setpoint influence from 0 to 100 %

12.19.1.2 Linear with Stop

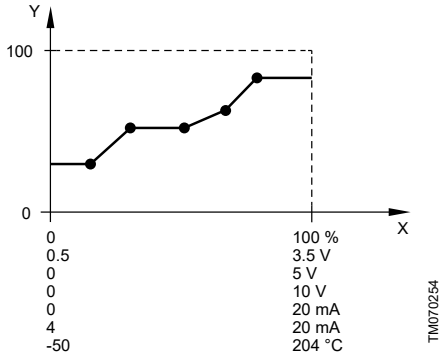
In the input signal range from 20 to 100 %, the setpoint is influenced linearly. If the input signal is below 10 %, the motor changes to the **Stop** operating mode. If the input signal increases over 15 %, the operating mode changes back to **Normal**.



Pos.	Description
X	External input signal from 0 to 100 %
Y	Setpoint influence from 0 to 100 %
A	**Normal**
B	**Stop**

12.19.1.3 Influence table

The setpoint is influenced by a curve made of two to eight points. There is a straight line between the points and a horizontal line before the first point and after the last point.



Pos.	Description
X	External input signal from 0 to 100 %
Y	Setpoint influence from 0 to 100 %

12.20 Predefined setpoints

Pump variant	Predefined setpoints
TPE2, TPE2 D	•
TPE3, TPE3 D	•

You can set and activate seven predefined setpoints by combining the input signals with digital inputs 2, 3 and 4 as shown in the table below. Set the digital inputs 2, 3 and 4 to **Predefined setpoints** if all seven predefined setpoints are to be used. You can also set one or two of the digital inputs to **Predefined setpoints**. However, this limits the number of predefined setpoints available.

Digital inputs			Setpoint
2	3	4	
0	0	0	Normal setpoint or **Stop**
1	0	0	**Predefined setpoint 1**
0	1	0	**Predefined setpoint 2**
1	1	0	**Predefined setpoint 3**
0	0	1	**Predefined setpoint 4**

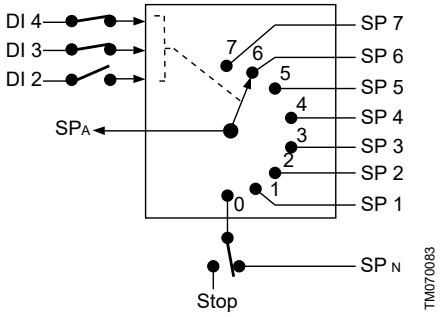
Digital inputs			Setpoint
2	3	4	
1	0	1	**Predefined setpoint 5**
0	1	1	**Predefined setpoint 6**
1	1	1	**Predefined setpoint 7**

0: Open contact

1: Closed contact

Example

The figure shows how you can use the digital inputs to set seven predefined setpoints. Digital input 2 is open, and digital inputs 3 and 4 are closed. If you compare with the table above, you can see that **Predefined setpoint 6** is activated.



Pos.	Description
DI	**Digital input**
SP	**Setpoint**
SP_A	**Actual setpoint**
SP_N	Normal setpoint
Stop	**Stop**

If all digital inputs are open, the motor stops or runs at the normal setpoint. Set the desired action with Grundfos GO or with the HMI 300 or 301 operating panel.

12.21 Limit-exceeded function

Pump variant	Limit-exceeded function
TPE2, TPE2 D	●
TPE3, TPE3 D	●

Use this function to monitor a measured parameter or one of the internal values such as speed, motor load or motor current. If a set limit is reached, a selected action can take place. You can set up to four limit-exceeded functions, meaning that you can monitor four different parameters or two to four limits of the same parameter simultaneously.

Note that, if using **Limit 1-3 exceeded** in a multipump system, the selected action will have an impact on the system, e.g. if Action is set to Stop, then the system will stop.

Limit 4 exceeded in a multipump system will be a local function. The selected action will only have an impact on the single pump, e.g. if Action is set to Stop, then only the single pump will stop.

For **Limit 4 exceeded** in a multipump system, the action will always lead to a pump stop, alarm and stop or a warning. [12)]

The function requires setting of the following parameters:

Measured

Set the measured parameter to be monitored.

Limit

Set the limit which activates the function.

Hysteresis band

Set the hysteresis band for when the function must be deactivated again.

Limit exceeded when

Set the function to be activated when the selected parameter exceeds or drops below the set limit.

- **above limit**

  The function is activated if the measured parameter exceeds the set limit.
- **below limit**

  The function is activated if the measured parameter drops below the set limit.

Action

If the value exceeds a limit, you can set an action. The following actions are available:

- **Not active**

12) Only available if an advanced functional module, type FM310 or FM311, is fitted.

The pump remains in its current state. Use this setting if you only want to activate a signal relay output when the limit is reached.

- **Stop**

  The pump stops.
- **Min.**

  The pump reduces the speed to minimum speed.
- **Max.**

  The pump increases the speed to maximum speed.
- **User-defined speed**

  The pump runs at a speed set by the user.
- **Alarm and Stop**

  An alarm is given, and the pump stops.
- **Alarm and Min**

  An alarm is given, and the pump decreases the speed to a minimum.
- **Alarm and Max**

  An alarm is given, and the pump increases the speed to maximum.
- **Alarm and User-defined speed**

  An alarm is given, and the pump runs at the speed set by the user.

Alarm and Warning text

The **Limit-exceeded** function will automatically define a relevant alarm/warning text based on the **Measured parameter** and **Limit exceeded when** functions.

The auto defined text can be overwritten by selecting the alarm/warning text option **Limit X exceeded**.

The following list shows the auto defined texts:

- Limit X exceeded
- Low inlet pressure
- High discharge pressure
- High pressure
- Low pressure
- High temperature
- Low temperature
- High flow
- Low flow
- High level
- Low level
- High diff. pressure
- Low diff. pressure
- High conductivity

Detection in Stop

Enable this function to prevent the pump from monitoring the limit while the pump is in the **Stop** state.

Use the **Detection delay** function to allow the pump to start up and bring the value above the limit before detecting.

Detection delay

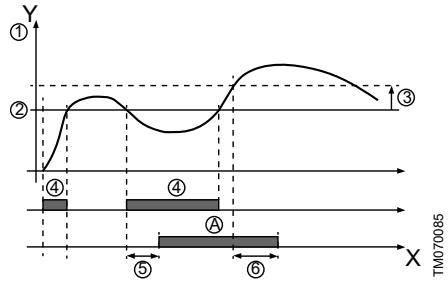
Setting the detection delay ensures that the monitored parameter stays above or below a set limit in a set time before the function is activated.

Resetting delay

The resetting delay is the time from when the measured parameter differs from the set limit, including the set hysteresis band, and until the function is reset.

Example

The function is to monitor the outlet pressure from the pump. If the pressure is below 5 bar for more than 5 seconds, a warning is indicated. If the pressure is above 7 bar for more than 8 seconds, reset the limit-exceeded warning.



X: Time in seconds

Y: Pressure in bar

Pos.	Parameter	Setting
1	**Measured**	**Discharge pressure**
2	**Limit**	5 bar
3	**Hysteresis band**	2 bar
4	**Limit exceeded when**	**below limit**
5	**Detection delay**	5 seconds
6	**Resetting delay**	8 seconds
A	**Limit-exceeded function active**	-
-	**Action**	**Warning**

12.22 Ramps

Pump variant	Ramps
TPE2, TPE2 D	●
TPE3, TPE3 D	●

The ramps determine how quickly the product can accelerate and decelerate during start and stop or setpoint changes.

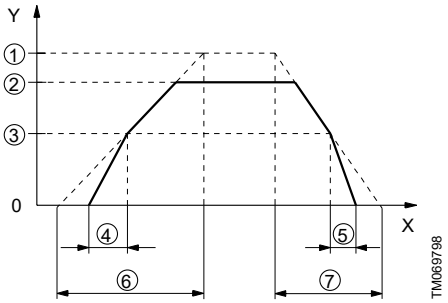
You can make the following settings:

- acceleration time, 0.1 to 300 s
- deceleration time, 0.1 to 300 s.

The times apply to the acceleration from 0 rpm to a fixed maximum speed and the deceleration from a fixed maximum speed to 0 rpm, respectively.

At short deceleration times, the deceleration of the product may depend on load and inertia as there is no possibility of actively braking the product.

If the power supply is switched off, the deceleration of the product only depends on the load and inertia.



Pos.	Description
Y	Speed
X	Time
1	Fixed maximum
2	User-set maximum
3	User-set minimum
4	Fixed initial ramp
5	Fixed final ramp
6	Ramp time up
7	Ramp time down

12.23 Skip band

Pump variant	Skid band
TPE2, TPE2 D	●
TPE3, TPE3 D	●

Use this function to select a skip band within the range from user-set minimum speed to user-set maximum speed if continuous operation is not required. The upper and lower speeds are stated in percentage of rated speed.

The purpose of the skip band is to avoid certain speeds which may cause noise or vibrations. If no skip band is required, select **-**.

12.24 Standstill heating

Pump variant	Standstill heating
TPE2, TPE2 D	●
TPE3, TPE3 D	●

Use this function to avoid condensation in humid environments.

When you set the function to **Active** and the product is in operating mode **Stop**, a low AC voltage is applied to the motor windings. The voltage is not high enough to make the motor rotate, but ensures that sufficient heat is generated to avoid condensation in the product, including the electronic parts in the drive.

Remember to remove the drain plugs and fit a cover over the product.

12.25 Alarm handling

Pump variant	Alarm handling
TPE2, TPE2 D	●
TPE3, TPE3 D	●

This setting determines how the pump must react in case of a sensor failure.

Alarm or warning types:

- **Warning**

  A warning. There is no change in the operating mode.
- **Stop**

  The pump stops.
- **Min.**

  The pump reduces the speed to minimum.
- **Max.**

  The pump increases the speed to maximum.

- **User defined speed**

  The pump runs at the speed set by the user.

Affected inputs:

- **Analog input 1**
- **Analog input 2**
- **Analog input 3**
- **Built-in Grundfos sensor**
- **Pt100/1000 input 1**
- **Pt100/1000 input 2**

12.26 Motor bearing monitoring

Pump variant	Motor bearing monitoring
TPE2, TPE2 D	•
TPE3, TPE3 D	•

Use this function to select whether or not you want to monitor the motor bearings.

You can make the following settings:

- **Active**
- **Not active**

When the function is set to **Active**, a counter in the controller starts counting the running hours of the bearings. The running hours are calculated on the basis of the motor speed. When a predefined limit is reached, a warning indicates that the bearings must be replaced or relubricated.



If you change the function to **Not active**, the counter continues to count. However, no warning is given when it is time to replace the bearings. If you change the function to **Active** again, the accumulated running hours are used to recalculate the replacement time.

12.27 Service

Pump variant	Service
TPE2, TPE2 D	•
TPE3, TPE3 D	•

12.27.1 Motor bearing service

This display shows when to replace the motor bearings. The controller monitors the operating pattern of the motor and calculates the period between bearing replacements.

Displayable values:

- **in 2 years**
- **in 1 year**
- **in 6 months**
- **in 3 months**
- **in 1 month**
- **in 1 week**
- **Now!**

12.27.2 Bearing replacements

The display shows the number of bearing replacements made during the lifetime of the motor.

12.27.3 Motor bearing maintenance

When the bearing monitoring function is active, a warning is given when the motor bearings must be replaced.

1. Replace the motor bearings.
2. Press **Bearings replaced**.

12.27.4 Bearings relubricated

When the bearing monitoring function is active, a warning is given when the motor bearings must be relubricated.



Bearings can be relubricated 5 times before they must be replaced.

The amount of grease can be found on the bearing nameplate on the motor.

1. When you have relubricated the bearings, press **Bearings relubricated**.

12.28 Communication

Pump variant	Communication
TPE2, TPE2 D	•
TPE3, TPE3 D	•

Use this function to set the communication of the product, both wired and wireless communication. The product contains built-in fieldbus protocols on the AYB terminals (RS-485).

12.28.1 Pump number

Use this function to allocate a unique number to the pump. This makes it possible to distinguish between pumps in connection with GENIbus communication.

12.28.2 Enable/disable radio comm.

Use this function to set the radio communication to **Enabled** or **Disabled**. Select **Disabled** in areas where radio communication is not allowed.



Bluetooth communication remains active.

12.28.3 Enable/disable Bluetooth comm.

Use this function to set the Bluetooth communication to **Enabled** or **Disabled**. Select **Disabled** in areas where Bluetooth communication is not allowed.

Radio communication remains active.

12.28.4 Initiate Bluetooth connection

Use this function if Grundfos GO is installed on Huawei smartphones with BLE version 5.0 or older. This function is used to establish a Bluetooth connection to Grundfos GO. Open the Grundfos GO app on your device and select **Initiate Bluetooth connection**. Then select **Yes** and follow the instructions on the device.

12.28.5 Setup of AYB terminals

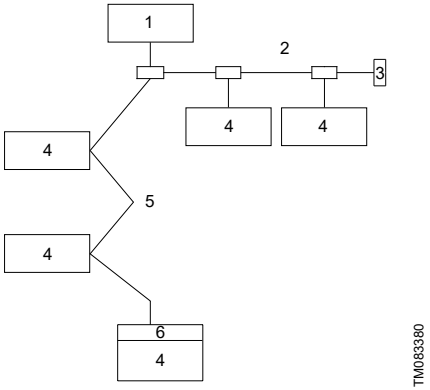
12.28.5.1 Protocol selection

Use this function to select which fieldbus protocol that must be active on the AYB terminals (RS-485).

Select between the following:

- Modbus RTU
- GENIbus.

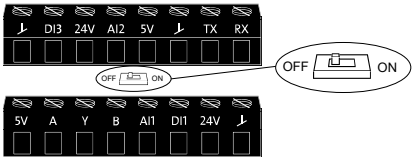
12.28.5.2 Modbus RTU settings



Example of Modbus network with termination

Pos.	Description
1	Master
2	Passive tap
3	Line termination
4	Slave
5	Daisy chain
6	BLT (BLT = Built-in line termination (dip switch))

Remember to set the AYB BUS termination dip switch to ON in case the pump is the first or the last pump on a daisy chain of pumps. The termination resistor has a value of 150 ohm.



Modbus RTU address

Use this function to allocate a unique number to the pump. This makes it possible to distinguish between pumps in connection with Modbus RTU communication.

Select a number between 1 and 247.

Baud rate

Use this function to select the baud rate at which Modbus RTU is to communicate.

Select between the following baud rates:

- 9600 bps
- 19200 bps
- 38400 bps
- 115200 bps.

Parity

Use this function to set the parity of the Modbus RTU channel.

Select between these values:

- None
- Odd
- Even.

Stop bits

Use this function to set the number of stop bits on the Modbus RTU channel.

Select between these values:

- 1 bit
- 2 bits.

12.28.6 Setup of Ethernet



The product is equipped with an Ethernet port with a GENI GDP protocol that can be accessed from Grundfos iSOLUTION Cloud and other cloud based solutions.

Grundfos will support the product with security updates for at least 2 years from production of the unit.

12.28.6.1 IP Settings

Use this function to set the Ethernet communication.

12.28.6.2 DHCP

Use this function to select if DHCP should be activated or deactivated.

If activated, the E-pump will receive network configuration from the DHCP server on the network.

If deactivated, IP address, Subnet mask, Gateway and Primary DNS must be configured manually.

12.28.6.3 IP address

Use this function to manually set the IP address. IP address format:

Example: 192.168.0.10

12.28.6.4 Subnet mask

Use this function to manually set the subnet mask. Subnet mask format:

Example: 255.255.255.0

12.28.6.5 Gateway

Use this function to manually set the gateway address. Gateway address format:

Example: 192.168.1.1

12.28.6.6 Primary DNS

Use this function to manually set the primary DNS address.

Example of primary DNS address format: 8.8.8.8

12.28.6.7 Secondary DNS

Use this function to manually set the secondary DNS address.

Example of secondary DNS address format: 4.4.4.4

12.28.7 Language

The function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to select the desired language from the list.

12.28.8 Set date and time

The function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to set the date and time as well as how you want them to be viewed on the display.

- **Select date format**:
  - **YYYY-MM-DD**
  - **DD-MM-YYYY**
  - **MM-DD-YYYY**.
- **Select time format**:
  - **HH:MM 24-hour clock**
  - **HH:MM am/pm 12-hour clock**.
- **Set date**.
- **Set time**.

12.28.9 Units

This function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to select SI or US units. You can make the setting for all parameters or customize for each individual parameter.

12.28.10 Enable/disable settings

Use this function to disable the adjusting of settings for protective reasons.

- If you use Grundfos GO and set the buttons to **Not active**, the buttons on the HMI 200 or HMI 201 operating panel are disabled, except the **Radio communication** button.
- If you disable the buttons on pumps fitted with the HMI 300 or HMI 301 operating panel via **Enable/ disable settings**, you can still use the buttons to navigate through the menus but you cannot make changes directly on these operating panels. A lock symbol appears on the display. However, you can unlock the motor temporarily and allow settings by pressing the **Up** and **Down** buttons simultaneously for at least 5 seconds.

12.28.11 Delete history

This function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to delete the following historical data:

- **Delete operating log**
- **Delete energy consumption**.

12.28.12 Define Home display

This function is only available in the HMI 300 and HMI 301 operating panels.

Set the **Home** display to show up to four user-defined parameters.

12.28.13 Display settings

This function is only available in the HMI 300 or HMI 301 operating panels.

Use this function to adjust the display brightness. You can also set whether or not the display is to switch off if no buttons are activated for a period of time.

12.28.14 Store actual settings

Use this function to store the current settings to enable the user to go back to a previous set of settings.

12.28.15 Recall stored settings

Grundfos GO

In this menu, you can recall stored settings from a number of previously stored settings that the pump then uses.

Advanced operating panel

In this menu, you can recall the last stored settings that the pump then uses.

12.28.16 Undo

This function is only available in the Grundfos GO.

Use this function to undo all settings made with the Grundfos GO in the current communication session. Once you recall settings, you cannot undo it.

12.28.17 Pump name

This function is only available in the Grundfos GO.

Use this function to give the motor a name. The selected name then appears in the Grundfos GO.

12.28.18 Connection code

Use the connection code to enable automatic connection between Grundfos GO and the product. Thus, you do not need to press **OK** or the **Radio communication** button each time.

You can also use the connection code to restrict remote access to the product.

You can only set the connection code with Grundfos GO.

12.28.18.1 Setting a connection code in the product by using the Grundfos GO

1. Connect Grundfos GO to the product.
2. Go to **Settings** > **General** > **Connection code**.
3. Enter a connection code and press **OK**.

   You can change the code in the **Connection code** menu at any time. The old code is not required.

12.28.19 Run start-up guide

The function is only available in the HMI 300 and HMI 301 operating panels.

The startup guide automatically starts when you start the product for the first time. You can always run it later. It guides you through the general settings of the product.

To run the startup guide, go to **Settings** > **General settings** > **Run start-up guide**.

12.28.20 Alarm log

This function contains a list of logged alarms from the product. The log shows the alarm code, name of the alarm, when the alarm occurred and when the alarm was reset.

12.28.21 Warning log

This function contains a list of logged warnings from the product. The log shows the warning code, name of the warning, when the warning occurred and when the warning was reset.

12.28.22 Assist

This menu consists of a number of different assist functions.

Assist functions are small guides that take you through the steps needed to set the product.

12.28.23 Assisted pump setup

This function guides you through the following:

Setting the motor

- Selection of control mode
- Configuration of feedback sensors
- Adjustment of the setpoint
- Controller settings
- Summary of settings.

With Grundfos GO, access the **Assisted pump setup** menu.

With the HMI 300 or 301 operating panel, access the **Assisted pump setup** menu.

Related information

12.6.6 Constant differential pressure

12.6.7 Constant differential temperature

12.11 Grundfos Direct Sensor

12.28.24 Setup, analog inputs

This function is only available in the HMI 300 and HMI 301 operating panels.

- **Analog inputs**, follow on-screen instructions.
- **Pt100/1000 inputs**, follow on-screen instructions.

12.28.25 Setting of date and time

This function is only available in the HMI 300 and HMI 301 operating panels.

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Setting of date and time
FM110	-
FM310	●
FM311	●

The function guides you through the following settings:

- **Select date format**
- **Set date**
- **Select time format**
- **Set time**.

12.28.26 Multipump function

The function **Multi-pump function** enables the control of two motors connected in parallel without the use of external controllers. The pumps or motors in a system communicate with each other via the wireless GENIair connection or the wired GENI connection.

You can set a multipump system via the master motor, which is the first selected motor.

If several pumps or motors in the system have sensors, they can all function as the master and take over the master function if the other fails. This provides additional redundancy in the multimotor system.

You can choose between the following multimotor functions:

Alternating operation

Alternating operation functions as a duty and standby operating mode and is possible with two pumps or two motors of the same size and type connected in parallel. The main purpose of the function is to ensure an even amount of running hours and to ensure that the other pump or motor starts if the duty pump or motor stops due to an alarm.

You can choose between two alternating operating modes:

- **Alternating operation, time**

  The change from one pump or motor to the other is based on time.
- **Alternating operation, energy**

  The change from one pump or motor to the other is based on energy consumption.

If the duty pump or motor fails, the other pump or motor starts.

Backup operation

Backup operation is possible with two motors of the same size and type connected in parallel. One motor is operating continuously. The backup motor is operated for a short time each day to prevent seizing up. If the duty motor stops due to a fault, the backup motor starts.

Cascade operation

This function is available with up to 4 motors installed in parallel. The motors must be of the same size and if used with pumps, the pumps must be of the same model.

- The performance is adjusted to the demand through cutting pumps in or out and through parallel control of the pumps in operation.
- The controller maintains a constant process value through continuous adjustment of the speed of the pumps.
- Pump changeover is automatic and depends on load, operating hours and fault detection.
- All pumps in operation run at the same speed.
- The number of pumps in operation also depends on the energy consumption of the pumps. If only one pump is required, two pumps will run at a lower speed if this results in a lower energy consumption.
- If several pumps or motors in the system have a sensor, they can all function as the master and take over the master function if the other fails.

Related information

9.2 Alternating operation of twin-head pumps

12.28.26.1 Alternating operation, time

The **Alternating operation, time** menu sets the interval of alternation between two pumps.

This setting is only available in alternating mode.

12.28.26.2 Time for pump changeover

The **Time for pump changeover** menu sets the time of day for pump changeover to take place.

This setting is only available in alternating operation.

12.28.26.3 Sensor to be used

This function defines the sensor for controlling the pump system.

Select **Master pump sensor** if the sensor is placed in a way where it can measure the output from all the pumps in the system, for example, in the manifold.

Select **Running pump sensor** if the sensor is placed on or across individual pumps, for example, if the sensor is installed behind non-return valves, or if it is not able to measure the output from all pumps.

This setting is only available in alternating operation and cascade operation.

12.28.26.4 Setting up a multipump system

You can set up a multipump system in the following ways:

- Grundfos GO and wireless motor connection
- Grundfos GO and wired motor connection
- HMI 300 or HMI 301 operating panel and wireless motor connection
- HMI 300 or HMI 301 operating panel and wired motor connection.

Grundfos GO and wireless motor connection

1. Power on both motors.
2. Establish contact to one of the motors with the Grundfos GO.

3. Set the needed analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
4. Assign a name to the motor using the Grundfos GO.
5. Disconnect the Grundfos GO from the motor.
6. Establish contact to the other motor.
7. Set the needed analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
8. Assign a name to the motor using the Grundfos GO.
9. Select the **Assist** menu and **Setup of multi-pump system**.
10. Select the desired multimotor function.
11. Press the **Right** button to continue.
12. Set the time when the alternation between the two motors is to take place.



This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.

13. Press the **Right** button to continue.
14. Select **Radio** as the communication method between the two motors.
15. Press the **Right** button to continue.
16. Select pump 2 (motor 2).
17. Select the pump from the list.



Use **OK** or the **Radio communication** button to identify the pump.

18. Press the **Right** button to continue.
19. Confirm the setting by pressing **Send**.
20. When you finish the setup and the dialog box disappears, wait for the green indicator light in the middle of **Grundfos Eye** to light up.

Grundfos GO and wired motor connection

1. Connect the two motors with each other with a 3-core screened cable between the GENIbus terminals A, Y, B.
2. Power on both motors.
3. Establish contact to one of the motors with the Grundfos GO.
4. Set the required analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
5. Assign a name to the motor using the Grundfos GO.
6. Assign motor number 1 to the motor.
7. Disconnect the Grundfos GO from the motor.
8. Establish contact to the other motor.
9. Set the analog and digital inputs according to the connected equipment and the required functionality by the Grundfos GO.
10. Assign a name to the motor using the Grundfos GO.
11. Assign motor number 2 to the motor.
12. Select the **Assist** menu and **Setup of multi-pump system** (**multimotor setup**).
13. Select the desired multimotor function.
14. Press the **Right** button to continue.
15. Set the time when the alternation between the two motors is to take place.



This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.

16. Press the **Right** button to continue.
17. Select **Bus** as the communication method between the two motors.
18. Press the **Right** button to continue.
19. Select pump 2 (motor 2).
20. Select the additional motor from the list.



Use **OK** or the **Radio communication** button to identify the pump.

21. Press the **Right** button to continue.
22. Confirm the setting by pressing **Send**.
23. When you have finish the setup and the dialog box disappears, wait for the green indicator light in the middle of the **Grundfos Eye** to light up.

HMI 300 or HMI 301 and wireless motor connection

1. Power on both motors.

2. On both motors, set the analog and digital inputs according to the connected equipment and the required functionality.
3. Select the **Assist** menu on one of the motors and **Setup of multi-pump system**.
4. Press the **Right** button to continue.
5. Select **Wireless** as the communication method between the two motors.
6. Press the **Right** button to continue.
7. Select the desired multimotor function.
8. Press the **Right** button three times to continue.
9. Press **OK** to search for other motors.
   The green indicator light in the middle of the **Grundfos Eye** flashes on the other motors.
10. Press **OK** or the **Radio communication** button on the motor that is to be added to the multimotor system.
11. Press the **Right** button to continue.
12. Set the **Time for pump changeover**.
    This is the time when the alternation between the two motors is to take place.



This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.

13. Press the **Right** button to continue.
14. Press **OK** to confirm the setting.
    The multipump function icons appear at the bottom of the operating panels.

HMI 300 or HMI 301 and wired motor connection

1. Connect the two motors with each other with a 3-core screened cable between the GENIbus terminals A, Y, B.
2. Set the needed analog and digital inputs according to the connected equipment and the required functionality.
3. Assign motor number 1 to the first motor.
4. Assign motor number 2 to the other motor.
5. Select the **Assist** menu on one of the motors and the **Setup of multi-pump system**.
6. Press the **Right** button to continue.
7. Select **Wired GENIbus** as the communication method between the two motors.
8. Press the **Right** button twice to continue.
9. Select the desired multimotor function.
10. Press the **Right** button to continue.
11. Press **OK** to search for other motors.
12. Select the additional motor from the list.
13. Press the **Right** button to continue.
14. Set the **Time for pump changeover**.
    This is the time when the alternation between the two motors is to take place.



This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.

15. Press the **Right** button to continue.
16. Press **OK** to confirm the setting.
    The multipump function icons appear at the bottom of the operating panels.

12.28.26.5 Grundfos GO

1. Go to **Assist**.
2. Select the **Multi-pump setup** and press **Disable**.
3. Press the **Right** button to continue.
4. Confirm the setting by pressing **Send**.
5. Press **Finish**.

12.28.26.6 HMI 300 or HMI 301

1. Go to **Assist**.
2. Select the **Setup of multi-pump system**.
3. Press the **Right** button to continue.
4. Press **OK** to confirm **Disable**.
5. Press the **Right** button to continue.
6. Press **OK** to confirm.

12.28.27 Description of control mode

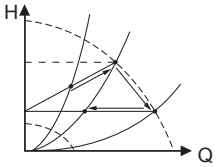
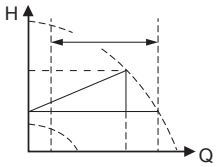
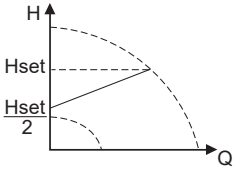
This function is only available in the HMI 300 and HMI 301 operating panels.

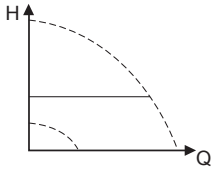
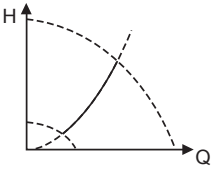
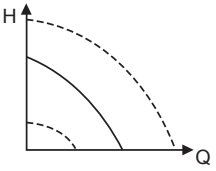
The function describes each of the control modes available for the product.

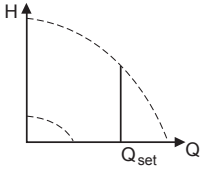
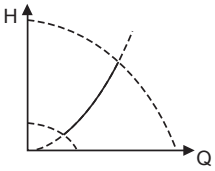
12.28.28 Assisted fault advice

This function provides guidance and corrective actions in the event of product failure.

13. Selecting control mode

System application	Select this control mode
Recommended for most heating systems, especially in systems with relatively large pressure losses in the distribution pipes. See the description under proportional pressure. In replacement situations where the proportional-pressure duty point is unknown. The duty point has to be within the AUTOADAPT operating range. During operation, the pump automatically adjusts to the actual system characteristic. This setting ensures minimum energy consumption and low noise level from valves, which reduces operating costs and increases comfort.	AUTOADAPT 
The FLOWADAPT control mode is a combination of AUTOADAPT and FLOWLIMIT. This control mode is suitable for systems where you want a maximum flow limit, FLOWLIMIT. The pump continuously monitors and adjusts the flow, thus ensuring that the selected FLOWLIMIT is not exceeded. Main pumps in boiler applications where a steady flow through the boiler is required. No extra energy is used for pumping too much liquid into the system. In systems with mixing loops, you can use the control mode to control the flow in each loop. **Benefits** • Enough water for all loops at peak load conditions if you have set each loop to the right maximum flow. • The dimensioned flow for each zone, required heat energy, is determined by the flow from the pump. You can set this value precisely in the FLOWADAPT control mode without the use of throttling valves. • When the flow is set lower than the balancing valve setting, the pump ramps down instead of losing energy by pumping against a balancing valve. • Cooling surfaces in air-conditioning systems can operate at high pressure and low flow.	FLOWADAPT 
In systems with relatively large pressure losses in the distribution pipes and in air-conditioning and cooling systems. • Two-pipe heating systems with thermostatic valves and the following: - very long distribution pipes - strongly throttled balancing valves - differential-pressure regulators - large pressure losses in those parts of the system through which the total quantity of water flows, for example boiler, heat exchanger and distribution pipe up to the first branching. • Primary circuit pumps in systems with large pressure losses in the primary circuit. • Air-conditioning systems with the following: - heat exchangers, fan coils - cooling ceilings - cooling surfaces.	Proportional pressure 

System application	Select this control mode
In systems with relatively small pressure losses in the distribution pipes. • Two-pipe heating systems with thermostatic valves and the following: - sized for natural circulation - small pressure losses in those parts of the system through which the total quantity of water flows, for example boiler, heat exchanger and distribution pipe up to the first branching or modified to a high differential temperature between flow pipe and return pipe, for example district heating. • Underfloor heating systems with thermostatic valves. • One-pipe heating systems with thermostatic valves or balancing valves. • Primary circuit pumps in systems with small pressure losses in the primary circuit.	Constant differential pressure 
In systems with a fixed system characteristic. **Examples** • one-pipe heating systems • boiler shunts • systems with three-way valves • domestic hot-water circulation. You can use FLOWLIMIT with advantage to control the maximum circulation flow.	Constant temperature and constant differential temperature 
If an external controller is installed, the pump is able to change from one constant curve to another, depending on the external signal. You can also set the pump to operate according to the maximum or minimum curve, like an uncontrolled pump: • Use the maximum-curve mode in periods in which a maximum flow is required. This operating mode is for instance suitable for hot-water priority. • Use the minimum-curve mode in periods in which a minimum flow is required. This operating mode is for instance suitable for manual night setback if you do not want automatic night setback.	Constant curve 

System application	Select this control mode
In systems requiring a constant flow, independently of pressure drop. **Examples** • chillers for air-conditioning • heating surfaces • cooling surfaces.	Constant flow rate 
In systems requiring a constant tank level, independently of the flow rate. **Examples** • process water tanks • boiler condensate tanks.	Constant level 
In systems with pumps connected in parallel. The multipump function enables the control of two to four single-head pumps connected in parallel and twin-head pumps without the use of external controllers. The pumps in a multipump system communicate with each other via the wireless GENIair connection or the wired GENI connection.	**Assist** menu Multipump setup

14. Priority of settings

With Grundfos GO, you can set the motor to operate at maximum speed or to stop.

If two or more functions are enabled at the same time, the motor operates according to the function with the highest priority.

If you have set the motor to maximum speed via the digital input, the motor operating panel or Grundfos GO can only set the motor to **Manual** or **Stop**.

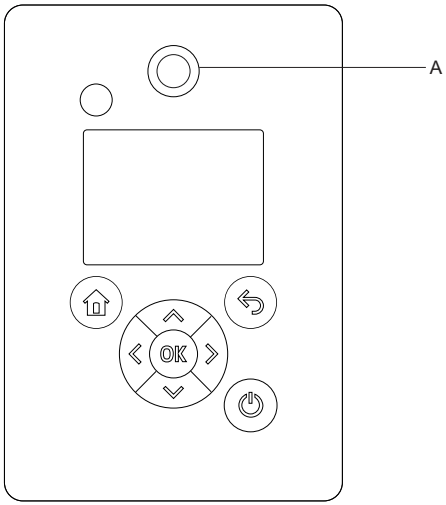
The priority of the settings appears from the table below:

Priority	Start/stop button	Grundfos GO or operating panel on motor	Digital input	Bus communication
1	**Stop**			
2		**Stop** [13)]		
3		**Manual**		
4		**Maximum speed** / **User defined speed** [13)]		
5			**Stop**	
6			**User defined speed**	
7				**Stop**
8				**Maximum speed** / **User defined speed**
9				**Minimum speed**
10				**Start**
11			**Maximum speed**	
12		**Minimum speed**		
13			**Minimum speed**	
14			**Start**	
15		**Start**		

13) **Stop** and **Maximum speed** settings made with Grundfos GO or on the motor operating panel can be overruled by another operating-mode command sent from a bus, for example **Start**. If the bus communication is interrupted, the motor resumes its previous operating mode, for example **Stop**, that was selected with Grundfos GO or the motor operating panel.

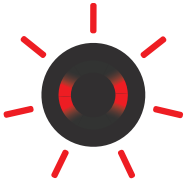
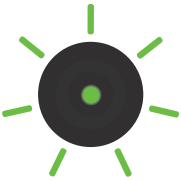
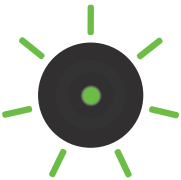
15. Grundfos Eye

The operating condition of the motor is indicated by Grundfos Eye on the motor operating panel. See figure below (A).



Grundfos Eye indicator light

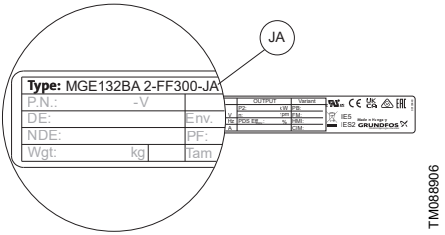
Indicator light	Indication	Description
	No lights are on.	**Power off** The motor is not running.
	Two opposite green indicator lights are rotating.	**Power on** The motor is running. The indicator lights are rotating in the direction of rotation of the motor when seen from the non-drive end.
	Two opposite green indicator lights are permanently on.	**Power on** The motor is not running.
	One yellow indicator light is rotating.	**Warning** The motor is running. The indicator light is rotating in the direction of rotation of the motor when seen from the non-drive end.

Indicator light	Indication	Description
	One yellow indicator light is permanently on.	**Warning** The motor has stopped.
	Two opposite red indicator lights are flashing simultaneously.	**Alarm** The motor has stopped.
	The green indicator light in the middle flashes quickly four times.	Grundfos Eye flashes four times when you press the Grundfos Eye symbol next to the motor name in Grundfos GO.
	The green indicator light in the middle is flashing continuously.	You have selected the motor in Grundfos GO, and the motor is ready to be connected.
	The green indicator light in the middle flashes quickly for a few seconds.	The motor is controlled by Grundfos GO or exchanging data with Grundfos GO.
	The green indicator light in the middle is permanently on.	The motor is connected to Grundfos GO.

16. Identification of the motor model

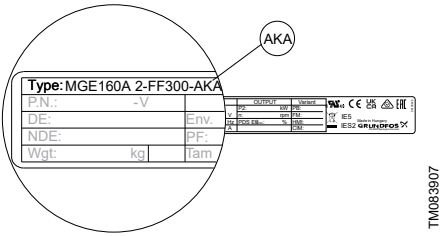
Identify the motor model by means of the nameplate on the terminal box.

Model J



Supply voltage	Speed	Power P2					
[V]	[rpm]	[kW]					
3-phase		2.2	3	4	5.5	7.5	11
380-500	1450-2200	•	•	•	•	•	-
	2900-4000/ 4000-5900	-	•	•	•	•	•
200-240	3400-4000	•	•	•	•	-	-

Model K



Supply voltage	Speed	Power P2					
[V]	[rpm]	[kW]					
3-phase		7.5	11	15	18.5	22	26
380-480	1450-2200	-	•	•	•	•	-
	2900-4000	-	-	•	•	•	-
400-480	3500-4000	-	-	-	-	-	•
200-240	3400-4000	•	•	-	-	-	-

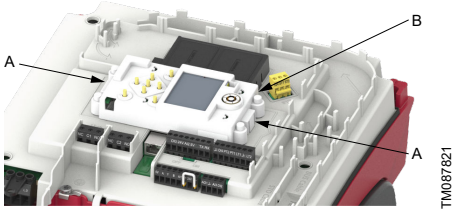
17. Changing the position of the operating panel

WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box.

You can turn the operating panel 180°. Follow the instructions.

1. Loosen the four screws of the terminal box cover.
2. Remove the terminal box cover.
3. Press and hold in the two locking tabs (A) while gently lifting the plastic cover (B).

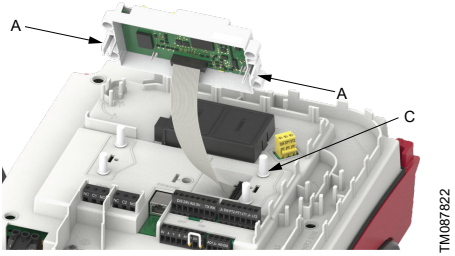


Lifting the plastic cover, shown on a Model J motor

4. Turn the plastic cover 180°.

Do not twist the cable more than 90°.

5. Position the plastic cover correctly over the four rubber pins (C). Make sure that the locking tabs (A) are placed correctly.



Positioning the plastic cover, shown on a Model J motor

6. Tighten the four screws with 5 Nm.

Make sure that the terminal box cover is aligned with the orientation of the operating panel.

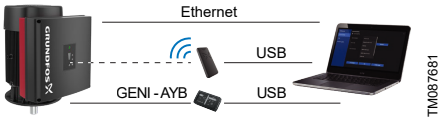
18. Grundfos GO Link

The product is designed for wired or wireless communication with the Grundfos GO Link.

The Grundfos GO Link enables you to set functions and gives you access to status overviews, configuration and current operating parameters.

Use the Grundfos GO Link together with the following interfaces:

- Ethernet cable (Only FM310 and FM311)
- Grundfos MI 301 - USB - Wired/wireless (Only HMI 100, HMI 200 and HMI 300)
- Grundfos PC Tool Link - USB - Wired.



Grundfos GO Link setup

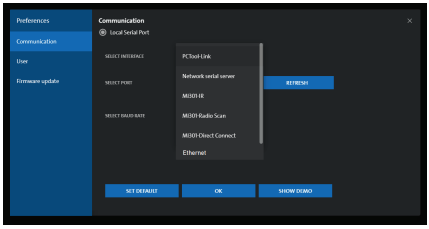
Pos.	Description
1	Ethernet cable: Standard Ethernet cable CAT5/CAT6
2	Grundfos MI 301: Separate radio equipment enabling radio communication [14]
3	Grundfos PC Tool Link: Separate module enabling wired connection to the pump [14]

14) Use the module together with a USB cable to connect to a laptop.

18.1 Communication

When Grundfos GO Link initiates communication with the product, it is done using different verification methods.

Select the interface connected to the pump:



18.2 Ethernet

Wired connection can take place using an Ethernet cable connected directly between a laptop and the RJ45 interface in the pump or via a local network having both the pump and the laptop connected to the same network.

To establish a secure connection between the laptop and the pump, the user will have to go through a verification process.

Connecting to a pump can either happen by scanning for a connected product, which can be a direct Ethernet connection, or the pump is connected to a local network or a connection via the pump IP address.

Initiate connection from Grundfos GO Link and follow onscreen instructions.

18.3 Grundfos MI 301

Radio communication can take place at distances up to 30 metres. The first time Grundfos GO Link communicates with the product, you enable communication by pressing the **Radio communication** button or **OK** on the operating panel. Select either MI301-Direct connect or MI301-Radio. When communication takes place, the product is recognized by Grundfos GO Link, and you can connect using Direct connect or Radio scan without having to run a verification.

18.4 Grundfos PC Tool Link

Wired connection can take place using Grundfos PC Tool connected to the AYB terminals of the pump. Since Grundfos GO Link is wired to the pump within a short distance, no verification is needed. A direct connection will be established.

19. Servicing the product



DANGER
Electric shock
Death or serious personal injury

- Switch off the power supply to the motor and to the signal relays. Wait at least 5 minutes before starting any work on the motor. Make sure that the power supply cannot be accidentally switched on.



DANGER
Magnetic field
Death or serious personal injury

- Do not handle the motor or rotor if you have a pacemaker.



WARNING
Hazardous liquids
Death or serious personal injury

- If the pump is used for a liquid which is injurious to health, it will be classified as contaminated. In such cases, take proper precautions to avoid injury to health when operating or working on the pump.
- Wear personal protection equipment.



CAUTION
Hot or cold liquids
Minor or moderate personal injury

- Wear personal protection equipment.



CAUTION
Pressurised system
Minor or moderate personal injury

- Do not remove the ultrasonic flow sensor during service and maintenance.



CAUTION
Pressurised system
Minor or moderate personal injury

- Wear personal protective equipment.

19.1 Maintenance

19.1.1 Pump

If the pump is to be drained for a long period of inactivity, inject a few drops of silicone oil on the shaft between the motor stool and the coupling. This prevents the shaft seal faces from sticking.

19.1.2 Motor

The motor must be checked at regular intervals. It is important to keep the motor clean in order to ensure adequate ventilation. If the pump is installed in a dusty environment, both pump and motor must be cleaned and checked regularly.

19.2 Service

If Grundfos is requested to service the pump, you must clean it before returning it. If you cannot clean the pump properly, then provide Grundfos with all relevant information about the pumped liquid.

If the above is not fulfilled, Grundfos can refuse to accept the pump for service.

Possible costs of returning the pump are to be paid by the customer.

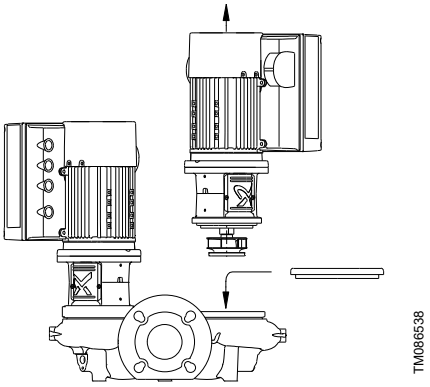
19.2.1 Integral shaft and coupling

TPE2 and TPE3 pumps have integral shaft and coupling. We recommend that you do not remove the motor.

If you have removed the motor, you must remove the motor stool in order to refit the motor correctly. Otherwise the shaft seal may be damaged.

19.2.2 Blanking flanges

For twin-head pumps, a blanking flange with a pump housing gasket is available.



Fitting the blanking flange

If one pump requires service, fit the blanking flange to allow the other pump to continue operating.

20. Cleaning the product

WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the motor and the signal relays. Confirm that the terminal box cover is intact before spraying water on the product.

To avoid condensation in the motor, let the motor cool down before spraying it with cold water.

21. Fault finding



DANGER
Electric shock
Death or serious personal injury

- Switch off the power supply to the motor and to the signal relays. Wait at least 5 minutes before starting any work on the motor. Make sure that the power supply cannot be accidentally switched on.



DANGER
Magnetic field
Death or serious personal injury

- Do not handle the motor or rotor if you have a pacemaker.



WARNING
Hazardous liquids
Death or serious personal injury

- If the pump is used for a liquid which is injurious to health, it will be classified as contaminated. In such cases, take proper precautions to avoid injury to health when operating or working on the pump.
- Wear personal protection equipment.



CAUTION
Hot or cold liquids
Minor or moderate personal injury

- Wear personal protection equipment.

21.1 The motor does not run when started

Cause	Remedy
Supply failure.	Check the electrical connection. Remedy, if necessary.
The fuses are blown.	Replace the fuse.
The motor is defective.	Replace the motor.

21.2 Motor alarm when the power supply is switched on

Cause	Remedy
Supply failure.	Check the electrical connection. Remedy, if necessary.
The cable connection is loose or faulty.	Repair cable connection.
The motor winding is defective.	Repair the motor.
The pump is mechanically blocked.	Clean the pump.

21.3 Occasional motor alarm

Cause	Remedy
The supply voltage is periodically too low or too high.	Make sure the input supply voltage stable.

21.4 No motor alarm, but the pump does not run

Cause	Remedy
Check the power supply.	Make sure the power supply connected.
Check the fuses.	Replace the fuse, if defective.

21.5 The pump performance is not constant

Cause	Remedy
The pump inlet pressure is too low.	Increase the inlet pressure.
The inlet pipe or pump is partly is blocked by impurities.	Clean the inlet pipe or pump.

Cause	Remedy
The pump draws in air.	1. Increase the liquid level on the inlet side. 2. Keep the liquid level as constant as possible. 3. Increase the inlet pressure.

21.6 The pump runs but delivers no water

Cause	Remedy
The inlet pipe or pump is blocked by impurities.	Clean the inlet pipe or pump.
The foot or non-return valve is blocked in closed position.	Repair the foot or non-return value.
There is a leakage in the inlet pipe.	Check the gaskets and bolts torque. Replace gaskets, if defective.
There is air in the inlet pipe or pump.	Vent the inlet pipe or the pump.

21.7 The pump runs backwards when switched off

In twin-head pump installations, the backup pump often rotates slowly.

Cause	Remedy
There is a leakage in the inlet pipe.	Check the gaskets and bolts torque. Replace gaskets, if defective.
The foot or non-return valve is defective.	Replace the foot or non-return valve, if defective.
The foot or non-return valve is blocked in open or partly open position.	Repair the foot or non-return value.

21.8 Leakage in shaft seal

Cause	Remedy
The shaft seal is defective.	Replace the shaft seal.

21.9 Noise

Cause	Remedy
The pump is cavitating.	Increase inlet pressure.
The pump does not rotate freely (frictional resistance) because of incorrect pump shaft position.	Check the motor bearing, wear ring and impeller, shaft seal. Replace them, if worn out or blocked.
There is resonance in the installation.	Fasten the support on foundation. Add damping joints. Make sure the pipes are properly supported.
There are foreign bodies in the pump.	Clean the pump.

22. Technical data

22.1 Technical data, three-phase motors



WARNING
Electric shock
Death or serious personal injury
- Use the recommended fuse size.

Supply voltage

- 3 × 380-500 V -10 % / +10 %, 50/60 Hz, PE
- 3 × 380-480 V -10 % / +10 %, 50/60 Hz, PE
- 3 × 400-480 V -10 % / +10 %, 50/60 Hz, PE
- 3 × 200-240 V -10 % / +10 %, 50/60 Hz, PE

Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Recommended size of fuse

For recommended size of fuses see the tables below.

3 × 380-500 V, Model J

Motor size [kW]	Recommended [A]	Maximum [A]
3	10	16
4	13	16
5.5	16	32
7.5	20	32
11	32	32

3 × 380-480 V, Model K

Motor size [kW]	Recommended [A]	Maximum [A]	Fuse type
11	35	63	gG
15	50	80	gG
18.5	60	80	gG
22	70	80	gG

3 × 200-240 V, Model J

Motor size [kW]	Recommended [A]	Maximum [A]
2.2	13	35
3	16	35
4	25	35
5.5	32	35

3 x 200-240 V, Model K

Motor size [kW]	Recommended [A]	Maximum [A]	Fuse type
7.5	60	80	gG
11	70	80	gG

3 × 400-480 V, Model K

Motor size [kW]	Recommended [A]	Maximum [A]	Fuse type
26	80	80	gG



For recommended size of fuses see the appendix concerning installation in the USA and Canada.

22.1.1 Leakage current (AC)

The leakage currents are measured without any load on the shaft and in accordance with EN 61800-5-1:2007.

WARNING
Electric shock
Death or serious personal injury



- If the leakage current is greater than 3.5 mA, use a PE cable with a minimum cross-section of at least 10 mm^2, or use 2 separate PE cables with the same cross section as the power cable.

3 × 380-500 V, 50/60 Hz, Model J

Speed [rpm]	Power [kW]	Mains voltage [V]	Leakage current (I_L) [mA]
1450-2200	2.2 - 4	≤ 400	< 3.5
		> 400	< 3.5
	5.5 - 7.5	≤ 400	< 3.5
		> 400	3.5 < I_L < 5.0
2900-4000	3 - 5.5	≤ 400	< 3.5
		> 400	< 3.5
	7.5 - 11	≤ 400	< 3.5
		> 400	3.5 < I_L < 5.0
4000-5900	3 - 5.5	≤ 400	< 3.5
		> 400	< 3.5
	7.5 - 11	≤ 400	< 3.5
		> 400	3.5 < I_L < 5.0

3 × 380-480 V, 50/60 Hz, Model K

Speed [rpm]	Power [kW]	Mains voltage [V]	Leakage current (I_L) [mA]
1450-2200	11-22	≤ 400	3.5 < I_L < 20
		> 400	3.5 < I_L < 30
2900-4000	15-22	≤ 400	3.5 < I_L < 20
		> 400	3.5 < I_L < 30

3 × 400-480 V, 50/60 Hz, Model K

Speed [rpm]	Power [kW]	Mains voltage [V]	Leakage current (I_L) [mA]
3500-4000	26	≤ 400	3.5 < I_L < 20
		> 400	3.5 < I_L < 30

3 × 200-240 V, 50/60 Hz, Model J

Speed [rpm]	Power [kW]	Mains voltage [V]	Leakage current (I_L) [mA]
3400-4000	2.2 - 5.5	200-240	< 3.5

3 × 200-240 V, 50/60 Hz, Model K

Speed [rpm]	Power [kW]	Mains voltage [V]	Leakage current (I_L) [mA]
3400-4000	7.5 - 11	> 200	$3.5 < I_L < 20$

22.2 Inputs and outputs

Signal reference

All voltages refer to signal ground (GND). All currents return to signal ground.

Absolute maximum voltage and current limits

Exceeding the following electrical limits may result in severely reduced operating reliability and motor life.

Relay 1:

- Maximum contact load: 250 VAC, 2 A or 30 VDC, 2 A.

Relay 2:

- Maximum contact load: 30 VDC, 2 A.

GENI terminals: -5.5 to +9.0 VDC or less than 25 mADC.

Other input and output terminals: -0.5 to +26 VDC or less than 15 mADC.

Digital inputs

Internal pull-up current greater than 10 mA at V_i equal to 0 VDC.

Internal pull-up to 5 VDC. Currentless for V_i greater than 5 VDC.

Input activated level: V_i less than 1.5 VDC.

Input deactivated level: V_i from 3.0 VDC to 24 VDC.

Hysteresis: No.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

Safe Torque Off (STO) terminals

S24:

24 V output voltage. Only for use with ST1 and ST2 inputs.

- Output voltage: 24 V -5 % to +5 %
- Maximum current: 50 mADC
- Overload protection: Yes.

ST1 and ST2:

- STO activated: V_{in} lower than 1.25 V
- STO deactivated: V_{in} greater than 21.6 V and lower than 25 V
- Input current greater than 10 mA at V_{in} equal to 24 V.

When the internal voltage source (connection S24) is used, the input voltage for ST1 and ST2 is within accepted limits.

When an external voltage source is used to drive the STO inputs, the following conditions must be met:

In operational state, the input voltage of ST1 and ST2 with reference to GND must be within:

- V_{min}: 21.6 V
- V_{max}: 25.0 V.

In the safe state, the input voltage of ST1 and ST2 with reference to GND must be as follows:

- V_{max}: 1.25 V.

In the operating state, the current flow into ST1 and ST2 must be within:

- Minimum contact current: 10 mA
- Maximum contact current: 25 mA.

Input source rating: SELV

Bus input (Ethernet)

Protocols TC/IP GENI, GDP.

Cable type, Standard CAT5, CAT5e or CAT6.

Open-collector digital outputs (OC)

Current-sinking capability: 75 mADC, no current sourcing.

Load types: Resistive and/or inductive.

Low-state output voltage at 75 mADC: Maximum 1.2 VDC.

Low-state output voltage at 10 mADC: Maximum 0.6 VDC.

Overcurrent protection: Yes.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

Analog inputs (AI)

Voltage signal ranges:

- 0.5 - 3.5 VDC, AL AU
- 0-5 VDC, AU
- 0-10 VDC, AU.

Voltage signal:

- R_i greater than 100 kΩ at 25 °C.

Leak currents may occur at high operating temperatures. Keep the source impedance low.

Current signal ranges:

- 0-20 mADC, AU
- 4-20 mADC, AL AU.

Current signal: R_i is equal to 292 Ω.

Current overload protection: Yes. Change to voltage signal.

Measurement tolerance: +/- 2 % of full scale.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m, excluding potentiometer.

Potentiometer connected to +5 V, GND, any AI: Use maximum 10 kΩ.

Maximum cable length: 100 m.

Analog output (AO)

Current sourcing capability only.

Voltage signal:

- Range: 0-10 VDC
- Minimum load between AO and GND: 1 kΩ

- Short-circuit protection: Yes.

Current signal:

- Ranges: 0-20 and 4-20 mADC
- Maximum load between AO and GND: 500 Ω
- Open-circuit protection: Yes.

Tolerance: +/- 4 % of full scale.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

Pt100 or Pt1000 inputs (Pt)

Temperature range:

- Minimum -50 °C (80 Ω/803 Ω).
- Maximum 204 °C (177 Ω/1773 Ω).

Measurement tolerance: +/- 1.5 °C.

Measurement resolution: less than 0.3 °C.

Automatic range detection (Pt100 or Pt1000): Yes.

Sensor fault alarm: Yes.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Use Pt100 for short wires.

Use Pt1000 for long wires.

LiqTec sensor inputs

Use a Grundfos LiqTec sensor only.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Grundfos Digital Sensor input and output (GDS)

Use Grundfos Digital Sensor only.

Power supplies, +5 V, +24 V

+5 V

- Output voltage: 5 VDC -5 % to +5 %
- Maximum current: 60 mADC, sourcing only
- Overload protection: Yes.

+24 V

- Output voltage: 24 VDC -5 % to +5 %
- Maximum current: 200 mADC, sourcing only
- Overload protection: Yes.

Digital outputs, relays

Potential-free changeover contacts.

Minimum contact load when in use: 5 VDC, 10 mA.

Screened cable: 0.5 - 2.5 mm^2 / 28-12 AWG.

Maximum cable length: 500 m.

Bus input

Grundfos GENIbus protocol, RS-485.

Grundfos Modbus protocol, RS-485.

Screened 3-core cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

23. Other technical data

23.1 Ecodesign Directive

This product is out of scope of Directive 2009/125/EC and Commission Regulation (EU) 2019/1781 due to Article 2 (3a), as the variable speed drive (VSD) is integrated into a product and its energy performance cannot be tested independently from the product.

23.2 EMC (electromagnetic compatibility)

Standard used: EN 61800-3.

The table below shows the emission category of the motor.

C1 fulfils the requirements for residential areas.

C2 fulfils the requirements for residential areas if the system is operated and installed by qualified persons.

C3 fulfils the requirements for industrial areas.

In a residential environment, this product may cause radio interference in which case supplementary mitigation measures may be required.

Model J: When connected to a public network, 11-kW motors do not comply with the partial weighted harmonics (PWH) requirements of EN 61000-3-12. If required by the distribution network operator, compliance can be obtained in the following way:

The impedance of the mains cables between the motor and the point of common coupling (PCC) must be equivalent to the impedance of a 50 m cable.

Model K: This equipment complies with IEC 61000-3-12 provided that the short-circuit power S_{SC} is greater than or equal to the respective value described in the table below at the interface point between the user's supply and the public system. It is the responsibility of the installer or user of the equipment to ensure, by consultation with the distribution network operator if necessary, that the equipment is connected only to a supply with a short-circuit power S_{SC} greater than or equal to the respective value described in the table below.

When connected to a public network, 11-18.5 kW motors do not comply with the partial weighted harmonics (PWH) requirements of EN 61000-3-12. If required by the distribution network operator, compliance can be obtained in the following way:

The impedance of the mains cables between the motor and the point of common coupling (PCC) must be equivalent to the impedance of a 50 m cable.

3 × 380-480 V, 50/60 Hz, Model K

Speed [rpm]	Power P2 [kW]	Short-circuit power [MVA]
1450-2200	11	3.5
	15	4.6
	18.5	5.6
	22	6.6
2900-4000	15	4.6
	18.5	5.8
	22	6.6

3 × 400-480 V, 50/60 Hz, Model K

Speed [rpm]	Power P2 [kW]	Short-circuit power [MVA]
3500-4000	26	7.9

Model J

Speed [rpm]	Power P2 [kW]	Emission category
1450-2000	2.2 - 4	C1
	5.5 - 7.5	C3/C2 [15)]
2900-4000/ 4000-5900	2.2 - 5.5	C1
	7.5 - 11	C3/C2 [15)]

[15)] C2, if equipped with an external Grundfos EMC filter.

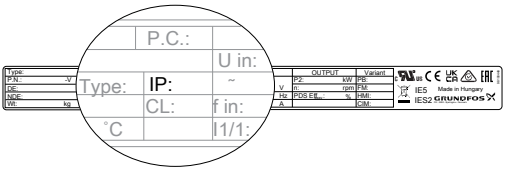
Model K

Speed [rpm]	Power P2 [kW]	Emission category
1450-2200	11 - 22	C2/C3 [16]
2900-4000	15 - 22	
3400-4000	7.5 - 11	
3500-4000	26	

16) Depending on product hardware configuration.

23.3 Enclosure class

Standard: IP55.
Optional: IP66.
IP ratings are evaluated in accordance with IEC 60034-5.
The IP rating can be found on the product nameplate:



TM084099

23.4 Insulation class

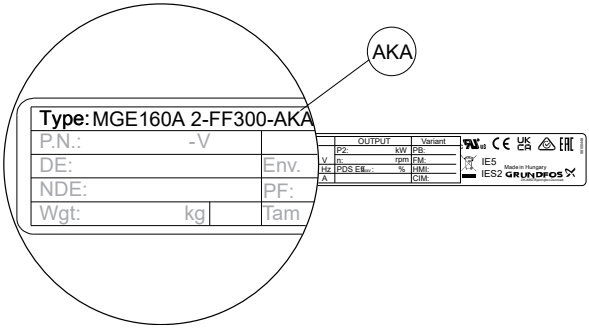
311 °F (155 °C).

23.5 Standby power consumption

5-10 W.

23.6 Cable entry sizes

Identify the motor model by means of the nameplate on the terminal box.



TM083907

Example shown on a Model K motor

Model	Number and size of cable entries
H, I	4 × M20
J	1 × M25 + 4 × M20 or 1 × M32 + 5 × M20
K	1 × M40 + 6 × M20

23.7 Torques

Torques for terminals

Terminal	Recommended torque [Nm]
L1, L2, L3	2.2
PE	6
NC, C1, C2, NO	0.5
DI1, DI2, DI3, DI4, AI1, AI2, AI3, AO1, PT1, PT2, LT1, LT2, GND, 24V, 5V, TX, RX, A, Y, B, S24, ST1, ST2	0.5

Torques for other parts

Part designation	Recommended torque [Nm]
Control box, upper part	6.5 - 7
Cover for mains	1.0 - 1.3
Cable glands:	
M20/M40	1 - 1.5
Blind plugs:	
M20	1 - 1.5
½" NPT	8 - 10

23.8 Sound pressure level

Pump type	Sound pressure level ISO 3743 [dB(A)]
TPE2 25-50	53
TPE2 25-80	53
TPE2 25-90	53
TPE2 32-50	53
TPE2 32-70	53
TPE2 32-90	53
TPE2 40-50	53
TPE2 40-70	53
TPE2 40-90	53
TPE2 125-150	59
TPE2 125-190	61
TPE2 125-230	64
TPE2 125-300	66
TPE2 125-340	70
TPE2 150-130	59
TPE2 150-160	61
TPE2 150-200	64
TPE2 150-220	66
TPE2 150-250	70
TPE2 150-260	66
TPE2 150-280	70
TPE2 200-160	64
TPE2 200-190	68
TPE2 200-200	70
TPE2 100-140	59
TPE2 100-200	59
TPE2 100-250	61
TPE2 100-330	64
TPE2 100-370	66
TPE2 100-410	70
TPE2 D, TPE3 D 32-390	64
TPE2 D, TPE3 D 32-510	67
TPE2 D, TPE3 D 32-610	69
TPE2 D, TPE3 D 32-710	71
TPE2 D, TPE3 D 40-310	61

Pump type	Sound pressure level ISO 3743 [dB(A)]
TPE2 D, TPE3 D 40-380	69
TPE2 D, TPE3 D 40-470	66
TPE2 D, TPE3 D 40-530	66
TPE2 D, TPE3 D 40-560	69
TPE2 D, TPE3 D 40-630	66
TPE2 D, TPE3 D 50-310	69
TPE2 D, TPE3 D 50-360	66
TPE2 D, TPE3 D 50-370	66
TPE2 D, TPE3 D 50-450	68
TPE2 D, TPE3 D 50-560	69
TPE2 D, TPE3 D 50-700	72
TPE2 D, TPE3 D 50-730	76
TPE2 D, TPE3 D 50-740	74
TPE2 D, TPE3 D 50-750	75
TPE2 D, TPE3 D 65-220	60
TPE2 D, TPE3 D 65-280	67
TPE2 D, TPE3 D 65-310	69
TPE2 D, TPE3 D 65-430	68
TPE2 D, TPE3 D 65-580	70
TPE2 D, TPE3 D 65-680	72
TPE2 D, TPE3 D 65-740	76
TPE2 D, TPE3 D 65-750	74
TPE2 D, TPE3 D 65-760	75
TPE2 D, TPE3 D 80-210	62
TPE2 D, TPE3 D 80-250	65
TPE2 D, TPE3 D 80-310	68
TPE2 D, TPE3 D 80-340	67
TPE2 D, TPE3 D 80-440	70
TPE2 D, TPE3 D 80-520	75
TPE2 D, TPE3 D 80-560	74
TPE2 D, TPE3 D 80-570	75
TPE2 D, TPE3 D 100-100	44
TPE2 D, TPE3 D 100-130	54
TPE2 D, TPE3 D 100-160	56
TPE2 D, TPE3 D 100-170	58
TPE2 D, TPE3 D 100-230	62
TPE2 D, TPE3 D 100-260	62

Pump type	Sound pressure level ISO 3743 [dB(A)]
TPE2 D, TPE3 D 100-350	67
TPE2 D, TPE3 D 100-460	73
TPE2 D, TPE3 D 100-540	73
TPE2 D, TPE3 D 100-570	72
TPE2 D, TPE3 D 125-110	50
TPE2 D, TPE3 D 125-120	51
TPE2 D, TPE3 D 125-160	61
TPE2 D, TPE3 D 125-180	62
TPE2 D, TPE3 D 125-290	66
TPE2 D, TPE3 D 125-310	70
TPE2 D, TPE3 D 125-380	71
TPE2 D, TPE3 D 125-430	72
TPE2 D, TPE3 D 125-460	73
TPE2 D, TPE3 D 150-120	58
TPE2 D, TPE3 D 150-170	60
TPE2 D, TPE3 D 150-230	68
TPE2 D, TPE3 D 150-240	68
TPE2 D, TPE3 D 150-250	67
TPE2 D, TPE3 D 150-270	77
TPE2 D, TPE3 D 150-280	67
TPE2 D, TPE3 D 200-130	59
TPE2 D, TPE3 D 200-150	63
TPE2 D, TPE3 D 200-190	66
TPE2 D, TPE3 D 200-200	67
TPE2 D, TPE3 D 200-50	48
TPE2 D, TPE3 D 200-70	58
TPE2 D, TPE3 D 200-90	58

24. Factory settings

●	Function is enabled.
❍	Function is disabled.
-	Function is not available.

Settings	TPE3, TPE3 D	TPE2, TPE2 D
Setpoint	Auto	67 %
Operating mode	**Normal**	**Normal**
Set user-defined speed	67 %	67 %
Control mode	AUTOADAPT	**Const. curve**
Buttons on product	●	●
Controller	●	●
Ti	1.0	0.5
Kp	8.0	0.5
Operating range	●	●
Min.	25 %	25 %
Max.	100 %	100 %
Ramps	❍	❍
Ramp-up time	20 seconds	20 seconds
Ramp-down time	20 seconds	20 seconds
Sensor to be used		
	Master pump sensor	Master pump sensor
	Running pump sensor	Master pump sensor
Number	1	1
Radio communication	●	●
Analog input 1	❍	❍
Analog input 2	❍	❍
Analog input 3	❍	❍
Grundfos Direct Sensor	●	-
Grundfos differential pressure sensor	Feedback sensor	-
Grundfos temperature sensor [17)]	❍	-
Sensor	❍	❍
Temperature	❍	❍
Dry-run detection	❍	❍
Dry-running detection delay	10 s	10 s
Pt100/1000 input 1	❍/ **Other function**, liquid temperature [17)]	❍

Settings	TPE3, TPE3 D	TPE2, TPE2 D
Pt100/1000 input 2	❍	❍
Digital input 1	❍	❍
Digital input 2	❍	❍
Digital input/output 3	❍	❍
Digital input/output 4	❍	❍
Predefined setpoint	❍	❍
Analog output	❍	❍
External setpoint function	❍	❍
Signal relay 1	❍	❍
Signal relay 2	❍	❍
Limit 1 exceeded	❍	❍
Limit 2 exceeded	❍	❍
Limit 3 exceeded [18)]	❍	❍
Limit 4 exceeded [18)]	❍	❍
Standstill heating	❍	❍
Motor bearing monitoring	❍	❍

17) Some pumps are fitted with a Grundfos temperature sensor and some pumps are fitted with an external Pt100/1000 temperature sensor.

18) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Related information

12.11 Grundfos Direct Sensor

12.17 Controller (Controller settings)

25. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way.

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

	The crossed-out wheelie bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

See also end-of-life information at *www.grundfos.com/product-recycling*.

26. Document quality feedback

To provide feedback about this document, use your smart device to scan the QR code.

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A.1. Appendix

Inlet pressure stated in bar relative pressure (pressure gauge value measured on the suction side of the pump).

Pump type	p [bar]						
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C	150 °C
TPE2 25-50	0.1	0.1	0.2	0.5	-	-	-
TPE2 25-80	0.1	0.1	0.1	0.3	-	-	-
TPE2 25-90	0.1	0.1	0.2	0.5	-	-	-
TPE2 32-50	0.1	0.1	0.1	0.2	-	-	-
TPE2 32-70	0.1	0.1	0.2	0.5	-	-	-
TPE2 32-90	0.1	0.1	0.2	0.5	-	-	-
TPE2, TPE3 32-390	0.1	0.1	0.13	0.85	1.41	3.02	4.17
TPE2, TPE3 32-510	0.1	0.1	0.22	0.94	1.50	3.11	4.26
TPE2, TPE3 32-610	0.1	0.1	0.32	1.04	1.60	3.21	4.36
TPE2, TPE3 32-710	0.1	0.1	0.34	1.06	1.62	3.23	4.38
TPE2 40-50	0.1	0.1	0.1	0.3	-	-	-
TPE2 40-70	0.1	0.1	0.2	0.5	-	-	-
TPE2 40-90	0.1	0.1	0.2	0.5	-	-	-
TPE2, TPE3 40-310	0.1	0.1	0.1	0.76	1.32	2.93	4.08
TPE2, TPE3 40-380	0.1	0.1	0.16	0.88	1.44	3.05	4.20
TPE2, TPE3 40-470	0.1	0.1	0.23	0.95	1.51	3.12	4.27
TPE2, TPE3 40-530	0.1	0.1	0.1	0.74	1.30	2.93	4.08
TPE2, TPE3 40-560	0.1	0.1	0.35	1.07	1.63	3.24	4.39
TPE2, TPE3 40-630	0.1	0.1	0.1	0.76	1.32	2.93	4.08
TPE2, TPE3 50-310	0.1	0.1	0.30	1.02	1.58	3.19	4.34
TPE2, TPE3 50-360	0.1	0.1	0.29	1.02	1.57	3.18	4.33
TPE2, TPE3 50-370	0.1	0.1	0.51	1.23	1.79	3.39	4.54
TPE2, TPE3 50-450	0.1	0.1	0.40	1.12	1.68	3.28	4.43
TPE2, TPE3 50-560	0.1	0.1	0.10	0.82	1.37	2.99	4.14
TPE2, TPE3 50-700	0.1	0.1	0.15	0.88	1.43	3.05	4.20
TPE2, TPE3 50-730	0.1	0.1	0.25	0.97	1.53	3.14	4.29
TPE2, TPE3 50-740	0.1	0.1	0.31	1.04	1.59	3.20	4.35
TPE2, TPE3 50-750	0.1	0.1	0.31	1.03	1.59	3.20	4.35
TPE2, TPE3 65-220	0.1	0.1	0.52	1.24	1.79	3.40	4.54
TPE2, TPE3 65-280	0.1	0.14	0.62	1.34	1.89	3.50	4.64
TPE2, TPE3 65-310	0.25	0.41	0.89	1.61	2.16	3.76	4.90
TPE2, TPE3 65-430	0.28	0.44	0.92	1.63	2.18	3.78	4.93

Pump type	p [bar]						
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C	150 °C
TPE2, TPE3 65-580	0.1	0.1	0.28	1.00	1.56	3.17	4.32
TPE2, TPE3 65-680	0.1	0.1	0.40	1.12	1.68	3.29	4.43
TPE2, TPE3 65-740	0.1	0.1	0.58	1.30	1.85	3.46	4.60
TPE2, TPE3 65-750	0.1	0.24	0.72	1.44	1.99	3.59	4.74
TPE2, TPE3 65-760	0.12	0.28	0.76	1.48	2.03	3.63	4.77
TPE2, TPE3 80-210	0.1	0.1	0.21	0.94	1.49	3.10	4.25
TPE2, TPE3 80-250	0.1	0.1	0.30	1.02	1.58	3.19	4.33
TPE2, TPE3 80-310	0.1	0.1	0.51	1.23	1.79	3.39	4.54
TPE2, TPE3 80-340	0.1	0.1	0.36	1.08	1.64	3.25	4.39
TPE2, TPE3 80-440	0.1	0.1	0.56	1.28	1.83	3.44	4.58
TPE2, TPE3 80-520	0.14	0.3	0.78	1.50	2.05	3.65	4.80
TPE2, TPE3 80-560	0.19	0.35	0.83	1.55	2.10	3.70	4.84
TPE2, TPE3 80-570	0.19	0.35	0.83	1.55	2.10	3.70	4.84
TPE2, TPE3 100-100	0.1	0.1	0.10	0.75	1.30	2.92	4.07
TPE2, TPE3 100-130	0.1	0.1	0.29	1.01	1.57	3.18	4.33
TPE2, TPE3 100-160	0.1	0.1	0.16	0.88	1.44	3.05	4.20
TPE2, TPE3 100-170	0.1	0.1	0.26	0.99	1.54	3.15	4.30
TPE2 100-200	0.1	0.1	0.5	1.2	1.8	3.4	4.7
TPE2, TPE3 100-230	0.1	0.1	0.30	1.02	1.58	3.19	4.33
TPE2 100-250	0.1	0.1	0.16	0.88	1.44	3.05	4.20
TPE2, TPE3 100-260	0.1	0.1	0.38	1.10	1.66	3.27	4.42
TPE2 100-330	0.3	0.5	1.0	1.7	2.3	3.9	5.2
TPE2, TPE3 100-350	0.28	0.44	0.92	1.63	2.18	3.78	4.92
TPE2 100-370	0.3	0.5	1.0	1.7	2.3	3.9	5.2
TPE2 100-410	0.5	0.7	1.2	1.9	2.5	4.1	5.4
TPE2, TPE3 100-460	0.38	0.54	1.02	1.73	2.28	3.88	5.02
TPE2, TPE3 100-540	0.43	0.59	1.06	1.77	2.32	3.92	5.06
TPE2, TPE3 100-570	1.04	1.19	1.65	2.35	2.90	4.48	5.62
TPE2, TPE3 125-110	0.1	0.10	0.20	0.92	1.48	3.09	4.24
TPE2, TPE3 125-120	0.1	0.10	0.32	1.04	1.59	3.20	4.35
TPE2 125-150	0.2	0.4	0.8	1.8	2.3	3.9	5.2
TPE2, TPE3 125-160	0.1	0.1	0.45	1.16	1.72	3.33	4.47
TPE2, TPE3 125-180	0.1	0.17	0.65	1.37	1.92	3.53	4.67
TPE2 125-190	0.1	0.1	0.2	0.9	1.5	3.1	4.4
TPE2 125-230	0.1	0.1	0.3	1.0	1.6	3.2	4.5
TPE2, TPE3 125-290	0.1	0.16	0.65	1.36	1.92	3.52	4.67
TPE2 125-300	0.3	0.5	1.0	1.7	2.3	3.9	5.2
TPE2, TPE3 125-310	0.17	0.33	0.81	1.52	2.08	3.68	4.82

Pump type	p [bar]						
	20 °C	60 °C	90 °C	110 °C	120 °C	140 °C	150 °C
TPE2 125-340	0.1	0.1	0.3	1.0	1.5	3.2	4.5
TPE2, TPE3 125-380	0.19	0.36	0.84	1.55	2.1	3.7	4.85
TPE2, TPE3 125-430	0.43	0.59	1.07	1.78	2.32	3.92	5.07
TPE2, TPE3 125-460	0.1	0.16	0.64	1.35	1.91	3.51	4.66
TPE2, TPE3 150-120	0.1	0.1	0.53	1.24	1.80	3.40	4.55
TPE2 150-130	0.1	0.1	0.4	1.1	1.6	3.3	4.6
TPE2 150-160	0.1	0.1	0.4	1.1	1.7	3.3	4.6
TPE2, TPE3 150-170	0.1	0.19	0.67	1.38	1.94	3.54	4.69
TPE2 150-200	0.1	0.1	0.4	1.1	1.7	3.3	4.6
TPE2 150-220	0.1	0.1	0.5	1.2	1.8	3.4	4.7
TPE2, TPE3 150-230	0.24	0.41	0.89	1.60	2.15	3.75	4.89
TPE2, TPE3 150-240	0.5	0.66	1.13	1.84	2.39	3.98	5.13
TPE2, TPE3 150-250	0.1	0.1	0.1	0.79	1.35	2.96	4.11
TPE2 150-260	0.1	0.1	0.5	1.2	1.8	3.4	4.7
TPE2, TPE3 150-270	0.46	0.62	1.1	1.81	2.35	3.95	5.09
TPE2, TPE3 150-280	0.1	0.1	0.1	0.81	1.37	2.98	4.13
TPE2, TPE3 200-130	0.1	0.1	0.29	1.01	1.57	3.18	4.33
TPE2, TPE3 200-150	0.1	0.1	0.21	0.93	1.49	3.1	4.25
TPE2 200-160	0.3	0.5	1.0	1.7	2.3	3.9	5.2
TPE2, TPE3 200-190	0.1	0.18	0.67	1.38	1.93	3.54	4.68
TPE2, TPE3 200-200	0.1	0.12	0.6	1.32	1.87	3.48	4.62
TPE2, TPE3 200-50	0.11	0.28	0.76	1.47	2.03	3.63	4.77
TPE2, TPE3 200-70	0.1	0.1	0.58	1.29	1.85	3.45	4.6
TPE2, TPE3 200-90	0.1	0.1	0.49	1.2	1.76	3.37	4.51

Appendix B

B.1. 中国 RoHS

产品中有害物质的名称及含有信息表

部件名称	有害物质									
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr6+)	多溴联苯 (PBBs)	多溴二苯醚 (PBDEs)	邻苯二甲酸二正丁酯（DBP）	邻苯二甲酸二异丁酯 (DIBP)	邻苯二甲酸丁基苄酯（BBP）	邻苯二甲酸二（2-乙基）己酯（DEHP）
泵壳	X	O	O	O	O	O	O	O	O	O
印刷电路板	X	O	O	O	O	O	O	O	O	O
紧固件	X	O	O	O	O	O	O	O	O	O
管件	X	O	O	O	O	O	O	O	O	O
定子	X	O	O	O	O	O	O	O	O	O
转子	X	O	O	O	O	O	O	O	O	O

注 1：O：表示该有害物质在该部件所有均质材料中的含量均不超出电器电子产品有害物质限制使用国家标准要求。
X：表示该有害物质至少在该部件的某一均质材料中含量超出电器电子产品有害物质限制使用国家标准要求。
注 2：以上未列出的部件，表明其有害物质含量均不超出电器电子产品有害物质限制使用国家标准要求。



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